

BT_Tuan_8

VuTienDat

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Bài 1:

*# . Tập dữ liệu USJudgeRatings {datasets} trong R gồm 12 chỉ số đánh giá cho 43
thẩm phán. Sử dụng dữ liệu trên, hãy cho biết:*

```
data(USJudgeRatings)
View(USJudgeRatings)
```

Thực hiện phân tích thành phần chính trên tập dữ liệu, hãy cho biết nếu dùng 5 thành phần chính đầu

```
pca = princomp(USJudgeRatings)
summary(pca)
```

Importance of components:

```
##              Comp.1      Comp.2      Comp.3      Comp.4      Comp.5
## Standard deviation  2.9944812  0.97631616  0.54821237  0.46436892  0.260897305
## Proportion of Variance 0.8476102  0.09010189  0.02840865  0.02038352  0.006434164
## Cumulative Proportion 0.8476102  0.93771205  0.96612070  0.98650423  0.992938391
##              Comp.6      Comp.7      Comp.8      Comp.9
## Standard deviation  0.172098999  0.127444176  0.107411902  0.0862216855
## Proportion of Variance 0.002799688  0.001535299  0.001090581  0.0007027259
## Cumulative Proportion 0.995738079  0.997273378  0.998363959  0.9990666848
##              Comp.10     Comp.11     Comp.12
## Standard deviation  0.0703843063  0.0550575991  0.0434546479
## Proportion of Variance 0.0004682789  0.0002865415  0.0001784947
## Cumulative Proportion 0.9995349638  0.9998215053  1.0000000000
```

Tỷ lệ dữ liệu của 5 thành phần chính đầu tiên là 99.3%. Trong đó PC1 chiếm tới 84.4%

*# Gọi 5 thành phần chính là Y1 đến Y5. Hãy cho biết biến "CONT" và "CFMG" đóng
góp bao nhiêu trong mỗi thành phần trên*

```
loadings_cov = pca$loadings[, 1:5]
loadings_cov[c("CONT", "CFMG"), ]
```

```
##              Comp.1      Comp.2      Comp.3      Comp.4      Comp.5
## CONT  0.00599117  0.9332488   0.3198540   0.11293271   0.09462326
## CFMG -0.27201855  0.1631993  -0.1893545  -0.02481803  -0.47964024
```

Đóng góp = Bình phương hệ số tải

```
contribution = loadings_cov[c("CONT", "CFMG"), ] ^ 2
```

```
contri = contribution * 100
contri
```

```
##          Comp.1    Comp.2    Comp.3    Comp.4    Comp.5
## CONT 0.003589412 87.095340 10.230659 1.27537972 0.8953561
## CFMG 7.399409222  2.663401  3.585513 0.06159348 23.0054762
```

#Nx: CONT đóng góp lớn nhất cho Y2 87% , trong khi đó CFMG đóng góp nhỏ nhất cho Y4

Biến CONT gần như không đóng góp vào CP1, Đóng góp rất mạnh vào CP2, có ảnh hưởng nhỏ CP3-5

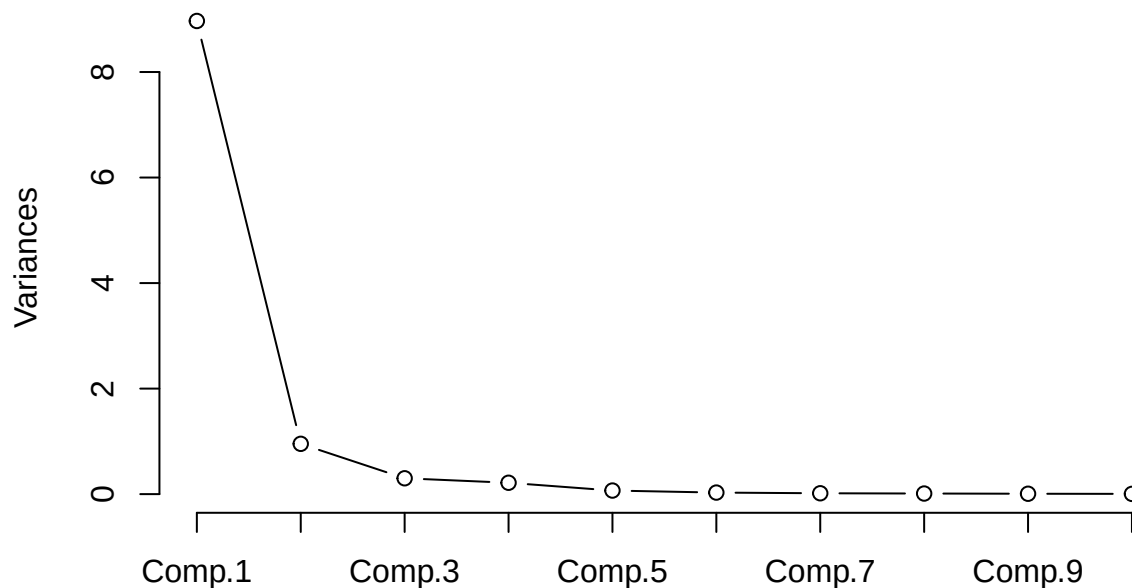
Biến CFMG đóng góp đáng kể vào PC1 theo chiều âm, PC5 theo chiều âm, nhưng yếu hơn so với CONT ở PC2

Vẽ biểu đồ tỉ lệ các thành phần trong trường hợp phân tích trên tập dữ liệu và sử dụng ma trận tương

Biểu đồ Scree plot cho ma trận HIỆP PHƯƠNG SAI

```
screepLOT(pca, type = "lines", main = "Scree Plot (Sử dụng Ma trận Hiệp phương sai)")
```

Scree Plot (S. d.ng Ma tr.n Hi.p phuong sai)



```
pca_cor <- princomp(USJudgeRatings, cor = TRUE)
# Thực hiện PCA với ma trận tương quan
summary(pca_cor)
```

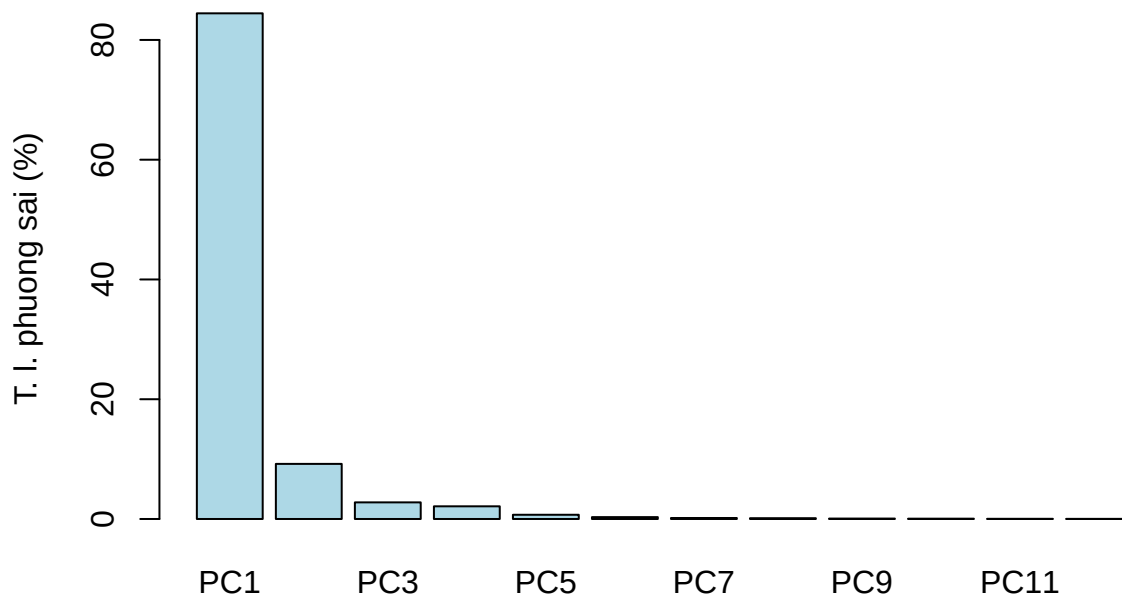
Importance of components:

```
##          Comp.1    Comp.2    Comp.3    Comp.4    Comp.5
## Standard deviation 3.1833165 1.05078398 0.5769763 0.50383231 0.290607615
```

```
## Proportion of Variance 0.8444586 0.09201225 0.0277418 0.02115392 0.007037732
## Cumulative Proportion 0.8444586 0.93647089 0.9642127 0.98536661 0.992404341
##                               Comp.6      Comp.7      Comp.8      Comp.9
## Standard deviation      0.193095982 0.140295449 0.124158319 0.0885069038
## Proportion of Variance 0.003107172 0.001640234 0.001284607 0.0006527893
## Cumulative Proportion 0.995511513 0.997151747 0.998436354 0.9990891437
##                               Comp.10     Comp.11     Comp.12
## Standard deviation      0.0749114592 0.0570804224 0.0453913429
## Proportion of Variance 0.0004676439 0.0002715146 0.0001716978
## Cumulative Proportion 0.9995567876 0.9998283022 1.0000000000
```

```
pcs <- (pca_cor$sdev^2) / sum(pca_cor$sdev^2) * 100
barplot(pcs,
        names.arg = paste0("PC", 1:length(pcs)),
        main = "Tỷ lệ phương sai của các thành phần chính",
        ylab = "Tỷ lệ phương sai (%)",
        col = "lightblue")
```

T. l. phương sai c.a các thành ph.n chính

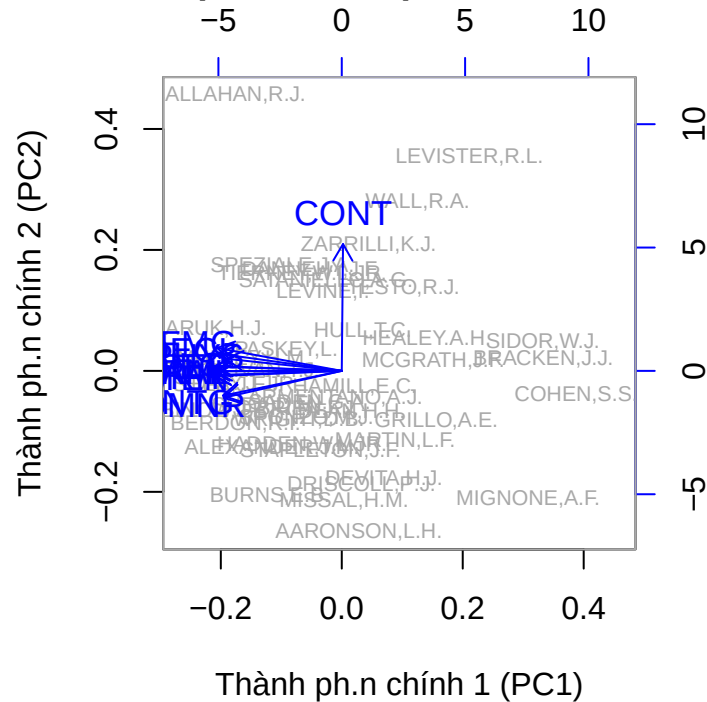


#4. Vẽ biểu đồ hai thành phần trong trường hợp sử dụng ma trận tương quan.

```
pca_cor <- princomp(USJudgeRatings, cor = TRUE)
biplot(pca_cor,
       cex = c(0.7, 1.1),
       col = c("darkgray", "blue"),
       xlab = "Thành phần chính 1 (PC1)",
       ylab = "Thành phần chính 2 (PC2)",
```

```
main = "Biểu đồ hai thành phần chính (dựa trên ma trận tương quan)")
```

Biểu đồ hai thành phần chính (dựa trên ma trận tương quan)



#5. Tiến hành phân tích nhân tố cho bộ dữ liệu trên với 4 nhân tố. Hãy cho biết tỉ lệ tổng phương sai giải thích bởi 4 nhân tố (Factor Analysis) với 4 nhân tố

```
fa4 <- factanal(USJudgeRatings, factors = 4, rotation = "varimax")
```

```
# Xem kết quả
```

```
print(fa4, digits = 3, cutoff = 0.3)
```

```
##
## Call:
## factanal(x = USJudgeRatings, factors = 4, rotation = "varimax")
##
## Uniquenesses:
##  CONT  INTG  DMNR  DILG  CFMG  DECI  PREP  FAMI  ORAL  WRIT  PHYS  RTEN
## 0.749 0.013 0.032 0.023 0.014 0.025 0.008 0.005 0.005 0.005 0.047 0.006
##
## Loadings:
##      Factor1 Factor2 Factor3 Factor4
## CONT                0.499
## INTG  0.726   0.596
## DMNR  0.666   0.616   0.305
## DILG  0.916   0.345
## CFMG  0.874   0.326
```

```
## DECI 0.886
## PREP 0.935
## FAMI 0.944
## ORAL 0.881 0.335 0.323
## WRIT 0.911
## PHYS 0.682 0.622
## RTEN 0.797 0.458 0.380
##
##
## Factor1 Factor2 Factor3 Factor4
## SS loadings 7.829 1.650 1.084 0.507
## Proportion Var 0.652 0.138 0.090 0.042
## Cumulative Var 0.652 0.790 0.880 0.922
##
## Test of the hypothesis that 4 factors are sufficient.
## The chi square statistic is 49.78 on 24 degrees of freedom.
## The p-value is 0.00151
```

```
# Tính tổng phương sai được giải thích bởi 4 nhân tố
total_var_4 <- sum(fa4$PVAL)
total_var_4
```

```
## [1] 0.001510792
```

```
# Bài 2:
# Bộ dữ liệu wdbc {kdevine} trong R được thu thập từ Bệnh viện Đại học Wisconsin, Madison, và được sử dụng để hỗ trợ
# install.packages("kdevine")
library(kdevine)
```

```
## Warning: package 'kdevine' was built under R version 4.4.3
```

```
## The kdevine package is no longer actively developed. Consider using
## - the 'kde1d' package for marginal estimation,
## - the functions vine() and vinecop() from the 'rvinecopulib'
## package as replacements for kdevine() and kdevinecop().
```

```
data(wdbc)
#1. Tiến hành tiền xử lý dữ liệu, bao gồm việc loại bỏ các cột không cần thiết và chuyển đổi các biến
# loại bỏ các cột không cần thiết (id, diagnosis)
# Giữ lại các cột đặc trưng
data = wdbc[, -c(1, 2)]
diag = as.numeric(as.factor(wdbc$diagnosis))

# 2. Chuẩn hóa dữ liệu để đảm bảo rằng tất cả các biến có cùng quy mô:
data_scale = scale(data)
data_scale
```

```
## mean texture mean perimeter mean area mean smoothness
## [1,] -2.071512e+00 1.26881726 0.983509520 1.567087457
## [2,] -3.533215e-01 1.68447255 1.907030269 -0.826235447
## [3,] 4.557859e-01 1.56512598 1.557513185 0.941382123
```

##	[4,]	2.535091e-01	-0.59216612	-0.763791736	3.280666839
##	[5,]	-1.150804e+00	1.77501133	1.824623802	0.280125349
##	[6,]	-8.346009e-01	-0.38680772	-0.505205927	2.235454519
##	[7,]	1.605082e-01	1.13712450	1.094332010	-0.123027974
##	[8,]	3.581350e-01	-0.07280278	-0.218772414	1.602638897
##	[9,]	5.883121e-01	-0.18391855	-0.383869508	2.199903080
##	[10,]	1.104467e+00	-0.32919213	-0.508615849	1.581308033
##	[11,]	9.184652e-01	0.44162208	0.406095932	-1.016791162
##	[12,]	-3.254213e-01	0.47866067	0.358357013	0.052596137
##	[13,]	1.281168e+00	1.66389556	1.330185001	0.073927000
##	[14,]	1.083542e+00	0.48277607	0.363187737	-0.878140548
##	[15,]	7.719888e-01	0.06712078	-0.217635773	1.190242199
##	[16,]	1.918224e+00	0.19593277	0.011113212	1.247124502
##	[17,]	1.953835e-01	0.11403633	0.084142391	0.164227657
##	[18,]	3.232597e-01	0.66385362	0.408937534	1.467543427
##	[19,]	6.650378e-01	1.56512598	1.719484517	0.138630620
##	[20,]	-1.146154e+00	-0.18556471	-0.251735001	0.101657123
##	[21,]	-8.322759e-01	-0.26087651	-0.383301188	0.792066077
##	[22,]	-1.592558e+00	-1.30166089	-1.082619515	0.429441395
##	[23,]	-1.169404e+00	0.43339128	0.140690277	0.777845502
##	[24,]	8.719647e-01	1.86143471	2.128675248	-0.147913982
##	[25,]	4.860112e-01	0.74204620	0.709579058	1.119139320
##	[26,]	-6.718494e-01	0.98897014	0.732596037	1.581308033
##	[27,]	5.208865e-01	0.22391748	-0.028669220	0.642750032
##	[28,]	2.232838e-01	1.24000947	1.247778534	-0.139381636
##	[29,]	1.390444e+00	0.42927589	0.220255141	0.841838093
##	[30,]	-9.857273e-01	0.94781615	0.853079974	0.150007081
##	[31,]	1.353244e+00	1.35112524	1.230728920	0.713852911
##	[32,]	-1.370945e-01	-0.57776223	-0.608924411	1.033815866
##	[33,]	1.090517e+00	0.85727737	0.694518566	1.659521200
##	[34,]	1.669447e+00	1.47870261	1.441007491	-0.167111759
##	[35,]	-3.277463e-01	0.61858424	0.432806994	0.543206001
##	[36,]	5.348366e-01	0.74616160	0.609838817	-0.018506742
##	[37,]	5.650619e-01	0.06835540	-0.062200127	0.132942390
##	[38,]	-2.021951e-01	-0.38516156	-0.372503099	-0.464321793
##	[39,]	1.374169e+00	0.14695952	0.124777304	-0.177066162
##	[40,]	3.558100e-01	-0.14687996	-0.271910377	0.372559092
##	[41,]	5.325116e-01	-0.23824182	-0.261112289	-1.048076429
##	[42,]	4.790361e-01	-0.82592078	-0.806415771	1.872829837
##	[43,]	1.283494e+00	1.49516420	1.276194557	-0.394640972
##	[44,]	2.302588e-01	-0.19132627	-0.311692810	0.550316289
##	[45,]	5.859871e-01	-0.26951885	-0.350622761	0.055440252
##	[46,]	-3.928469e-01	1.30585585	1.196629693	0.962712987
##	[47,]	-5.695485e-01	-1.65681982	-1.287214880	-0.736645819
##	[48,]	-1.463946e-01	-0.24647261	-0.341813794	1.382219972
##	[49,]	-1.083378e+00	-0.57323529	-0.584202471	0.479213410
##	[50,]	6.999132e-01	-0.20819940	-0.266795493	-0.628569443
##	[51,]	5.371617e-01	-0.70986653	-0.645012760	-0.710337754
##	[52,]	-6.857996e-01	-0.19585321	-0.236106188	-1.387237161
##	[53,]	-2.440455e-01	-0.66912408	-0.617449218	-0.977684579
##	[54,]	-1.370945e-01	1.16593229	1.074440794	1.311117093
##	[55,]	6.348126e-01	0.21774438	0.164559736	-0.412416691
##	[56,]	-1.254694e-01	-0.76665904	-0.698719043	-0.079655218
##	[57,]	-1.673198e-01	1.37993303	1.412591468	0.635639744

##	[58,]	5.348366e-01	0.14737106	0.005714168	1.232903927
##	[59,]	4.731766e-03	-0.38516156	-0.362841651	-1.120601365
##	[60,]	-1.743684e+00	-1.54858483	-1.222994668	0.082459346
##	[61,]	-1.025253e+00	-1.12840260	-0.974638627	1.211573063
##	[62,]	3.930103e-01	-1.53541555	-1.230666995	1.986594443
##	[63,]	6.650378e-01	0.18317503	-0.026111778	0.607198592
##	[64,]	-1.262405e+00	-1.34857644	-1.119560344	-1.361640125
##	[65,]	1.057966e+00	-0.38186924	-0.442974836	1.126249608
##	[66,]	1.081217e+00	0.22350594	0.038108434	1.481764003
##	[67,]	3.999854e-01	-1.31112631	-1.095406725	0.571647153
##	[68,]	-5.804381e-02	-0.83003618	-0.741058918	-1.064430091
##	[69,]	-4.556225e-01	-1.36544958	-1.149113008	0.728073486
##	[70,]	-6.509243e-01	-0.43619250	-0.433029228	0.138630620
##	[71,]	4.697360e-01	1.30174045	1.350076217	-0.445835045
##	[72,]	-1.081053e+00	-1.36544958	-1.167583423	0.104501238
##	[73,]	1.216068e+00	0.91489296	0.780050795	0.763624926
##	[74,]	-8.136758e-01	-0.06333736	-0.201154480	0.308566501
##	[75,]	-6.439492e-01	-0.52590820	-0.522823861	-0.329937352
##	[76,]	8.378249e-02	0.49923767	0.462643818	-0.332781467
##	[77,]	-1.941311e+00	-0.16704542	-0.271910377	2.327888262
##	[78,]	-7.299750e-01	1.16181689	0.997717532	0.720963198
##	[79,]	1.088192e+00	2.12893563	1.676860482	2.292336822
##	[80,]	-2.998460e-01	-0.36129224	-0.422231139	0.211866585
##	[81,]	3.906853e-01	-0.74731666	-0.720031061	0.984043850
##	[82,]	-7.974006e-01	-0.22548408	-0.383301188	0.813396941
##	[83,]	1.306744e+00	3.27301653	3.475594740	0.706742623
##	[84,]	1.627597e+00	1.52808739	1.355759422	1.787506382
##	[85,]	-8.462260e-01	-0.61809314	-0.601252084	0.061839511
##	[86,]	-1.789449e-01	1.19885548	1.193788091	0.169204858
##	[87,]	5.046114e-01	0.09387087	-0.019007772	-0.136537521
##	[88,]	1.232343e+00	1.23589407	1.196629693	-0.431614469
##	[89,]	5.836621e-01	-0.50162735	-0.536463552	-0.614348868
##	[90,]	-9.415519e-01	0.15642494	-0.008493844	1.197352487
##	[91,]	1.099817e+00	0.10704015	0.022195461	-0.470721052
##	[92,]	8.068641e-01	0.33873711	0.208320412	-0.310028546
##	[93,]	-1.053153e+00	-0.29750356	-0.293222395	-1.621876662
##	[94,]	-2.300954e-01	-0.22095714	-0.283560947	0.415220819
##	[95,]	1.256329e-01	0.34285251	0.144100200	0.536095713
##	[96,]	8.696397e-01	1.66389556	1.730850926	-0.396774058
##	[97,]	-3.370464e-01	-0.58352379	-0.579087586	0.578757441
##	[98,]	1.512081e-01	-1.22881833	-1.024082507	0.429441395
##	[99,]	-1.499557e+00	-0.72550505	-0.688489275	-0.464321793
##	[100,]	1.116827e-01	0.10333629	-0.035204906	0.082459346
##	[101,]	1.323019e+00	-0.16128386	-0.205132723	-0.105252255
##	[102,]	-1.362381e+00	-1.98275941	-1.453164455	1.467543427
##	[103,]	2.860593e-01	-0.60698156	-0.557491409	-1.154019719
##	[104,]	2.565696e-02	-1.15309499	-1.013284418	0.294345925
##	[105,]	8.172307e-05	-1.01070219	-0.905871851	0.250973169
##	[106,]	-8.671512e-01	-0.19585321	-0.354316844	3.088689066
##	[107,]	-2.231203e-01	-0.69134724	-0.688773435	1.268455366
##	[108,]	-1.742949e-01	-0.53331592	-0.534758590	-0.824102360
##	[109,]	8.843253e-02	2.50343693	2.427043490	2.576748338
##	[110,]	4.581109e-01	-0.80205147	-0.734239072	-0.623592242
##	[111,]	-5.346732e-01	-1.21276828	-1.036301397	0.521875137

## [112,]	3.418599e-01	-0.40409239	-0.495828639	0.211155557
## [113,]	8.378249e-02	0.24120216	-0.071009095	-1.279160785
## [114,]	2.093336e-01	-0.96008279	-0.911270896	1.126249608
## [115,]	-8.043757e-01	-1.48685385	-1.204808414	1.325337669
## [116,]	5.208865e-01	-0.63537781	-0.614607615	0.093835806
## [117,]	-8.206508e-01	-1.36750728	-1.164173501	-0.123739003
## [118,]	-6.090739e-01	0.27453689	0.078459187	1.410661124
## [119,]	8.417395e-01	0.56508405	0.362903577	1.360889109
## [120,]	1.674833e-01	0.91489296	0.929519076	-0.877429520
## [121,]	-1.969211e+00	-0.76665904	-0.714916176	-0.187020565
## [122,]	-5.044479e-01	1.21120168	1.199471295	0.642750032
## [123,]	2.116587e-01	3.05490039	3.143127271	3.437093173
## [124,]	-1.952936e+00	0.09510549	-0.040319790	0.976933562
## [125,]	-6.741745e-01	-0.24153414	-0.288107511	-1.792523571
## [126,]	-4.835227e-01	-0.14523380	-0.188083109	-0.605105493
## [127,]	1.255593e+00	-0.17321851	-0.233832907	-0.268788876
## [128,]	-8.826909e-02	1.29350966	1.372809036	-1.008969846
## [129,]	-6.741745e-01	0.31322164	0.055726368	1.325337669
## [130,]	1.355569e+00	1.58158758	1.526255560	0.365448804
## [131,]	-1.394931e+00	-0.53043514	-0.565732056	0.728073486
## [132,]	4.425713e-02	0.40046809	0.267141579	0.912940971
## [133,]	5.232115e-01	0.58566104	0.440195160	0.315676789
## [134,]	-1.246130e+00	0.41281429	0.303514089	-0.123739003
## [135,]	6.092373e-01	1.16181689	1.193788091	-0.146491924
## [136,]	7.394385e-01	-0.42178861	-0.422231139	-0.413127720
## [137,]	-6.090739e-01	-0.70986653	-0.657231650	0.621419168
## [138,]	-9.066766e-01	-0.77818215	-0.724861785	0.002113093
## [139,]	-3.998219e-01	0.20087125	0.065956136	1.446212563
## [140,]	-1.371681e+00	-0.78065139	-0.767485819	1.424881700
## [141,]	-1.701834e+00	-1.26462230	-1.041132121	-0.274477106
## [142,]	-2.882209e-01	0.54039166	0.449288287	0.060417453
## [143,]	-4.602725e-01	-0.75348976	-0.729976669	0.912940971
## [144,]	-7.834505e-01	-0.33865755	-0.405465686	-0.681896603
## [145,]	-1.004328e+00	-0.97572130	-0.851313087	-1.310446052
## [146,]	-1.078728e+00	-0.57035451	-0.631088909	1.339558245
## [147,]	-6.299991e-01	-0.53413900	-0.633362190	0.905830684
## [148,]	-1.208194e-01	0.24161370	0.098350403	-1.065141120
## [149,]	-9.555021e-01	0.08234776	-0.042024751	0.237463622
## [150,]	-3.207712e-01	-0.15840308	-0.198597038	-1.203080705
## [151,]	3.465099e-01	-0.34812297	-0.385006149	1.218683351
## [152,]	3.279097e-01	-1.59261960	-1.281531676	-0.164267644
## [153,]	-9.183017e-01	-1.16009117	-1.007885374	0.770735214
## [154,]	-1.443757e+00	-0.86830939	-0.775726466	0.083881403
## [155,]	-9.183017e-01	-0.27404579	-0.329594904	-0.179199249
## [156,]	-3.137962e-01	-0.56376987	-0.552944845	-0.698250265
## [157,]	3.372098e-01	1.04658572	0.877517754	1.076477593
## [158,]	3.960709e-02	0.67619982	0.640243962	-1.557884071
## [159,]	-1.522807e+00	-0.62262007	-0.586191592	-0.231104350
## [160,]	-1.471657e+00	-0.95802509	-0.818634661	-1.508112056
## [161,]	2.070086e-01	-0.65307403	-0.668029738	0.891610108
## [162,]	-7.788004e-01	1.41285622	1.426799479	-0.669809113
## [163,]	-2.649707e-01	1.59393378	1.588770811	1.112029032
## [164,]	6.813130e-01	-0.49874657	-0.541010116	0.344117940
## [165,]	6.394626e-01	2.47462914	2.930007098	-0.851121454

## [166,]	1.093577e-01	0.14531336	0.100339524	-0.863919973
## [167,]	-2.227289e+00	-0.95473277	-0.844777402	-0.029883203
## [168,]	-1.138443e-01	0.71323841	0.657577736	-0.548223190
## [169,]	1.253268e+00	0.99308554	0.936907242	0.607198592
## [170,]	-5.439733e-01	0.17494423	0.088120634	0.155695311
## [171,]	-1.604183e+00	-0.53990056	-0.542146756	0.457882546
## [172,]	7.913245e-02	-0.25223417	-0.254292443	-0.418104922
## [173,]	-1.720434e+00	0.43339128	0.233042352	2.086138474
## [174,]	-1.064778e+00	-0.89547102	-0.801869207	0.301456213
## [175,]	-9.624771e-01	-1.00740987	-0.867510220	-0.600128292
## [176,]	-1.125229e+00	-1.54529251	-1.215322342	-0.354112331
## [177,]	-2.858959e-01	-1.12634490	-1.001633849	0.044774820
## [178,]	1.907335e-01	0.71323841	0.505836173	0.138630620
## [179,]	6.813130e-01	-0.40985395	-0.365114933	-2.406852445
## [180,]	-1.448407e+00	-0.43948482	-0.415127134	-0.637812818
## [181,]	5.999372e-01	3.70924881	4.532670797	0.927161547
## [182,]	1.692697e+00	2.08778165	1.864406234	1.261345078
## [183,]	2.372339e-01	0.37989110	0.317437940	-0.027750117
## [184,]	-1.015953e+00	-0.75883978	-0.718610259	-0.410283605
## [185,]	7.254884e-01	0.28606000	0.158308211	-0.411705663
## [186,]	-9.717772e-01	-1.16091425	-0.958725654	-0.262389617
## [187,]	-1.649948e-01	1.09597051	1.097173612	-0.745178165
## [188,]	-4.881728e-01	-0.71151269	-0.666608937	0.098101979
## [189,]	-4.416723e-01	-0.68723184	-0.642171158	0.308566501
## [190,]	-7.881005e-01	-0.54072364	-0.543283397	-1.106380790
## [191,]	8.905649e-01	0.09880935	-0.127841141	0.792066077
## [192,]	4.929862e-01	-0.40944241	-0.419105377	-0.630702530
## [193,]	-2.486956e-01	-1.28561084	-1.042268762	-1.909843321
## [194,]	1.760123e+00	-0.44524638	-0.504353446	0.500544274
## [195,]	9.114901e-01	0.34696791	0.046917401	0.571647153
## [196,]	-6.881246e-01	-0.38845388	-0.393530956	-1.205213791
## [197,]	6.975881e-01	-0.05510657	-0.187514789	1.680852064
## [198,]	5.929622e-01	1.04658572	1.048866373	-1.610500201
## [199,]	7.440886e-01	1.46224101	1.401225059	-0.791395036
## [200,]	2.163087e-01	0.10374783	-0.034636585	0.167782800
## [201,]	6.285730e-02	-0.55265829	-0.550955724	-0.035571433
## [202,]	7.056787e-03	0.95193155	0.843134366	-0.474987225
## [203,]	1.715947e+00	2.75447627	2.927165495	1.261345078
## [204,]	1.037041e+00	-0.01683336	-0.162224529	2.555417474
## [205,]	-1.603448e-01	-0.44771562	-0.491566235	0.233908478
## [206,]	-6.067489e-01	0.28029845	0.175357825	-0.540401874
## [207,]	-4.695726e-01	-1.19548360	-1.021525065	0.891610108
## [208,]	2.256088e-01	0.72970001	0.708726577	-0.614348868
## [209,]	7.557137e-01	-0.20367247	-0.356590126	0.273015061
## [210,]	-1.483282e+00	0.25519451	0.200648086	-1.033855853
## [211,]	6.627128e-01	1.75854973	1.804732586	-0.388241713
## [212,]	-8.129402e-02	-0.67735488	-0.644728600	-0.543957017
## [213,]	-1.905700e-01	3.97263434	5.240229770	1.268455366
## [214,]	1.457870e+00	0.92723915	0.832904598	0.301456213
## [215,]	1.050991e+00	0.03707837	-0.125567859	-0.123027974
## [216,]	-5.486233e-01	-0.04152575	-0.215930812	0.443661971
## [217,]	-2.184703e-01	-0.60286616	-0.632793870	-0.194130853
## [218,]	-4.207471e-01	-1.10782561	-0.948211726	-1.124867538
## [219,]	5.278616e-01	1.55277979	1.634236448	-0.179910277

## [220,]	3.064460e+00	1.48281801	1.614345231	-0.864631001
## [221,]	-1.425156e+00	-0.16828004	-0.244346835	0.007090294
## [222,]	-1.253105e+00	-0.13906070	-0.265943013	0.621419168
## [223,]	-4.091220e-01	-1.10494483	-0.971228704	0.692522047
## [224,]	2.232838e-01	0.43750668	0.302377448	0.436551683
## [225,]	-5.276981e-01	-0.30532282	-0.308282887	-0.846855282
## [226,]	-1.353081e+00	0.02226293	-0.038898989	0.191957779
## [227,]	-8.904015e-01	-1.04321384	-0.924342266	0.635639744
## [228,]	-8.787763e-01	0.22556364	0.084142391	-0.899471412
## [229,]	1.088192e+00	-0.43701558	-0.450363002	-1.232232885
## [230,]	7.068882e-01	-0.27610349	-0.431040106	0.884499820
## [231,]	-4.874373e-02	0.88196977	0.682299676	1.261345078
## [232,]	1.811273e+00	-0.83168234	-0.736512354	-1.957482250
## [233,]	3.376013e+00	-0.87160171	-0.761802615	-1.319689426
## [234,]	1.981000e+00	1.74620354	1.887139053	-0.339180726
## [235,]	-7.857755e-01	-1.30701091	-1.066422381	-0.833345735
## [236,]	4.557859e-01	-0.08967592	-0.146311556	-0.402462288
## [237,]	1.785698e+00	2.53224473	2.884541461	-0.090320650
## [238,]	5.046114e-01	1.66801096	1.850198223	-0.910847873
## [239,]	1.990300e+00	0.02390909	-0.088058708	-1.004703673
## [240,]	4.647799e+00	0.88196977	0.755044695	0.125121073
## [241,]	-8.578512e-01	-0.18885703	-0.226160580	-0.151469126
## [242,]	-9.880524e-01	-0.54977751	-0.506910888	-1.215879223
## [243,]	-2.556706e-01	-0.74237818	-0.754414449	-0.031305261
## [244,]	1.041691e+00	-0.14111840	-0.184389026	-1.132688855
## [245,]	9.789157e-01	1.52808739	1.421116275	0.450772259
## [246,]	1.326079e-01	-1.03909844	-0.901325288	0.756514638
## [247,]	-4.323722e-01	-0.32260749	-0.321922578	-1.721420692
## [248,]	-1.204279e+00	-0.28886122	-0.405465686	-0.622881213
## [249,]	1.378819e+00	-0.98600980	-0.874898386	0.014911611
## [250,]	-1.013628e+00	-0.74484742	-0.706391370	0.351228228
## [251,]	9.928658e-01	1.93139649	2.015011156	0.308566501
## [252,]	-1.952201e-01	-0.76912828	-0.703265607	-0.206929371
## [253,]	1.233079e-01	1.59393378	1.566037992	0.699632335
## [254,]	-5.137480e-01	0.86550817	0.776640872	0.315676789
## [255,]	9.381808e-03	1.42108702	1.460898707	0.507654562
## [256,]	-5.207231e-01	-0.02218337	-0.149153158	0.941382123
## [257,]	2.204202e+00	1.71328034	1.568879595	-0.267366818
## [258,]	-4.695726e-01	0.46219908	0.165980537	2.640740929
## [259,]	9.091651e-01	0.75027700	0.337044996	1.033815866
## [260,]	3.317887e+00	0.48277607	0.255775170	0.706742623
## [261,]	1.806623e+00	1.68447255	1.799049381	0.258794485
## [262,]	8.766148e-01	0.78320019	0.790564724	-0.692562034
## [263,]	6.603878e-01	0.92312375	0.832336278	-0.452945332
## [264,]	2.100692e-02	0.33050631	0.294705122	-1.277027699
## [265,]	6.464377e-01	0.80789259	0.776925033	0.063972597
## [266,]	2.750582e+00	1.79970372	2.171299283	-0.118761802
## [267,]	-7.896900e-02	-0.93374423	-0.876603348	0.036953503
## [268,]	5.929622e-01	-0.19791091	-0.266795493	-1.194548360
## [269,]	-7.160249e-01	-0.39462698	-0.405465686	-0.150047068
## [270,]	2.558341e-01	-0.92469035	-0.880865751	0.841838093
## [271,]	-5.741986e-01	-0.06868738	-0.063336768	-2.280289321
## [272,]	-1.453057e+00	-0.81233996	-0.758392692	0.140763706
## [273,]	3.953354e-01	2.27708999	2.375894649	-0.167111759

## [274,]	-8.415760e-01	-1.25392227	-1.037153878	-0.425926238
## [275,]	1.206768e+00	0.95604695	0.977542155	-0.555333478
## [276,]	-4.486474e-01	-0.64895863	-0.623132422	1.858609261
## [277,]	-1.192654e+00	-0.83044772	-0.733954912	-0.182754393
## [278,]	1.605082e-01	1.19062468	1.270511353	-0.506983520
## [279,]	-3.370464e-01	-0.23577258	-0.234685387	-1.200236590
## [280,]	-9.555021e-01	-0.12259911	-0.191777192	-0.085343448
## [281,]	1.699672e+00	1.40874083	1.372809036	0.401000243
## [282,]	-1.225205e+00	-0.72962045	-0.646717721	-1.296225476
## [283,]	-2.579956e-01	1.44989481	1.392700252	0.521875137
## [284,]	-1.208194e-01	0.69266142	0.426839629	0.728073486
## [285,]	-8.346009e-01	-0.32466519	-0.392962636	-1.292670332
## [286,]	-2.068452e-01	-0.49956965	-0.471390859	-0.883828779
## [287,]	3.418599e-01	-0.58023147	-0.607787770	-0.733090675
## [288,]	-1.434456e+00	-0.41479243	-0.394951757	-1.906288177
## [289,]	1.558582e-01	-0.75102052	-0.741058918	-1.149042517
## [290,]	-9.291913e-02	-0.81480920	-0.735659873	-0.656299566
## [291,]	1.023827e-01	0.16712498	-0.011051286	-0.625014299
## [292,]	-4.409368e-02	0.20827896	0.092098878	-0.457922534
## [293,]	-7.602003e-01	-0.36334994	-0.401203282	0.294345925
## [294,]	-4.253972e-01	-0.67612026	-0.631373069	-0.898760383
## [295,]	-1.281005e+00	-0.41931937	-0.462581892	0.022021899
## [296,]	-1.399581e+00	-0.16087232	-0.205132723	-0.311450603
## [297,]	-1.613483e+00	-0.93950579	-0.827443628	-0.794950180
## [298,]	-2.672957e-01	-0.69834341	-0.635919632	0.236041564
## [299,]	-2.603207e-01	-0.03082571	-0.061915967	-2.175768089
## [300,]	8.835898e-01	-1.03374843	-0.911270896	0.365448804
## [301,]	-9.059411e-02	1.54454899	1.597295618	1.325337669
## [302,]	1.395830e-01	-0.47487725	-0.521687220	-0.842589109
## [303,]	1.062616e+00	1.75854973	1.682543687	0.827617517
## [304,]	-1.580197e-01	-1.03333689	-0.910986735	0.742294062
## [305,]	-2.626457e-01	-0.75637054	-0.715484497	-0.556755536
## [306,]	1.209093e+00	-0.73003199	-0.675417904	-1.537264236
## [307,]	-8.067007e-01	-0.32507673	-0.334141468	-0.799927382
## [308,]	-1.136854e+00	-1.46545377	-1.161047738	-1.870736738
## [309,]	-1.529782e+00	-0.25840727	-0.252019161	-1.606945057
## [310,]	-1.267055e+00	-0.38104616	-0.353180203	-0.912980959
## [311,]	-4.176866e-02	-0.72591659	-0.671155501	-0.584485658
## [312,]	-8.369260e-01	0.02925911	0.028446986	-1.434876090
## [313,]	-1.376331e+00	-0.39833083	-0.428482664	-0.598706234
## [314,]	-1.992462e+00	-0.75060898	-0.698434883	-0.738778906
## [315,]	-1.603448e-01	-1.55887333	-1.232371956	0.784955789
## [316,]	-5.672235e-01	-0.52590820	-0.492418716	-0.799927382
## [317,]	-1.211254e+00	-0.60574694	-0.549819083	-1.352396751
## [318,]	-9.756917e-02	1.10008591	1.057391180	0.078193173
## [319,]	-9.059411e-02	-1.31277247	-1.166162622	0.236041564
## [320,]	-5.323482e-01	-0.55018905	-0.504637606	-1.478248846
## [321,]	-7.229999e-01	-1.04732924	-0.939686919	0.692522047
## [322,]	8.610751e-02	1.61039537	1.759266949	-1.149042517
## [323,]	-1.387956e+00	-0.37651922	-0.426493543	1.211573063
## [324,]	5.162365e-01	1.80793452	1.730850926	1.467543427
## [325,]	-9.485270e-01	-0.57446991	-0.559764691	-0.684740718
## [326,]	-4.625975e-01	-0.44113098	-0.468833417	0.457882546
## [327,]	-1.490257e+00	-0.07979896	-0.109086565	-0.232526408

## [328,]	-3.161212e-01	-0.65348557	-0.593579758	-1.388659219
## [329,]	3.302348e-01	0.61446884	0.451277409	1.460433139
## [330,]	6.022623e-01	0.63916123	0.488502399	1.431991988
## [331,]	-8.787763e-01	0.56919945	0.393024562	-0.103119168
## [332,]	1.403185e-02	-0.30655744	-0.400350802	-0.040548635
## [333,]	1.326079e-01	-0.82427462	-0.760381813	0.642750032
## [334,]	-1.048503e+00	-0.84732086	-0.752709487	-0.945688283
## [335,]	-6.269385e-02	-0.57981993	-0.541294276	-0.940711082
## [336,]	3.976604e-01	0.81612338	0.749361490	1.104918745
## [337,]	-1.176379e+00	-0.32466519	-0.399498321	-0.123739003
## [338,]	4.976363e-01	1.27293266	1.242095330	-0.369754964
## [339,]	-4.091220e-01	-1.13416416	-0.977764390	0.308566501
## [340,]	1.157942e+00	2.59809111	3.103344838	0.749404350
## [341,]	-6.392991e-01	0.08975548	-0.038898989	0.081748317
## [342,]	-5.695485e-01	-1.24816071	-1.063864939	-0.821258245
## [343,]	-1.006653e+00	-0.84279392	-0.798459284	0.493433986
## [344,]	5.557618e-01	1.56101059	1.531938765	0.114455641
## [345,]	-8.927265e-01	-0.69710880	-0.666608937	1.325337669
## [346,]	-1.064778e+00	-1.06049852	-0.947075085	0.174893088
## [347,]	-9.059411e-02	-0.63002779	-0.595568880	-0.888805980
## [348,]	-1.057803e+00	0.11938635	0.039245075	-0.541112902
## [349,]	-7.578752e-01	-0.77982831	-0.716621138	-0.398196116
## [350,]	-1.006653e+00	-0.60657002	-0.648422683	1.382219972
## [351,]	-5.160730e-01	-0.75184360	-0.664619816	-1.475404731
## [352,]	-1.619343e-02	0.62269963	0.294705122	1.986594443
## [353,]	-4.253972e-01	3.38413230	3.850686244	1.318227381
## [354,]	1.499720e+00	0.24819833	0.175357825	0.429441395
## [355,]	-1.213579e+00	-0.85308241	-0.768054140	-1.679469994
## [356,]	-5.106875e-02	-0.41355781	-0.480483986	-0.622881213
## [357,]	-1.626698e-01	-0.28309967	-0.406034006	0.841838093
## [358,]	-7.160249e-01	-0.14194148	-0.173875098	-0.634968702
## [359,]	-8.834264e-01	-1.44981526	-1.176108230	-0.954931658
## [360,]	-2.254453e-01	-1.32306097	-1.069263984	0.322787076
## [361,]	-2.835709e-01	-0.51644278	-0.463150212	-1.564283330
## [362,]	5.301866e-01	-0.27692657	-0.309135368	-0.749444338
## [363,]	-1.045442e-01	-0.41561551	-0.449794682	0.028421158
## [364,]	-2.324204e-01	0.60212264	0.520612505	0.035531446
## [365,]	-5.439733e-01	-0.26704961	-0.291233273	-1.208057907
## [366,]	5.790120e-01	1.72151114	1.813257393	-0.345579985
## [367,]	1.753148e+00	1.71739574	1.645602857	0.191246750
## [368,]	-2.951960e-01	-0.56212371	-0.558343890	-0.287986653
## [369,]	-4.742226e-01	2.01370446	2.532182775	-0.179199249
## [370,]	6.069123e-01	2.27297460	2.350320228	0.706742623
## [371,]	9.300903e-01	0.70089221	0.527148190	0.075349058
## [372,]	-1.413531e+00	0.23379444	0.161718134	-1.189571158
## [373,]	-9.741022e-01	2.03016606	2.077526407	0.265904773
## [374,]	-4.509724e-01	1.76266513	1.932604689	-0.135115464
## [375,]	-7.485752e-01	-0.16992620	-0.215362492	-0.948532398
## [376,]	-7.485752e-01	0.58977644	0.379669031	0.173471031
## [377,]	2.163087e-01	-0.89794026	-0.899620326	-0.400329202
## [378,]	2.074001e+00	-0.25017647	-0.263669731	-1.506689998
## [379,]	-9.624771e-01	-0.15222998	-0.211100088	-0.972707377
## [380,]	-1.068693e-01	-0.76830520	-0.833410993	1.794616670
## [381,]	-1.471657e+00	-0.77406675	-0.763223416	1.943932716

## [382,]	-1.013628e+00	-0.87654019	-0.801869207	-1.172506467
## [383,]	7.975640e-01	-0.54401596	-0.588464874	-1.920508753
## [384,]	-4.207471e-01	-0.46623492	-0.545556679	0.557426577
## [385,]	-1.294955e+00	-0.25429187	-0.321354258	-0.905159642
## [386,]	9.300903e-01	0.08234776	0.027878666	-0.678341459
## [387,]	-1.208929e+00	-0.54278134	-0.548114121	-1.086471984
## [388,]	-7.276500e-01	-0.14811458	-0.165634451	-1.855805133
## [389,]	-8.811014e-01	-0.76501288	-0.747026283	-0.903737585
## [390,]	9.114901e-01	1.51985660	1.475106719	0.329897364
## [391,]	-1.643708e+00	-1.07901781	-0.947075085	0.255950370
## [392,]	-5.695485e-01	-1.51031162	-1.195146966	0.536095713
## [393,]	1.581832e-01	0.42927589	0.255206850	1.396440548
## [394,]	6.952631e-01	2.15774343	2.137200055	1.446212563
## [395,]	-3.649466e-01	-0.57200067	-0.593011438	0.464992834
## [396,]	-4.904978e-01	-0.09132208	-0.130114423	-1.131266797
## [397,]	-9.291913e-02	-0.15922616	-0.275036140	0.678301471
## [398,]	-4.253972e-01	-0.36705380	-0.416547935	-1.131977826
## [399,]	-1.036878e+00	-0.89135562	-0.786240394	-1.347419549
## [400,]	-4.718976e-01	-0.68764338	-0.633646351	-0.390374799
## [401,]	4.023104e-01	1.33466365	0.963618304	1.894160700
## [402,]	-1.948286e+00	-0.65142787	-0.602957046	-0.543245989
## [403,]	-2.324204e-01	-0.32054980	-0.368524856	-1.624720777
## [404,]	-7.253249e-01	-0.36170378	-0.418537056	0.172760002
## [405,]	-1.008978e+00	-0.56294679	-0.527938745	-0.678341459
## [406,]	-1.626698e-01	-0.88806330	-0.809541533	0.287235637
## [407,]	-1.029903e+00	0.50746846	0.412347457	-0.100275053
## [408,]	4.836862e-01	-0.38433848	-0.398930000	-1.482515019
## [409,]	3.186096e-01	1.06304732	0.957082619	0.514764850
## [410,]	-3.184462e-01	-0.55800831	-0.536463552	-0.676208372
## [411,]	-3.998219e-01	-0.80163993	-0.724861785	-0.553200392
## [412,]	-5.718735e-01	-0.86625169	-0.800448406	0.806286653
## [413,]	5.557618e-01	-1.32594175	-1.097111686	-1.185304985
## [414,]	6.557377e-01	0.22885596	0.110285132	-0.797083266
## [415,]	2.446004e+00	0.19510969	0.183598472	-0.935733880
## [416,]	4.371857e-01	-0.64113937	-0.628247306	0.097390950
## [417,]	5.604119e-01	-1.33211485	-1.090291841	0.571647153
## [418,]	4.162605e-01	0.44985288	0.421156424	1.112029032
## [419,]	-1.655333e+00	-0.45635796	-0.454341245	-0.605105493
## [420,]	4.929862e-01	-0.86501707	-0.780273029	0.386779668
## [421,]	-5.804381e-02	-0.73126661	-0.696729922	-0.775041374
## [422,]	-1.234505e+00	0.25725221	0.003440886	0.479213410
## [423,]	-7.602003e-01	-0.67941258	-0.700992325	0.884499820
## [424,]	-3.711862e-02	-0.10325673	-0.226160580	-0.411705663
## [425,]	-3.944364e-02	-1.23622605	-1.037722198	0.792066077
## [426,]	4.627610e-01	-1.18437203	-0.987709998	-1.080072724
## [427,]	-1.002002e+00	-1.00740987	-0.912975857	0.127965188
## [428,]	6.255125e-01	-0.95390969	-0.838241716	-0.593729033
## [429,]	-6.206990e-01	-0.88477099	-0.777999748	-1.055897746
## [430,]	-3.765717e-01	-0.45224256	-0.436439151	-1.237210087
## [431,]	7.533886e-01	0.41692969	0.085563192	0.221109960
## [432,]	-3.742467e-01	-0.43207711	-0.531632828	0.642750032
## [433,]	5.820726e-02	1.72151114	1.691068494	1.204462775
## [434,]	6.231874e-01	1.30585585	1.293244171	0.386779668
## [435,]	-5.462983e-01	0.12020943	0.053453086	-0.506272492

## [436,]	7.680743e-02	-0.03494111	-0.157393805	0.685411759
## [437,]	5.820726e-02	-0.38269232	-0.413990493	-0.355534388
## [438,]	-7.695003e-01	-0.09008746	-0.124147058	-0.837611907
## [439,]	7.215739e-02	-0.13535684	-0.177000861	-0.676919401
## [440,]	-8.439010e-01	-0.09790671	-0.137502588	-1.187438072
## [441,]	-4.858477e-01	-0.83291696	-0.805279130	-0.512671751
## [442,]	1.425320e+00	0.84081578	0.778345834	-0.927912564
## [443,]	-8.136758e-01	-0.14811458	-0.196039596	-0.582352572
## [444,]	-2.254453e-01	-1.03498305	-0.892232160	-1.062297005
## [445,]	-5.672235e-01	1.05070112	0.952251895	-0.489918829
## [446,]	1.302094e+00	-0.59093150	-0.606935289	0.472103122
## [447,]	2.032150e+00	1.04247032	0.928382435	0.256661399
## [448,]	-3.788968e-01	0.16095188	0.056578849	-0.324960150
## [449,]	1.170683e-02	0.09387087	0.013670654	-0.887383923
## [450,]	2.860593e-01	1.89847330	2.071843202	0.034109388
## [451,]	5.232115e-01	-0.62303161	-0.633362190	-2.149460023
## [452,]	1.327669e+00	1.47047181	1.523413958	0.486323698
## [453,]	2.078651e+00	-0.62550085	-0.603525366	-0.852543512
## [454,]	-1.234505e+00	0.07782082	-0.030374182	0.962712987
## [455,]	-4.974728e-01	-0.46705800	-0.460308610	-0.748733309
## [456,]	2.657581e+00	-0.23165718	-0.277593582	-0.278032250
## [457,]	2.325103e+00	-0.70369343	-0.681385269	-0.198397026
## [458,]	1.385794e+00	-0.32384211	-0.332436506	-0.600839321
## [459,]	1.357894e+00	-0.38516156	-0.382732867	-0.900893470
## [460,]	2.071676e+00	-1.24651455	-1.034312275	-1.174639553
## [461,]	1.827549e+00	0.79143099	0.784313199	0.186269549
## [462,]	1.622947e+00	3.90678796	5.245912975	0.856058668
## [463,]	1.790348e+00	0.01156290	-0.024975137	-1.877847026
## [464,]	-2.161453e-01	-0.74443588	-0.688205115	-0.802060468
## [465,]	-2.486956e-01	-0.31643440	-0.334141468	-1.542952466
## [466,]	1.953835e-01	-0.20984556	-0.318228495	-0.961330917
## [467,]	3.372098e-01	-0.24647261	-0.335278109	-0.683318660
## [468,]	-2.765958e-01	-1.27203002	-1.047383646	-0.942133139
## [469,]	9.393903e-01	1.11243210	0.925256673	-0.246746984
## [470,]	-2.579956e-01	-0.64155091	-0.699287364	1.503094866
## [471,]	-1.859200e-01	-1.25433381	-1.039427160	-0.490629858
## [472,]	2.057726e+00	-0.62220853	-0.582497509	-0.628569443
## [473,]	-1.013628e+00	0.18440965	0.090962237	-1.093582271
## [474,]	2.483205e+00	-0.59875076	-0.538452673	-1.377282758
## [475,]	-8.532011e-01	-0.88724022	-0.841083319	0.308566501
## [476,]	-8.276259e-01	-0.37363844	-0.420526178	-0.423793152
## [477,]	2.883844e-01	0.01814753	-0.103687521	-0.501295290
## [478,]	-6.206990e-01	-0.12342219	-0.157677965	-1.996588834
## [479,]	-1.092678e+00	-0.73990894	-0.710369613	0.585867728
## [480,]	5.123219e-02	0.73381540	0.457244774	0.443661971
## [481,]	-2.928710e-01	-0.56294679	-0.567152857	-0.390374799
## [482,]	-1.154338e-02	-0.13329914	-0.147732357	-1.169662352
## [483,]	-1.215904e+00	-0.19132627	-0.308567047	0.763624926
## [484,]	-3.835468e-01	-0.17321851	-0.238095310	0.223243046
## [485,]	-1.862260e+00	0.44573748	0.262310856	0.564536865
## [486,]	-6.695244e-01	-0.37528460	-0.506342567	-0.086765506
## [487,]	-5.672235e-01	0.09222472	0.031572749	-0.707493639
## [488,]	-1.091943e-01	1.48693340	1.455215502	0.891610108
## [489,]	-7.253249e-01	-0.67817796	-0.666040617	1.168911336

## [490,]	2.116587e-01	0.62269963	0.576023750	-1.520910574
## [491,]	7.324634e-01	-0.56747373	-0.535326911	-1.026745565
## [492,]	-1.408881e+00	0.93135455	0.958219260	-1.278449757
## [493,]	2.953594e-01	1.08773971	1.000559134	0.265904773
## [494,]	-1.501882e+00	-0.54072364	-0.504637606	-1.609789172
## [495,]	2.907094e-01	-0.32548827	-0.330163225	-1.636097238
## [496,]	2.139837e-01	0.17082883	0.073912623	-0.034860404
## [497,]	-2.603207e-01	-0.38186924	-0.481052307	0.799176365
## [498,]	-4.602725e-01	-0.47405417	-0.496681120	-0.503428376
## [499,]	-4.114471e-01	1.20708628	1.173896874	0.344117940
## [500,]	4.534609e-01	1.88612710	1.889980655	0.863168956
## [501,]	-5.927987e-01	0.27824075	0.098066242	0.175604117
## [502,]	1.209093e+00	0.01485522	-0.167623573	1.410661124
## [503,]	-6.904496e-01	-0.44113098	-0.507479208	1.382219972
## [504,]	1.256329e-01	2.47462914	2.918640689	-0.209062458
## [505,]	-1.492582e+00	-1.25433381	-1.154227893	4.766717009
## [506,]	-1.429806e+00	-1.14609882	-1.086597758	2.071917898
## [507,]	1.744583e-01	-0.51438508	-0.573404382	0.941382123
## [508,]	-5.044479e-01	-0.85267087	-0.819487141	1.638190336
## [509,]	-8.346009e-01	0.52393006	0.468611183	-0.148625011
## [510,]	1.083542e+00	0.48689147	0.217129379	1.559977170
## [511,]	-1.069428e+00	-0.64443169	-0.650411804	-1.092871243
## [512,]	-1.067103e+00	0.11074401	0.073344302	-0.827657504
## [513,]	2.860593e-01	-0.13700300	-0.279014383	1.012485002
## [514,]	-1.308905e+00	0.09551703	0.011113212	0.139341649
## [515,]	-5.106875e-02	0.21774438	0.133586271	-0.299363114
## [516,]	-1.580197e-01	-0.79052835	-0.749299564	0.607198592
## [517,]	3.000095e-01	1.18650928	1.128431238	0.742294062
## [518,]	2.256088e-01	1.58570298	1.588770811	0.521875137
## [519,]	-2.486956e-01	-0.30943822	-0.459740290	1.808837246
## [520,]	-6.020988e-01	-0.38927696	-0.457751168	1.147580472
## [521,]	-1.253105e+00	-1.31729941	-1.128369312	2.896711293
## [522,]	5.371617e-01	3.02609260	3.370455455	0.472103122
## [523,]	1.256329e-01	-0.85061317	-0.758108532	-0.799927382
## [524,]	-1.417446e-01	-0.13329914	-0.238379470	0.199068067
## [525,]	-8.392510e-01	-1.19219128	-1.027776590	-0.102408139
## [526,]	-1.439106e+00	-1.54076557	-1.232087796	0.514764850
## [527,]	-1.254694e-01	-0.18638779	-0.294927356	0.792066077
## [528,]	-1.632083e+00	-0.53619670	-0.529643706	-0.450101217
## [529,]	-1.422831e+00	-0.06827584	-0.172454297	2.022145883
## [530,]	-1.360056e+00	-0.58187763	-0.595853040	0.969823275
## [531,]	-4.021470e-01	-0.66171636	-0.659220771	0.777845502
## [532,]	1.698083e-01	-0.68970108	-0.678259507	0.372559092
## [533,]	-6.881246e-01	-0.17321851	-0.225592260	-0.255279329
## [534,]	3.209347e-01	1.75854973	1.830307006	-0.341313813
## [535,]	-3.881968e-01	-0.87160171	-0.822044583	0.036242474
## [536,]	3.651101e-01	1.88612710	1.855881427	0.585867728
## [537,]	7.580387e-01	0.07411696	-0.071293255	0.528985425
## [538,]	1.197468e+00	-0.64196245	-0.706107209	1.936822428
## [539,]	1.441595e+00	-1.81032420	-1.352855894	-1.093582271
## [540,]	1.429970e+00	-1.79550876	-1.376725353	-0.688295862
## [541,]	-1.127554e+00	-0.71274731	-0.716052817	0.247418025
## [542,]	1.325344e+00	0.15807110	0.004293366	-0.568131996
## [543,]	1.425320e+00	0.11239017	0.038960915	-0.967730176

## [544,]	2.039125e+00	-0.29174200	-0.331015705	-0.686162775
## [545,]	3.279097e-01	-0.09049900	-0.199165358	-0.041259664
## [546,]	9.161401e-01	-0.19667629	-0.232127945	-0.277321222
## [547,]	-6.834746e-01	-1.09712557	-0.937697797	-0.143647809
## [548,]	-6.299991e-01	-1.07490241	-0.949348367	-0.539690845
## [549,]	1.170683e-02	-1.27244156	-1.049088607	-0.814147957
## [550,]	1.143992e+00	-0.94979429	-0.833410993	-1.026745565
## [551,]	5.092614e-01	-0.96543280	-0.836536755	-1.567838474
## [552,]	7.324634e-01	-0.84279392	-0.785672074	-0.049792009
## [553,]	2.357653e+00	-0.43701558	-0.417684576	-0.967019147
## [554,]	6.162124e-01	-1.35639570	-1.110751377	-0.281587394
## [555,]	2.239077e+00	-0.38968850	-0.399498321	-1.075806552
## [556,]	1.934499e+00	-1.08231013	-0.947643405	-0.430903440
## [557,]	6.983236e-02	-1.12099488	-0.975206947	0.280125349
## [558,]	1.997275e+00	-1.34610720	-1.090007681	-1.075806552
## [559,]	7.882640e-01	0.18194041	0.006282488	-0.826946476
## [560,]	1.078892e+00	-0.71809733	-0.714347856	-0.266655790
## [561,]	1.827549e+00	-0.02424107	-0.154836363	0.208311441
## [562,]	2.343703e+00	-0.87654019	-0.764075896	-1.555039956
## [563,]	2.634331e+00	0.47042987	0.176210306	0.600088304
## [564,]	1.348594e+00	2.10012784	1.966703917	0.962712987
## [565,]	7.208383e-01	2.05897385	2.341795421	1.040926154
## [566,]	2.083301e+00	1.61451077	1.722326119	0.102368152
## [567,]	2.043775e+00	0.67208442	0.577444551	-0.839744994
## [568,]	2.334403e+00	1.98078127	1.733692528	1.524425730
## [569,]	1.220718e+00	-1.81279344	-1.346604368	-3.109348889
##	mean compactness	mean concavity	mean concave points	mean symmetry
## [1,]	3.280628064	2.650541786	2.530248864	2.215565542
## [2,]	-0.486643478	-0.023824892	0.547662271	0.001391139
## [3,]	1.051999895	1.362279789	2.035439783	0.938858720
## [4,]	3.399917422	1.914212875	1.450431130	2.864862154
## [5,]	0.538866307	1.369806149	1.427236955	-0.009552062
## [6,]	1.243241565	0.865540012	0.823930674	1.004517928
## [7,]	0.088217620	0.299808599	0.646366374	-0.064268069
## [8,]	1.139100062	0.060972100	0.281702390	1.402120910
## [9,]	1.682529360	1.218024551	1.149679986	1.963871911
## [10,]	2.561104951	1.737343408	0.940932405	0.796597103
## [11,]	-0.712914562	-0.700068361	-0.404329784	-1.034565253
## [12,]	0.470700959	0.134730431	0.441742202	0.110823153
## [13,]	2.678500827	1.476429586	1.620521752	2.135315399
## [14,]	-0.078408785	0.132723402	0.121662578	0.129061821
## [15,]	2.366076318	1.555456369	0.807437038	0.938858720
## [16,]	1.044425968	0.942058008	0.637088704	1.792428424
## [17,]	-0.612370674	-0.186268834	0.094602706	-0.822996694
## [18,]	1.852942729	1.046172658	1.388579995	1.285393429
## [19,]	-0.031071738	0.741355068	1.187048380	-0.837587629
## [20,]	-0.436466208	-0.277964990	-0.028584138	0.267675705
## [21,]	0.429044358	-0.540885842	-0.459222666	0.566789875
## [22,]	-0.746429191	-0.743094054	-0.725697973	0.012334341
## [23,]	2.066906181	1.491482307	1.253538350	2.587634387
## [24,]	-0.040539147	0.262176798	0.963868867	-0.155461413
## [25,]	0.783125469	0.799057163	1.102518495	0.668926420
## [26,]	2.333887126	1.682150100	2.349849720	4.480808217
## [27,]	1.561346520	0.673617825	1.002783540	1.606394001

## [28,]	0.042774055	0.755153395	0.731669398	-0.418098245
## [29,]	1.237561119	0.997251316	0.994536722	0.417232790
## [30,]	0.215080906	0.124820723	0.788881698	-0.264893426
## [31,]	1.597322676	1.795045504	1.945240211	1.354700370
## [32,]	0.894840900	0.413958397	0.074758800	1.785132956
## [33,]	0.856971262	1.917976055	1.839577855	1.591803066
## [34,]	1.279217721	0.964637089	0.696104995	0.150948224
## [35,]	0.976260620	0.584555896	0.737081372	0.679869622
## [36,]	0.554014162	0.577029535	0.290206921	0.307800777
## [37,]	0.103365475	0.540652128	0.181967434	0.267675705
## [38,]	-1.262592351	-0.792517153	-0.507415009	-1.257077014
## [39,]	-1.004132074	-0.813089204	-0.513600122	-0.899599104
## [40,]	0.400642130	0.219527423	0.140991057	-0.334200368
## [41,]	-0.833718706	-0.723776396	-0.737295061	-0.100745407
## [42,]	0.330583300	0.195693949	0.200265062	0.304153043
## [43,]	2.171047684	1.529114108	1.306369528	1.817962560
## [44,]	0.743362349	0.121308422	0.326286750	0.592324011
## [45,]	0.006797899	-0.077889246	0.092798715	-0.239359290
## [46,]	1.216732819	1.362279789	1.339614513	0.347925848
## [47,]	-0.850381346	-0.914695068	-1.108221473	-0.155461413
## [48,]	0.355198565	0.423993544	0.630903590	1.154075013
## [49,]	-0.254123903	-0.286996622	-0.552257082	-0.498348388
## [50,]	-0.518075277	-0.517930443	-0.388609287	-0.009552062
## [51,]	-1.035374525	-0.906039753	-0.973360227	-1.154940468
## [52,]	-0.828416956	-0.880951886	-0.816670684	-1.672918664
## [53,]	-1.076084386	-0.866526362	-0.913055370	0.205664231
## [54,]	0.836142961	1.108892327	1.471048175	1.022756597
## [55,]	-0.634903109	-0.454959895	-0.401494940	-0.713564681
## [56,]	-0.939374994	-0.733058907	-0.673639935	0.395346387
## [57,]	0.423363912	0.545669701	1.057160996	0.384403186
## [58,]	0.608925136	0.508037900	0.832692918	0.785653901
## [59,]	-1.258237342	-1.105212334	-1.153321259	0.026925276
## [60,]	-0.977433980	-0.855362261	-1.059719874	-0.469166518
## [61,]	-0.449341885	-0.977916494	-0.928260441	3.397431286
## [62,]	-0.278549819	-0.737574723	-1.022093767	0.059754880
## [63,]	1.826433983	1.564237123	0.969280842	0.501130667
## [64,]	-0.318691635	-0.362761982	-0.698895815	1.931042307
## [65,]	0.413896503	0.301062993	0.510551590	0.340630380
## [66,]	0.824782070	0.475423672	1.066180953	0.515721602
## [67,]	-0.503874163	-0.841438494	-0.873109845	-0.345143570
## [68,]	-1.085551795	-0.648638233	-0.686010162	-1.078338059
## [69,]	0.699812266	2.812358532	-0.133215642	1.092063539
## [70,]	-0.984629211	-0.655662835	-0.522620080	-0.808405759
## [71,]	-0.027284774	0.240852111	0.788366272	-0.837587629
## [72,]	0.923243128	-0.034361796	-0.520558375	0.329687179
## [73,]	1.489394209	1.008540857	0.786562280	0.420880524
## [74,]	0.447979177	-0.136845735	0.045637224	-0.545768927
## [75,]	-0.682618852	-0.690911289	-0.675186213	-0.334200368
## [76,]	-0.380608493	0.111524153	0.449989020	-0.049677134
## [77,]	0.006797899	-0.251246411	0.428856549	2.157201801
## [78,]	2.087734482	0.998505710	1.522590788	1.241620623
## [79,]	4.564408776	3.595099999	2.873007238	3.992011891
## [80,]	-0.168159826	-0.626059152	-0.664104551	-0.341495836
## [81,]	-0.202999892	-0.538000737	-0.685237022	0.110823153

## [82,]	0.930817055	0.352493121	0.539930879	0.475596530
## [83,]	3.070451576	3.074526748	3.494095720	0.063402613
## [84,]	1.415548416	1.315867234	2.525094603	-0.647905473
## [85,]	-0.618997861	-0.593194045	-0.780590856	0.975336058
## [86,]	0.018158791	0.560722422	1.005876097	1.168665948
## [87,]	-0.092231203	0.396396889	0.011876813	0.960745123
## [88,]	0.307861518	0.727556741	0.870834452	0.515721602
## [89,]	-0.187283993	-0.359375120	-0.295574871	0.431823725
## [90,]	0.559694607	0.136235703	0.559774785	1.110302207
## [91,]	-0.346147122	-0.724779910	-0.498652765	-0.461871050
## [92,]	-0.014030401	0.293536632	0.667756558	-0.345143570
## [93,]	-1.018522537	-0.704835056	-0.578286101	-1.552543450
## [94,]	-0.429649673	-0.615396808	-0.544267977	-0.633314538
## [95,]	0.964899729	1.018576004	1.011030358	0.158243692
## [96,]	0.510464079	0.723793560	0.977012234	1.033699798
## [97,]	-0.639447465	-0.801548785	-0.502776174	0.322391712
## [98,]	-0.971942883	-1.028230212	-1.056163434	-1.683861866
## [99,]	-0.550832513	-0.587549275	-0.397371531	-0.698973746
## [100,]	0.184785196	0.063731766	0.244076283	0.245789302
## [101,]	-0.364135200	-0.031978449	-0.103836352	-0.739098817
## [102,]	-0.542690541	-1.113892736	-1.260710292	0.431823725
## [103,]	-1.211089644	-0.814970794	-0.804558170	-0.264893426
## [104,]	-0.139568250	-0.341939052	-0.480097424	0.486539732
## [105,]	-0.351448872	-0.738201920	-0.951196903	1.478723319
## [106,]	1.366317887	1.483955946	1.213592825	0.413585056
## [107,]	-0.050006557	-0.227036619	-0.362580268	-0.038733932
## [108,]	-0.685269726	-0.782356566	-0.765643498	-0.764632954
## [109,]	3.265480209	4.234840621	3.437398846	2.715305069
## [110,]	-0.730713292	-0.470012616	-0.771313185	-1.184122338
## [111,]	-0.384395456	-0.570238647	-0.802496466	-0.830292162
## [112,]	0.313541963	0.222036210	0.290980060	-0.279484362
## [113,]	2.252467405	2.653050573	0.748936173	-0.392564109
## [114,]	0.491529260	-0.301547585	-0.470046615	0.402641855
## [115,]	-0.422833138	-0.595577393	-0.764870359	-0.593189466
## [116,]	-0.489483700	-0.696430620	-0.743222462	-0.450927849
## [117,]	0.377920347	0.048051849	-0.665908543	-1.848009886
## [118,]	1.146673989	1.006032070	1.038863369	1.259859292
## [119,]	1.341702622	1.561728336	1.182151832	1.037347532
## [120,]	-0.702879108	-0.199063646	0.181452008	1.157722747
## [121,]	-0.709884991	-0.673349782	-0.584728928	-0.527530258
## [122,]	0.107152439	0.713758413	0.972373399	0.563142141
## [123,]	3.452934915	4.239858194	3.924476535	3.076430713
## [124,]	0.105258957	-0.004758112	0.228355786	0.161891425
## [125,]	-0.588702151	-0.098837616	-0.539113716	-1.421225034
## [126,]	-0.813837146	-0.935768876	-0.966659687	-0.720860149
## [127,]	-0.487022174	-0.450945836	-0.465665493	-0.184643283
## [128,]	-0.455590375	0.049055363	0.189441113	0.490187465
## [129,]	1.445844126	0.313606926	0.938612987	0.690812823
## [130,]	1.033065077	2.078538407	1.700412801	1.424007312
## [131,]	-0.175165709	-0.755763427	-0.517981244	0.249437036
## [132,]	0.340050710	0.725047954	0.823415248	0.435471459
## [133,]	0.455553104	0.194439556	0.185833130	1.270802494
## [134,]	-0.184065074	-0.218883062	0.268301311	0.015982074
## [135,]	-0.137296071	0.332422827	0.503851050	-0.436336914

## [136,]	-0.884842716	-0.522948016	-0.563854170	-0.826644428
## [137,]	-0.821600422	-0.663314635	-0.590656328	-1.723986937
## [138,]	-0.671257961	-0.674478736	-0.519785236	-0.283132095
## [139,]	0.495316224	0.816618670	0.961807163	0.530312537
## [140,]	0.175317786	-0.532481406	-0.024718442	-0.148165946
## [141,]	-1.198971360	-1.113892736	-1.260710292	0.333334913
## [142,]	0.177211268	0.071132687	0.270878441	0.180130094
## [143,]	-0.179520717	-0.859125441	-0.781106282	-0.607780402
## [144,]	-0.175165709	-0.499992617	-0.464892354	-0.122631809
## [145,]	-1.002617289	-0.831528787	-1.057761255	-1.505122911
## [146,]	0.478274887	-0.648512793	-0.486797964	0.668926420
## [147,]	1.243241565	0.967145875	0.650232070	3.160328590
## [148,]	0.234015725	0.021333270	-0.342736362	-0.246654758
## [149,]	-0.042432629	-0.049289077	0.164958372	-0.319609433
## [150,]	-0.768393581	-0.752502004	-0.918209631	-1.235190611
## [151,]	-0.538714229	-0.720514973	-0.579059240	2.656941329
## [152,]	0.495316224	0.543160914	-0.701988371	1.496961988
## [153,]	1.051999895	4.039155254	0.764141244	2.686123199
## [154,]	-1.007540342	-0.865271969	-0.800434761	0.067050347
## [155,]	-0.366596727	0.051815029	-0.363095694	0.037868477
## [156,]	-0.711021080	-0.626560909	-0.659981142	0.577733076
## [157,]	1.176969699	1.213006977	1.455585392	0.581380810
## [158,]	-0.608015666	-0.467880147	-0.546587394	0.118118620
## [159,]	-0.983303774	-0.866526362	-0.754819549	-0.808405759
## [160,]	-1.271681064	-1.075131980	-1.090928926	-1.348270358
## [161,]	0.184785196	-0.255511348	-0.297378863	0.661630953
## [162,]	0.268098398	0.382598562	1.230601888	-0.257597959
## [163,]	1.178863181	2.032125852	2.053479698	0.785653901
## [164,]	-0.053793521	-0.440283493	-0.533444028	-0.950667377
## [165,]	0.192359123	0.546924094	1.239621845	-0.038733932
## [166,]	-0.962286125	-0.869662345	-0.761004663	-1.081985793
## [167,]	-0.889576421	-0.796405772	-0.823113511	-1.570782119
## [168,]	-0.237082566	-0.057442634	0.434010810	0.296857575
## [169,]	1.059573823	1.594342564	1.427236955	-0.998087916
## [170,]	-0.482667166	-0.787499579	-0.286297201	-0.115336342
## [171,]	-0.653837927	-0.613766097	-0.307171959	0.537608004
## [172,]	-0.785056221	-0.379069096	-0.374692781	-0.779223889
## [173,]	0.968686693	1.435034605	1.566402009	0.563142141
## [174,]	-0.888250984	-0.817479581	-0.595037450	-0.895951370
## [175,]	-1.161101722	-1.113892736	-1.260710292	0.424528257
## [176,]	-1.166024775	-1.113892736	-1.260710292	-0.326904901
## [177,]	0.474487923	0.525599407	-0.303048550	-0.520234791
## [178,]	0.970580175	1.135234588	1.024173724	-0.064268069
## [179,]	-1.608720838	-1.093885161	-1.212981833	-1.519713846
## [180,]	-1.261077565	-0.998576353	-0.917951918	-1.260724748
## [181,]	1.648446686	2.487470647	3.579141030	-0.042381666
## [182,]	3.386663049	2.005783591	2.594677130	2.128019931
## [183,]	-0.309602922	-0.286871183	0.076562792	-0.706269213
## [184,]	-0.431543155	-0.338552190	-0.652249750	-2.351397147
## [185,]	0.016265309	-0.439656296	-0.419792568	-0.308666232
## [186,]	-1.086687884	-1.093860074	-1.198756072	-0.396211842
## [187,]	-0.372277172	-0.089178787	0.237633456	-0.695326012
## [188,]	-0.812890405	-0.636094299	-0.425977681	-1.078338059
## [189,]	-0.922523006	-0.818733974	-0.860481905	-0.341495836

## [190,]	-0.602335220	-0.631703922	-0.834452886	-0.527530258
## [191,]	2.593294143	1.371060543	0.444834759	2.087894860
## [192,]	-0.725790239	-0.723525517	-0.522620080	-0.429041446
## [193,]	-1.531845474	-1.113892736	-1.260710292	-0.578598531
## [194,]	0.586203354	0.247124077	-0.085023299	0.479244264
## [195,]	1.773416490	1.014812824	1.027266281	-0.272188894
## [196,]	-0.959635250	-0.628066181	-0.648126341	0.063402613
## [197,]	0.423363912	0.623442090	0.421125157	0.081641282
## [198,]	-0.339330588	0.269703158	0.228355786	-0.151813679
## [199,]	0.728214494	0.283501485	0.484522570	-0.162756881
## [200,]	0.307861518	0.366291448	0.280413825	0.504778400
## [201,]	-0.444418832	-0.588678229	-0.202282742	0.610562680
## [202,]	0.292713663	0.185658802	0.669045123	-1.114815397
## [203,]	1.970338605	3.305335129	2.914241328	1.416711845
## [204,]	1.371998332	0.840452144	1.104064774	1.602746267
## [205,]	0.027626200	-0.109750838	-0.275988678	0.413585056
## [206,]	-0.160207202	-0.166825737	-0.209498708	-0.793814824
## [207,]	-0.606311532	-0.893621259	-0.757654393	0.446414660
## [208,]	-0.592678463	-0.242089339	0.128363117	0.782006168
## [209,]	0.832355998	-0.021943302	0.054141755	0.140005023
## [210,]	-0.796038416	-0.374804158	-0.447110152	-1.651032262
## [211,]	0.576735944	0.943312401	1.203284303	-0.170052348
## [212,]	-0.669175131	-0.779095144	-0.901715995	-1.016326585
## [213,]	0.894840900	2.901420462	2.849813062	-0.596837200
## [214,]	0.194252605	0.995996923	0.439422784	-1.837066684
## [215,]	0.497209705	0.284755879	0.404631521	1.544382527
## [216,]	0.896734381	0.128082146	0.182998287	1.073824870
## [217,]	0.209400460	-0.281477291	-0.450975848	0.566789875
## [218,]	-0.857197881	-0.389606000	-0.984699602	0.555846673
## [219,]	0.497209705	0.481695639	0.979073938	1.030052064
## [220,]	0.163956895	0.322387680	0.449731307	-1.399338631
## [221,]	-0.326265563	-0.626184591	-0.600191712	-1.647384528
## [222,]	0.281352772	-0.127939542	-0.113629449	0.548551206
## [223,]	-0.365839334	-0.892115987	-0.767189776	0.358869049
## [224,]	0.304074554	0.324896467	0.404631521	0.450062394
## [225,]	-1.030072776	-0.668081330	-0.627767009	-1.151292734
## [226,]	-0.532087043	-0.395877967	-0.074457063	0.960745123
## [227,]	-0.513530920	-1.030563384	-0.947331207	-0.086154471
## [228,]	0.099578511	-0.297909845	-0.286554914	0.253084770
## [229,]	-0.550075120	-0.431753618	-0.736006496	-1.085633526
## [230,]	1.430696271	1.012304037	0.507459033	1.135836344
## [231,]	1.000875885	1.281998613	1.548362094	1.165018214
## [232,]	-1.253692986	-0.909050297	-1.180174960	0.209311965
## [233,]	-1.298947203	-1.051587017	-1.094897707	0.121766354
## [234,]	0.057921910	0.835434571	0.888616653	-1.326383956
## [235,]	-1.201811582	-0.906666950	-0.831102616	-0.950667377
## [236,]	-0.660654462	-0.930500424	-0.772086325	-1.074690325
## [237,]	1.209158891	1.332174348	1.927200297	0.355221316
## [238,]	-0.394998955	0.020329755	0.291237773	-1.257077014
## [239,]	-0.008349956	0.269703158	-0.124711110	-1.713043736
## [240,]	0.482061850	0.663582678	1.009999506	-0.009552062
## [241,]	-0.720299141	-0.523700652	-0.299182854	-0.345143570
## [242,]	-1.333219225	-0.981805113	-0.975164218	-0.968906045
## [243,]	0.533185861	0.827908211	-0.525197210	0.884142713

## [244,]	-0.686784512	-0.524704167	-0.656630872	-0.140870478
## [245,]	0.974367138	1.456359292	1.029327985	0.606914946
## [246,]	-0.845079597	-0.507895296	-0.469531189	-0.272188894
## [247,]	-1.119066424	-0.569987768	-0.975937357	-1.184122338
## [248,]	0.572948981	0.609643763	-0.235012301	-0.786519357
## [249,]	-0.605932836	-0.815472552	-0.844503695	0.311448510
## [250,]	-0.497246976	-0.570991283	-0.505868731	0.260380237
## [251,]	1.065254268	2.288022101	2.115330833	1.434950514
## [252,]	-0.841292633	-0.782983763	-0.727501965	0.081641282
## [253,]	1.525370364	1.917976055	1.249414941	-0.286779829
## [254,]	-0.004562992	0.474169279	0.891966923	0.005038873
## [255,]	0.273778844	0.615915730	0.953302632	-0.129927277
## [256,]	0.446085695	0.114032940	0.091252436	0.351573582
## [257,]	1.930575486	1.123945047	1.687527148	0.296857575
## [258,]	2.349034981	1.956862249	1.940085950	2.138963133
## [259,]	3.920624938	2.870060627	2.287998585	2.492793309
## [260,]	1.127739170	1.082550066	0.903821724	1.019108863
## [261,]	0.084430656	0.791530803	1.144525725	0.008686607
## [262,]	-0.784677525	-0.751247611	-0.529578332	-0.903246837
## [263,]	0.434724804	0.102492521	0.673941672	1.081120337
## [264,]	-0.912298203	-0.585918564	-0.527001202	-0.965258312
## [265,]	-0.272490677	0.022713102	0.421382870	0.202016497
## [266,]	0.188572160	0.600863010	0.967476850	-0.155461413
## [267,]	0.196146087	-0.312711686	-0.579832380	0.402641855
## [268,]	-0.411850943	-0.603103753	-0.708688911	-0.644257739
## [269,]	-0.798121246	-0.624679319	-0.844503695	0.723642427
## [270,]	0.465020514	-0.054181212	-0.521846940	-0.523882525
## [271,]	-1.469171223	-1.022949216	-1.099639628	-1.107519929
## [272,]	-0.535116614	-0.704333298	-0.550710803	-0.155461413
## [273,]	1.737440335	1.639500725	1.543207833	-0.330552634
## [274,]	-1.087823973	-0.975533146	-0.898108012	0.982631525
## [275,]	-0.645127911	-0.399013951	-0.038119521	-0.998087916
## [276,]	-0.610477192	-0.370162903	0.647397226	0.741881096
## [277,]	-1.242521443	-1.095239906	-1.174814528	0.519369336
## [278,]	-0.861552889	-0.107869248	0.245107135	-0.954315110
## [279,]	-1.208438769	-0.863390379	-0.941661520	-0.870417233
## [280,]	-0.519968759	-0.552049943	-0.304337115	1.088415805
## [281,]	0.775551541	1.295796940	1.229828748	0.329687179
## [282,]	-1.153906491	-0.832281423	-0.548649099	1.055586201
## [283,]	0.754723241	0.925750894	1.178286136	0.296857575
## [284,]	1.436376716	1.329665561	1.072108354	0.234846101
## [285,]	-0.161721988	0.284755879	-0.387063009	-1.384747696
## [286,]	-1.177385666	-1.090561019	-1.185354992	-0.418098245
## [287,]	-0.061367448	-0.289254530	-0.283720070	-0.815701227
## [288,]	-1.269598234	-0.830399833	-0.958928295	-1.731282405
## [289,]	0.260524471	0.049431681	0.179390304	2.857566687
## [290,]	-1.027421902	-0.812963765	-0.700699806	0.734585628
## [291,]	1.197798000	0.594591043	0.440711350	-0.356086771
## [292,]	-0.115710378	-0.368783070	-0.018791042	0.245789302
## [293,]	-0.471684971	-0.341813613	-0.392217270	-0.297723030
## [294,]	-0.907375151	-0.776711796	-0.673124508	0.231198367
## [295,]	-0.386478286	-0.952326869	-0.764870359	-0.633314538
## [296,]	-0.797742550	-0.980550720	-0.766674350	-0.801110292
## [297,]	-1.081764831	-0.958849714	-0.907901109	-1.322736222

## [298,]	-0.855872444	-0.777088114	-0.354848876	-0.702621480
## [299,]	-0.987280086	-0.803430375	-0.906612543	-0.644257739
## [300,]	-0.688677994	-0.800921588	-0.777498299	-0.425393713
## [301,]	1.133419616	1.642009512	1.476202437	-0.071563536
## [302,]	-0.055687002	-0.257142060	-0.462057510	-0.111688608
## [303,]	1.504542064	1.749887342	2.038016914	1.595450800
## [304,]	-0.711210428	-0.825758577	-0.801981040	-1.202361007
## [305,]	-0.518832670	-0.694423591	-0.873625271	-1.461350105
## [306,]	-0.898665134	-0.866275483	-0.922333040	0.450062394
## [307,]	-0.981410292	-1.095566049	-1.176670062	-0.655200941
## [308,]	-1.385668673	-1.067718516	-1.171232316	-0.086154471
## [309,]	-1.291373275	-1.079296566	-1.146826889	-1.629145859
## [310,]	-1.268462144	-1.056704942	-1.033175429	-1.308145287
## [311,]	-0.981031596	-0.915322264	-0.964855696	0.453710128
## [312,]	-1.310118746	-0.932382014	-0.776982873	-0.655200941
## [313,]	-0.470738230	-0.605612540	-0.604057408	-0.768280688
## [314,]	-0.845458293	-0.942417161	-1.031165267	0.077993549
## [315,]	-0.868558772	-1.113892736	-1.260710292	1.281745695
## [316,]	-1.249716674	-1.057783720	-1.095181192	-2.176305926
## [317,]	-1.367491247	-0.973024360	-1.130539424	-0.505643856
## [318,]	0.139341631	0.303571779	0.788108559	-0.016847530
## [319,]	1.758268635	1.363534182	0.004145421	1.890917236
## [320,]	-1.321668985	-0.945553145	-0.822855798	-1.238838345
## [321,]	0.127980740	-0.270187751	-0.238877997	-0.250302491
## [322,]	-0.354099746	0.334931614	0.730380832	0.424528257
## [323,]	-0.302975736	-0.637223253	-0.384485878	-0.979849247
## [324,]	1.574600893	2.103626274	2.615294175	2.762725608
## [325,]	-0.736393737	-0.863766697	-0.824659789	-0.633314538
## [326,]	-0.524513115	-0.713364931	-0.717708868	-0.381620907
## [327,]	-0.970996142	-0.892492305	-0.556380491	-1.599963989
## [328,]	-1.238734479	-1.094499814	-1.116597147	-1.567134385
## [329,]	0.521824970	0.740100674	0.926758186	0.497482933
## [330,]	0.453659622	1.142760948	0.796097664	0.209311965
## [331,]	0.620286027	0.396396889	0.553847384	-0.108040874
## [332,]	0.154489486	-0.222395363	-0.500456756	-0.184643283
## [333,]	-0.692086261	-1.051097803	-1.065286476	0.468301063
## [334,]	-1.131563405	-1.101678707	-1.184916880	-0.140870478
## [335,]	-1.180036541	-1.016601986	-1.040752193	-0.994440182
## [336,]	0.023839236	0.777732476	1.299411275	-0.308666232
## [337,]	-0.088822935	-0.645000492	-0.720028286	-0.582246265
## [338,]	0.678983966	0.215764243	0.308762261	0.515721602
## [339,]	-0.588512803	-0.798914559	-0.803269605	0.285914374
## [340,]	0.453659622	1.781247177	2.373043896	-0.053324867
## [341,]	0.180998232	-0.109499959	-0.172388027	0.366164517
## [342,]	-0.228372550	-0.057442634	-0.670031952	0.818483505
## [343,]	-0.253177162	-0.436896631	-0.399690949	-0.129927277
## [344,]	0.559694607	1.223042124	1.581864792	0.986279259
## [345,]	-0.597033471	-0.611382749	-0.423142837	0.719994693
## [346,]	-0.241437575	-0.664694468	-0.735748783	-0.651553207
## [347,]	-0.878594226	-1.019687793	-1.041963444	-0.936076441
## [348,]	-0.502548725	-0.535868268	-0.351498606	-1.060099390
## [349,]	-0.861174193	-0.789381169	-0.662300560	-0.647905473
## [350,]	0.307861518	-0.967003271	-0.800177048	2.361474893
## [351,]	-1.288343704	-1.009702822	-0.961247713	-0.512939323

## [352,]	2.500513531	2.541409563	1.940085950	2.055065256
## [353,]	2.498620049	3.110904155	3.669340602	0.526664803
## [354,]	-0.125935180	0.435283084	0.428083410	-0.600484934
## [355,]	-0.827470215	-0.548788520	-0.881614376	-0.443632382
## [356,]	-0.010243437	0.178132442	-0.129092232	-1.016326585
## [357,]	0.493422742	0.090701223	0.183256000	0.814835772
## [358,]	-0.935777379	-0.925482851	-0.722605416	-1.413929566
## [359,]	-0.518075277	-0.521693623	-0.647095489	0.431823725
## [360,]	-0.847919820	-0.773952131	-0.898365725	-1.114815397
## [361,]	-1.473904928	-1.098915279	-1.120282444	-1.034565253
## [362,]	-0.768961625	-0.694423591	-0.636013827	0.012334341
## [363,]	-0.469980837	-0.776711796	-0.801723327	-0.191938751
## [364,]	-0.372277172	-0.378567339	-0.014667633	-1.154940468
## [365,]	-0.897150348	-0.840309540	-0.881098950	-0.589541733
## [366,]	0.165850377	0.115287334	0.745585903	-0.706269213
## [367,]	1.184543627	0.944566795	1.999359954	0.231198367
## [368,]	-0.617104379	-0.562963165	-0.738325913	-0.425393713
## [369,]	-0.354478443	0.351238728	0.920830786	-0.345143570
## [370,]	1.724185962	1.956862249	2.607562783	0.045163945
## [371,]	0.858864744	1.157813669	1.000206409	1.325518500
## [372,]	-0.662737292	-0.688277063	-0.575966684	-0.330552634
## [373,]	0.892947418	1.309595267	1.973588648	0.588676277
## [374,]	0.061708874	0.801565950	1.043502204	-0.877712701
## [375,]	-0.768772277	-0.793269789	-0.737295061	0.220255166
## [376,]	0.747149313	-0.279595701	0.130167109	0.650687752
## [377,]	1.167502290	1.746124162	0.270363015	1.372939039
## [378,]	-1.080818091	-0.954459338	-0.972844801	-1.424872768
## [379,]	-0.546477505	-0.580900990	-0.623901313	-0.071563536
## [380,]	2.102882337	1.004777677	0.380148780	1.402120910
## [381,]	0.127980740	-0.122921969	0.169597207	0.752824297
## [382,]	-0.635281805	-0.669084845	-0.726213399	0.698108291
## [383,]	0.056028428	-0.117528077	-0.493240790	-2.220078731
## [384,]	0.480168369	-0.374804158	-0.518496671	-0.118984075
## [385,]	-0.352016916	-0.477037219	-0.522620080	-0.709916947
## [386,]	-0.719163052	-0.061456693	0.097695263	-0.673439609
## [387,]	-0.494406753	-0.256013106	-0.607665390	-0.604132668
## [388,]	-1.060936531	-0.857369290	-1.041473789	-0.746394285
## [389,]	0.133661185	0.149281394	-0.550195377	-0.005904328
## [390,]	0.519931488	1.214261371	1.370540081	0.647040018
## [391,]	-0.547613594	-0.872672890	-0.753530984	-0.042381666
## [392,]	-0.569199287	-1.113892736	-1.260710292	0.632449083
## [393,]	0.981941066	1.258165139	1.087828851	0.428175991
## [394,]	1.976019051	2.410952652	2.764767751	1.278097961
## [395,]	-0.128018010	-0.513916384	-0.403556645	0.457357861
## [396,]	-0.960581991	-0.777589872	-0.422885124	-0.622371337
## [397,]	0.196146087	-0.037623219	0.126043700	-0.020495263
## [398,]	-0.291425496	-0.186896031	-0.208467856	-0.866769500
## [399,]	-1.072865467	-0.773701252	-1.073971407	-1.009031117
## [400,]	-0.795659720	-0.756014306	-0.838576295	0.129061821
## [401,]	2.901931689	2.886367741	1.826692202	1.099359006
## [402,]	-0.983114426	-0.786997822	-0.797857631	-0.768280688
## [403,]	-0.480016291	-0.604985343	-0.775436594	0.227550633
## [404,]	-0.302597039	-0.700444679	-0.644776071	-0.279484362
## [405,]	-1.110167060	-0.849341173	-0.731367661	-0.877712701

## [406,]	-0.563140145	-0.493720651	-0.505095591	-1.187770072
## [407,]	-0.366028682	-0.423976379	-0.093785543	-0.279484362
## [408,]	-0.401058097	-0.345451353	-0.779560003	-0.844883097
## [409,]	0.493422742	0.392633709	1.013349776	0.657983219
## [410,]	-0.739991353	-0.711107023	-0.576997536	0.563142141
## [411,]	-0.969670704	-0.764795059	-0.719512860	-0.768280688
## [412,]	-0.498004369	-0.731804513	-0.621581896	-0.356086771
## [413,]	-0.829553045	-0.645376810	-1.128555033	-1.961089633
## [414,]	-0.034858702	-0.253504319	-0.261814460	0.482891998
## [415,]	-1.103729221	-0.526084000	-0.554834212	0.147300490
## [416,]	-0.438170342	-0.793395228	-0.699153528	0.756472031
## [417,]	-0.809482138	-0.857118412	-0.936764972	0.778358434
## [418,]	0.998982403	0.795293983	0.924954195	0.997222460
## [419,]	-0.878594226	-0.817855899	-0.641683515	-0.833939896
## [420,]	-0.843754160	-1.001561809	-0.983411036	-0.717212415
## [421,]	-0.513530920	-0.425857969	-0.892696038	0.800244836
## [422,]	1.500755100	0.704977660	0.362882005	1.000870194
## [423,]	0.235909207	-0.223649757	-0.101774648	0.271323439
## [424,]	0.196146087	0.097474948	-0.020595033	0.132709555
## [425,]	-0.397839178	-1.001825232	-0.753788697	2.649645861
## [426,]	-1.234947515	-1.082909219	-1.127756123	-0.662496408
## [427,]	-0.057580484	-0.319234532	-0.689102718	0.413585056
## [428,]	-0.888250984	-0.660554970	-0.898881152	0.745528830
## [429,]	-1.249716674	-0.942166283	-0.907643396	-1.096576728
## [430,]	-1.119445121	-0.938026784	-0.787806821	-1.286258884
## [431,]	2.237319550	2.314364362	1.241941262	0.836722174
## [432,]	0.516144524	-0.142866823	-0.539371429	-0.002256595
## [433,]	0.843716889	1.561728336	1.983897171	-0.319609433
## [434,]	0.654368701	0.885610306	0.992732731	0.479244264
## [435,]	-0.636228546	-0.694172712	-0.519269810	-0.870417233
## [436,]	0.169637341	0.298554206	0.404889234	-0.520234791
## [437,]	-0.483045862	-0.888478246	-0.722089990	0.180130094
## [438,]	-0.859470059	-0.670590117	-0.502003035	-0.356086771
## [439,]	-0.777103597	-0.945553145	-0.669774239	-0.936076441
## [440,]	-0.918925390	-0.852100838	-0.577255249	-0.812053493
## [441,]	0.131767703	0.072387080	-0.329592995	-1.176826871
## [442,]	0.124193776	0.396396889	0.217531837	-1.257077014
## [443,]	-0.703636501	-0.981554235	-1.004620821	-1.483236508
## [444,]	-1.130806012	-0.863892136	-0.974391079	2.044122054
## [445,]	0.357092047	0.253396044	0.351027204	-0.334200368
## [446,]	-0.230266032	-0.431377300	-0.159244661	0.030573009
## [447,]	0.512357560	1.016067217	0.876504139	-0.359734505
## [448,]	-0.292372237	-0.603480071	-0.678278770	0.296857575
## [449,]	-0.498761762	-0.007894096	-0.506899583	-1.235190611
## [450,]	0.249163580	0.858013652	1.715875585	-0.939724175
## [451,]	0.038987091	-0.012911669	-0.645806924	-1.687509599
## [452,]	-0.106621665	0.962128302	1.074943198	-0.542121194
## [453,]	-0.754381815	-0.605236222	-0.759458384	-0.717212415
## [454,]	-0.225721675	-0.248988503	0.413136052	-0.589541733
## [455,]	-0.947516966	-0.741839660	-0.675186213	-0.046029400
## [456,]	-0.569577984	-0.760279243	-0.419534854	-1.592668521
## [457,]	-0.352206264	-0.215747078	-0.740903044	-0.046029400
## [458,]	-0.990120309	-0.766174892	-0.727759678	-0.702621480
## [459,]	-1.015114269	-0.962612895	-0.806619875	-0.527530258

## [460,]	-1.099752909	-0.920590716	-0.991915567	-0.695326012
## [461,]	0.126087258	0.149281394	0.396642416	-0.067915803
## [462,]	1.788564345	3.445827187	3.092063341	0.909676850
## [463,]	-0.986712041	-0.677865598	-0.813062702	-0.381620907
## [464,]	-0.867043987	-0.691538486	-0.802754179	-1.078338059
## [465,]	-0.840724589	-0.504382994	-0.521073801	-1.304497553
## [466,]	0.340050710	0.153044574	-0.530609185	-0.768280688
## [467,]	0.086324138	0.247124077	-0.356137441	-0.910542305
## [468,]	-0.947895663	-0.928367955	-1.112035626	-0.480109719
## [469,]	1.818860055	1.565491516	1.321574599	-0.421745979
## [470,]	0.832355998	0.165588508	0.173205190	0.530312537
## [471,]	-0.790736667	-0.744097568	-0.870532715	1.555325728
## [472,]	-0.839588499	-0.816977824	-0.648126341	0.154595958
## [473,]	-0.356939969	-0.419084245	-0.430616516	-0.454575583
## [474,]	-1.332272484	-1.113892736	-1.260710292	-0.403507310
## [475,]	0.048454501	-0.472270524	-0.855843070	0.180130094
## [476,]	-0.409957462	-0.381954201	-0.467469484	-0.388916375
## [477,]	0.122300294	-0.478793369	-0.472623745	-1.114815397
## [478,]	-0.968534615	-0.834915649	-0.915632501	0.005038873
## [479,]	-0.417720737	-0.448060732	-0.753273271	-0.118984075
## [480,]	1.608683567	1.690930854	1.108703609	1.237972890
## [481,]	-0.491566530	-0.748111627	-0.867182445	-1.268020215
## [482,]	-0.967209178	-0.738201920	-0.727244252	-0.848530831
## [483,]	0.211293942	-0.388100728	0.096406698	-0.118984075
## [484,]	-0.469034096	-0.543394628	-0.446337013	-0.290427563
## [485,]	0.483955332	0.380089775	0.339945542	-0.100745407
## [486,]	0.885373490	0.822890637	-0.011832789	0.986279259
## [487,]	-0.707423464	-0.462611695	-0.541433133	-1.468645573
## [488,]	0.766084132	1.716018721	1.816383680	0.041516211
## [489,]	-0.221745363	-0.577137810	-0.453552979	0.150948224
## [490,]	-0.629033315	-0.656164593	-0.666166256	0.125414088
## [491,]	-0.991067050	-0.898889711	-0.935734120	-0.976201513
## [492,]	-0.798499943	-0.556314880	-0.183985115	-2.158067257
## [493,]	0.465020514	0.353747514	0.739658503	1.110302207
## [494,]	-1.210142903	-1.023915099	-0.964597983	-0.724507882
## [495,]	-0.976865935	-0.888101928	-0.937022685	-0.359734505
## [496,]	-0.395566999	-0.257894696	0.015227083	-1.184122338
## [497,]	0.550227198	-0.108245566	0.046925790	-0.622371337
## [498,]	-0.530950954	-0.661182166	-0.650188046	-1.041860721
## [499,]	0.518038006	0.756407788	1.105868765	0.074345815
## [500,]	1.137206580	1.630719971	1.628253144	0.132709555
## [501,]	0.607031654	-0.145375610	0.322163341	-0.523882525
## [502,]	1.207265409	0.588319076	0.481172301	1.690291878
## [503,]	0.078750211	-0.370288342	-0.415669158	0.479244264
## [504,]	0.438511767	0.988470563	1.324151730	-1.118463130
## [505,]	2.263828296	0.106632019	0.092798715	2.066008457
## [506,]	2.197556431	0.376326595	0.553074245	0.895085915
## [507,]	0.205613496	-0.088426151	-0.702503798	1.139484078
## [508,]	0.052241464	-0.604232707	-0.160790939	0.519369336
## [509,]	-0.704772590	-0.420714956	-0.084765586	-0.367029972
## [510,]	1.565133484	1.432525818	0.935005004	-0.016847530
## [511,]	-0.146384784	-0.270187751	-0.580605519	-1.140349533
## [512,]	-1.025907116	-0.685391958	-0.605861399	-0.556712129
## [513,]	0.805847251	0.698705693	0.845320859	1.110302207

## [514,]	-0.287070488	-0.082530502	-0.139916181	-0.264893426
## [515,]	-0.347851256	-0.174853854	-0.143524164	-0.914190039
## [516,]	-0.366407378	-0.574252705	-0.592202607	0.420880524
## [517,]	0.387387757	0.854250472	1.174935866	0.176482361
## [518,]	0.504783633	0.656056318	1.169781605	-0.035086199
## [519,]	1.169395772	-0.508647932	0.105942081	-0.374325440
## [520,]	0.139341631	-0.627188106	-0.488859668	1.124893143
## [521,]	0.343837673	-0.695928863	-0.636786967	1.405768643
## [522,]	2.011995207	1.783755963	2.530248864	0.654335485
## [523,]	-1.140084073	-1.050332623	-1.114741613	-0.636962272
## [524,]	0.050347983	-0.438401903	-0.285781775	-0.356086771
## [525,]	-0.381555233	-0.821619079	-0.638075532	-1.548895716
## [526,]	-0.530572257	-0.792140835	-0.871563567	-0.487405187
## [527,]	0.179104750	-0.586922078	-0.448398717	-0.323257167
## [528,]	-0.781458606	-0.742843175	-0.578543814	-0.447280115
## [529,]	-0.128586055	0.153044574	0.444061620	0.599619479
## [530,]	-0.269839803	-0.639606600	-0.539629142	-0.564007596
## [531,]	-0.136538679	-0.451322154	-0.116464292	-0.779223889
## [532,]	-0.185769207	-0.587047518	-0.704823215	0.172834627
## [533,]	-0.601956524	-0.894123016	-0.776209734	-0.658848674
## [534,]	0.510464079	0.796548376	1.355077297	1.292688896
## [535,]	-0.129154099	-0.453705502	-0.542206272	-0.702621480
## [536,]	1.317087358	1.501517454	2.146256400	1.150427279
## [537,]	0.209400460	0.721284774	0.321390202	0.417232790
## [538,]	0.963006247	-0.547534126	-0.093012404	1.165018214
## [539,]	-1.052037166	-1.113892736	-1.260710292	0.212959698
## [540,]	0.294607145	0.046672016	-0.909189674	0.822131239
## [541,]	0.145022076	-0.268807918	-0.592202607	0.023277542
## [542,]	0.353305083	0.151790181	-0.258206477	0.220255166
## [543,]	-0.609719800	-0.598964255	-0.480612850	0.103527685
## [544,]	-0.673530139	-0.739205434	-0.416700011	-0.669791876
## [545,]	-0.048113075	-0.651272459	-0.650188046	-0.698973746
## [546,]	-0.698145403	-0.740836146	-0.631117279	-0.538473460
## [547,]	-1.030072776	-0.986948126	-1.119096964	0.267675705
## [548,]	-0.448395144	-0.567228102	-0.632405844	-0.520234791
## [549,]	-1.023256241	-0.820741004	-1.012919182	-0.844883097
## [550,]	-0.725600890	-0.919712641	-1.050416433	0.599619479
## [551,]	-1.175302836	-1.113892736	-1.260710292	-0.549416661
## [552,]	-0.424158576	-0.508773371	-0.679051909	0.796597103
## [553,]	-1.173977399	-0.863390379	-0.874398411	-0.994440182
## [554,]	-0.914381034	-0.612637143	-0.930322145	-0.436336914
## [555,]	-0.872913780	-0.336796039	-0.656888585	-0.895951370
## [556,]	-0.525649204	-0.361382149	-0.555091926	-0.797462558
## [557,]	-0.554808825	-1.050859469	-0.973102514	-0.075211270
## [558,]	-1.034427785	-1.113892736	-1.260710292	-0.253950225
## [559,]	0.542653271	0.176878048	-0.297894289	-1.304497553
## [560,]	-0.042432629	0.280992699	-0.202798168	-1.545247982
## [561,]	0.156382968	-0.554182411	-0.151513269	-1.001735649
## [562,]	-1.301976774	-1.113892736	-1.260710292	-2.741704661
## [563,]	1.976019051	2.084810374	1.169266179	1.154075013
## [564,]	2.258147851	2.867551840	2.537980256	1.230677422
## [565,]	0.218867870	1.945572709	2.318924153	-0.312313966
## [566,]	-0.017817365	0.692433726	1.262558307	-0.217472887
## [567,]	-0.038645666	0.046546577	0.105684368	-0.808405759

## [568,]	3.269267173	3.294045588	2.656528265	2.135315399
## [569,]	-1.149740831	-1.113892736	-1.260710292	-0.819348961
##	mean fractal dimension	worst radius	worst texture	worst perimeter
## [1,]	2.253763811	2.4875451492	-0.5647681227	2.830540289
## [2,]	-0.867888807	0.4988156954	-0.8754732789	0.263095470
## [3,]	-0.397658013	1.2275957908	-0.7793975888	0.850180232
## [4,]	4.906601993	0.3260865190	-0.1103119811	0.286341454
## [5,]	-0.561955519	1.2694258211	-0.7895489825	1.272070124
## [6,]	1.888343495	-0.2548460559	-0.5921406306	-0.321021720
## [7,]	-0.761661971	0.1497513054	-0.8042322483	0.155273668
## [8,]	1.658893530	0.6430571790	0.2903055190	0.489620171
## [9,]	1.571079345	-0.3565363017	-0.3894753070	-0.227543186
## [10,]	2.780648917	-0.3879088244	0.6927357680	-0.409059705
## [11,]	-0.825398072	-0.0925743869	-0.0541167662	-0.197867461
## [12,]	-0.280100315	0.3628680972	-0.4204733127	0.345198309
## [13,]	2.153202406	1.9845029755	4.2620382918	4.057631523
## [14,]	-1.333870527	-0.0067507042	-0.2517063929	0.018270737
## [15,]	1.986072184	-0.6962249954	-0.0867462458	-0.398178605
## [16,]	1.129175708	-0.1268317392	-0.3332800920	0.006400447
## [17,]	-0.506717565	0.2435082696	0.0419589239	0.162692600
## [18,]	1.524339538	0.5914908486	-0.2607701373	0.488630981
## [19,]	-1.253138132	1.2730318581	-0.3622840739	1.483262368
## [20,]	-0.727669383	-0.4877960517	-0.7763159157	-0.399662392
## [21,]	0.752424530	-0.7932273930	-0.8504573445	-0.733514299
## [22,]	0.885562164	-0.4611113773	-0.4351565785	-0.473357109
## [23,]	1.065439607	0.1212636124	-0.9195230764	0.256171134
## [24,]	-1.418851996	1.0332303918	-0.1628816983	0.710704324
## [25,]	0.071156423	1.4482852606	-0.5712940186	1.280478246
## [26,]	1.605071933	2.3108493319	-0.4366067776	2.181136504
## [27,]	0.912472963	-0.5433290229	-0.4235549857	-0.373943430
## [28,]	-0.822565357	1.6145235703	1.1459229854	1.368021635
## [29,]	0.368591563	0.1219848198	-0.3713478183	0.312555011
## [30,]	-0.185204341	0.7036386020	-0.7148637290	0.884801911
## [31,]	-0.117219166	1.5344695470	0.4516401684	1.339335101
## [32,]	2.151786048	0.2788474331	-0.3387183386	0.301179317
## [33,]	0.144807029	0.7058022243	0.3283732453	0.560347316
## [34,]	-0.026572267	0.5431699516	-1.1069613095	0.327392874
## [35,]	0.333182618	-0.2566490744	-0.9628477744	-0.337837964
## [36,]	-0.883468743	0.2031206543	-0.5386645389	0.070203256
## [37,]	-0.219196929	-0.4297388546	-0.3586585762	-0.103399736
## [38,]	-0.590282676	-0.7979152412	2.0396081779	-0.838863124
## [39,]	-1.098755130	2.9166635627	1.7604448521	2.577307434
## [40,]	0.197212268	-0.6929795620	-1.1337899927	-0.653389842
## [41,]	-0.981197432	-0.6010256163	-0.7076127336	-0.640035765
## [42,]	0.835989641	-0.6078770867	0.3827557114	-0.516386911
## [43,]	0.089569074	2.0768175250	0.8141899423	2.949737785
## [44,]	0.711350154	-0.1253893244	-0.7105131318	-0.217156682
## [45,]	-0.145546323	-0.7622154741	-1.0959035414	-0.757749475
## [46,]	-0.326840123	0.8067712627	-1.0034533490	0.705758369
## [47,]	0.316186324	-0.8974418649	-0.4715928308	-0.876452375
## [48,]	0.704268364	-0.4257722138	-0.5857960095	-0.479292254
## [49,]	-0.335338270	-0.5105140854	-0.8836306488	-0.503527430
## [50,]	-0.795654558	-0.6179739906	0.2467995462	-0.559416712
## [51,]	-0.554873730	0.0037068033	-0.0124235422	-0.114280835

## [52,]	-0.462810473	-0.7860153188	-0.5319573681	-0.700871002
## [53,]	-0.240442296	-0.6414132316	-1.0585609147	-0.665754727
## [54,]	0.042829266	1.5452876583	0.6818592748	0.994602094
## [55,]	-0.843810724	-0.3413909460	-0.6941983919	-0.380373170
## [56,]	-0.527962932	-0.2894640119	-0.4672422335	-0.337837964
## [57,]	-0.451479610	1.1623265196	-0.0432402729	0.974818277
## [58,]	0.677357566	0.0628458116	-0.1211884743	-0.064821293
## [59,]	-1.103004204	-0.0042264783	-0.0051725467	-0.134064652
## [60,]	1.284975067	-0.8988842797	-1.1551804294	-0.900192956
## [61,]	0.963461844	0.3989284681	0.4063214467	0.220560264
## [62,]	0.675941208	-0.1693829769	1.5411022389	-0.184513384
## [63,]	1.433692638	1.0761422332	0.0927158923	1.239921422
## [64,]	0.967710917	0.0166885368	1.9000265150	-0.127634911
## [65,]	0.439409454	0.0733033191	-0.0704315060	0.030141027
## [66,]	0.530056354	-0.1711859954	0.1162816276	-0.205780988
## [67,]	0.877064018	-0.6132861424	1.4395883022	-0.596511368
## [68,]	-0.867888807	-0.4776991479	-0.4966087652	-0.511935552
## [69,]	2.501626428	-0.2804489192	-0.0414275241	-0.485227399
## [70,]	-0.887717816	-0.6071558793	-0.6229573614	-0.689989903
## [71,]	-1.159658516	1.3833765930	-0.7601824508	1.295810704
## [72,]	3.824504624	0.4364312538	-0.6610250876	0.149338523
## [73,]	0.293524599	0.6690206460	-0.3187781011	0.414936263
## [74,]	0.405416866	-0.4560629253	-1.0810390006	-0.449616529
## [75,]	-0.518048427	-0.5577531712	-0.3477820830	-0.556943735
## [76,]	-1.258803563	1.2340866576	-0.3640968228	1.069780598
## [77,]	0.511643702	0.0177703480	-0.3677223206	-0.105872713
## [78,]	0.556967152	2.0750145064	-1.2079314215	1.703851925
## [79,]	2.637596778	1.8986792928	1.2111819447	2.860216014
## [80,]	-0.400490729	-0.5036626149	-0.2208896621	-0.538149109
## [81,]	1.027197946	-0.2887428045	1.7350663679	-0.390265079
## [82,]	0.881313091	-0.4297388546	-0.3640968228	-0.658335796
## [83,]	0.711350154	1.7746316170	0.4661421593	2.233563618
## [84,]	1.337380306	0.4104677868	3.0692495357	1.451608261
## [85,]	-0.441565105	-0.6421344391	0.0691501570	-0.704827765
## [86,]	-0.365081784	1.0620786885	0.4679549082	0.947615529
## [87,]	-0.911795899	0.0549125300	1.8184528159	0.215119714
## [88,]	-0.921710404	0.5204519180	-1.0029095244	0.093449241
## [89,]	0.175966901	-0.3871876170	0.5168991277	-0.327946056
## [90,]	0.093818148	0.3834225086	-0.8694912076	0.468847164
## [91,]	-0.586033602	-0.1192590614	-0.1918856802	-0.290356804
## [92,]	-0.258854948	-0.3327364569	-0.6807840503	-0.391254270
## [93,]	-1.362197683	0.0019037848	-0.1157502277	-0.081637537
## [94,]	-0.806985421	-0.3972845208	0.2830545235	-0.379383980
## [95,]	0.006003963	0.2582930217	-0.4576346645	0.415430859
## [96,]	-0.893383247	1.2708682359	0.5295883698	0.834847774
## [97,]	0.503145555	-0.1408952839	0.5332138675	-0.225564804
## [98,]	0.864316797	-0.2530430373	1.4975962660	-0.363062331
## [99,]	0.428078591	-0.6262678759	-1.2285967586	-0.688011521
## [100,]	0.156137891	-0.4171177248	1.1495484831	-0.242381048
## [101,]	-0.578951813	0.1850904688	0.1325963674	-0.002502270
## [102,]	2.178696847	-0.6529525503	0.5277756209	-0.649433078
## [103,]	-0.853725229	-0.7672639260	0.6419787997	-0.832433383
## [104,]	0.059825560	-0.8108969748	0.0093294443	-0.661797964
## [105,]	0.285026452	-0.1809222956	0.5749070915	-0.278981109

## [106,]	2.000235762	-0.0518261678	-0.5312322686	-0.225564804
## [107,]	0.340264407	-0.3576181128	0.7978752024	-0.351686636
## [108,]	-0.302762040	-1.0287016148	-0.5845270853	-0.997925010
## [109,]	1.075354111	2.9202695997	0.5948473291	3.553144196
## [110,]	0.353011627	-0.6158103683	-0.4182980140	-0.627670880
## [111,]	1.112179414	-0.0078325154	0.3755047159	-0.058886148
## [112,]	1.119261203	-0.2263583629	1.0625365374	-0.076691583
## [113,]	2.109295314	-0.1527952063	0.4951461413	0.263590065
## [114,]	2.127707965	-0.2580914892	1.1658632229	-0.408070514
## [115,]	1.916670651	-0.8606602866	-1.1428537371	-0.747857567
## [116,]	-0.121468240	-0.3367030977	-0.5332262923	-0.428348926
## [117,]	1.250982480	-0.3316546458	-0.4313498059	0.204733210
## [118,]	0.691521144	0.0772699599	-0.4857322720	0.060805943
## [119,]	1.488930592	0.5294670107	-0.2625828861	0.362014553
## [120,]	-1.777190522	0.5244185588	-0.0051725467	0.242817057
## [121,]	-0.236193223	-0.9533354397	-1.3707162700	-0.872001017
## [122,]	-0.094557441	1.1093177744	0.6601062884	1.003504812
## [123,]	0.845904146	3.9804445035	3.4499267983	3.432957509
## [124,]	0.173134185	-0.4048571987	-0.6523238930	-0.463959796
## [125,]	-0.646936988	-0.8700359830	-0.1393159630	-0.812649566
## [126,]	-0.552041015	-0.6731463580	-0.6539553670	-0.678119612
## [127,]	-0.212115140	-0.6280708944	-0.3840370604	-0.551008590
## [128,]	-1.750279724	1.0256577139	0.2268593086	1.162269941
## [129,]	0.265197443	0.0927759194	-0.2698338816	-0.034650973
## [130,]	-0.236193223	0.3250047078	-0.0323637797	-0.049983431
## [131,]	0.270862874	-0.7380550256	-0.7261027720	-0.441703002
## [132,]	-0.685178649	0.2492779290	-0.7812103377	0.112738462
## [133,]	-0.550624657	0.1010698047	0.0872776457	-0.010910392
## [134,]	-0.788572769	-0.3370637014	-0.7275529711	-0.442197598
## [135,]	-0.782907338	0.6877720388	-0.0269255331	0.445106584
## [136,]	-0.304178398	-0.6075164830	0.2957437656	-0.696914238
## [137,]	-0.474141335	0.1576845870	2.3405244903	0.193852111
## [138,]	-0.587449960	-0.8267635379	-0.4043398477	-0.852217200
## [139,]	-0.090308368	3.2123586038	0.4262616842	2.746459068
## [140,]	-0.294263893	-0.2407825112	0.2286720575	-0.502043643
## [141,]	0.201461341	-0.7441852886	-1.3067262349	-0.815122544
## [142,]	-0.044984918	1.0808300814	0.2087318199	0.824461270
## [143,]	0.399751435	-0.4358691176	1.2528751686	-0.459508437
## [144,]	-0.063397570	-0.6882917138	-0.8078577460	-0.582168101
## [145,]	-0.838145293	-0.5505410970	0.0401461750	-0.524300437
## [146,]	2.208440361	-0.0323535675	-1.0206744633	0.076632997
## [147,]	1.545584905	-0.3082154047	0.3791302136	-0.289367613
## [148,]	0.302022746	0.0606821893	1.2546879175	0.200281851
## [149,]	-0.281516673	-0.5934529384	-0.8655031601	-0.368997476
## [150,]	-0.991111936	-0.5595561898	-0.8328736804	-0.639541170
## [151,]	-0.273018526	0.0541913226	0.1906043312	0.003432875
## [152,]	2.806143357	-0.7632972852	1.3507636076	-0.802757658
## [153,]	4.272073693	1.5121121171	2.6233133139	0.596947377
## [154,]	-0.247524085	-0.6493465132	-0.7891864327	-0.710762910
## [155,]	-0.103055588	-0.4838294109	-0.7688836453	-0.517870697
## [156,]	-0.073312074	-0.6677373024	-0.4251864597	-0.683560162
## [157,]	-0.161126259	1.4645124275	0.3319987430	1.322518857
## [158,]	-1.433015574	0.2658656996	1.5284129968	0.303157698
## [159,]	-0.527962932	-0.8040455043	-0.8852621228	-0.838368528

## [160,]	-0.760245613	-0.4448842104	-0.8256226850	-0.523311246
## [161,]	0.245368433	0.3484439489	0.8631341617	0.524241851
## [162,]	-1.563320492	2.1449716259	-1.0572919905	2.030284901
## [163,]	-0.280100315	1.1944202496	-0.3060888590	0.952561483
## [164,]	0.681606639	-0.3976451245	0.7960624536	-0.450605720
## [165,]	-1.029353597	0.9340643719	-0.6539553670	0.859082949
## [166,]	-1.435848289	-0.7975546375	-0.2752721282	-0.781490055
## [167,]	0.170301469	-0.8379422529	-1.4691485336	-0.860625322
## [168,]	-1.056264396	0.6989507538	0.3156840032	0.624644721
## [169,]	0.120728946	2.4623028896	0.3501262317	2.211306824
## [170,]	-0.891966890	-0.4827475998	0.0002656999	-0.481270636
## [171,]	-0.459977757	-0.6100407090	-0.9992840266	-0.591565414
## [172,]	-0.862223376	0.2326901584	-0.1266267209	0.136479042
## [173,]	1.117844846	0.0567155485	-1.0125170934	-0.030199614
## [174,]	0.551301721	-0.7135339734	1.0661620351	-0.736481872
## [175,]	-0.431650601	-0.2678277894	1.2836918994	-0.351686636
## [176,]	0.629201400	-0.6662948875	-0.7786724892	-0.707795338
## [177,]	2.600771475	0.0934971268	1.8927755195	0.131533088
## [178,]	0.061241918	-0.3659119981	0.1217198742	-0.189953934
## [179,]	-1.481171739	-0.8368604418	-0.1356904653	-0.872990208
## [180,]	-0.207866066	-0.4192813470	-0.4114095683	-0.538149109
## [181,]	-0.722003952	1.5539421473	0.4788314014	1.461005574
## [182,]	1.583826566	0.8100166961	-0.8229035617	0.765604415
## [183,]	-1.035019029	-0.1271923430	-0.1211884743	-0.227543186
## [184,]	-0.088892010	-0.2552066596	-0.1973239268	-0.476819277
## [185,]	0.052743771	-0.7203854439	-1.3074513344	-0.752803521
## [186,]	-0.328256480	0.0696972820	0.0927158923	-0.092024041
## [187,]	-1.210647397	-0.5317897042	-1.3435250369	-0.518859888
## [188,]	-0.261687663	-0.5772257715	-0.8181904146	-0.555954544
## [189,]	-0.707840374	-0.7907031671	1.2855046483	-0.917503795
## [190,]	-1.141245865	-0.6021074274	-0.6912979937	-0.583157292
## [191,]	1.787782091	-0.4297388546	1.6190504403	-0.372954239
## [192,]	0.010253037	1.1753082531	0.9628353496	1.113799590
## [193,]	0.236870287	-0.1848889364	6.6494285527	-0.314591980
## [194,]	0.930885614	0.0004613699	1.0734130306	-0.110818667
## [195,]	0.555550794	-0.4528174920	-0.4616227120	0.358552385
## [196,]	-0.867888807	-0.7607730592	-0.5587860514	-0.679108803
## [197,]	0.845904146	0.7714320993	1.6226759380	1.008945361
## [198,]	-1.331037811	0.8330953335	0.1597876004	0.715155682
## [199,]	-1.063346185	0.1100848974	-0.2607701373	0.478244477
## [200,]	0.263781085	-0.7066825029	-1.0259314350	-0.702354788
## [201,]	-0.377829004	-0.1866919549	0.1978553267	-0.276013537
## [202,]	-1.117167782	-0.0291081342	-0.7045310605	0.109770890
## [203,]	-0.071895716	0.5363184812	0.6220385621	0.890737056
## [204,]	1.616402795	0.5756242854	1.2927556438	0.515833729
## [205,]	0.132059808	-0.0327141713	-0.3133398545	-0.182535003
## [206,]	-0.416070665	-0.4834688072	-1.5494533085	-0.441208407
## [207,]	0.007420321	-0.6904553360	0.2268593086	-0.667238513
## [208,]	-1.496751675	0.6513510643	-0.6552242912	0.613269026
## [209,]	1.459187078	-0.7647397000	-0.5339513919	-0.680097994
## [210,]	-1.067595258	-0.6915371472	-1.5481843843	-0.663281750
## [211,]	-1.778606880	1.6404870374	0.4770186526	2.058971435
## [212,]	-0.315509260	-0.6598040208	-0.6374593523	-0.703343979
## [213,]	-1.069011616	8.8990790909	0.4697676571	9.453667543

## [214,]	-0.586033602	0.4486917799	0.8160026911	0.445601179
## [215,]	0.217041277	0.0559943411	1.1386719899	0.330360447
## [216,]	0.901142100	-0.5368381561	-0.0414275241	-0.461486819
## [217,]	0.048494698	-0.3925966726	0.6274768087	-0.385319125
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## [220,]	-1.369279472	1.2045171535	0.1887915823	0.917939804
## [221,]	0.090985432	-0.7030764658	-1.4198417643	-0.729557536
## [222,]	0.032914762	-0.5346745338	-1.3029194623	-0.422908376
## [223,]	0.889811238	-0.5714561121	0.0002656999	-0.605908681
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## [226,]	-1.178071168	0.4212858981	-0.7337163173	0.443622797
## [227,]	0.241119360	-0.7712305668	-0.5694812697	-0.820068498
## [228,]	-0.527962932	-0.6251860647	-1.3056385856	-0.291840590
## [229,]	-0.369330857	-0.5779469789	-0.2734593794	-0.702849383
## [230,]	1.379871041	-0.3572575091	-0.2680211327	-0.301237903
## [231,]	0.064074633	-0.3940390874	-0.9749931918	-0.352675827
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## [233,]	-0.639855199	-0.6536737577	0.7797477137	-0.681087185
## [234,]	-0.974115643	0.4284979722	-0.0504912684	0.445601179
## [235,]	0.174550543	-0.6850462804	-0.7010868376	-0.816606330
## [236,]	-0.629940694	-0.5274624597	0.5187118766	-0.593049200
## [237,]	0.041412908	2.3541217770	-0.4592661385	2.166793237
## [238,]	-1.561904135	1.0177244323	-0.3187781011	1.126659071
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## [246,]	0.226955782	-0.1199802688	2.5290503727	-0.172643094
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## [249,]	0.069740065	-0.5606380009	0.5005843879	-0.677130422
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## [251,]	-0.540710152	2.1593957743	-0.7179454021	1.734022245
## [252,]	-0.489721271	-0.0449746974	-0.6778836521	-0.090045659
## [253,]	0.590959740	1.3015195511	-0.7919055560	0.617720385
## [254,]	-0.944372129	-0.3457181905	-0.6526864428	-0.332892010
## [255,]	-0.896215963	0.6877720388	-1.0562043411	0.460439042
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## [257,]	-0.067646643	1.5773813883	-0.0323637797	2.122774244
## [258,]	1.864265412	0.9160341865	-0.2861486214	0.591012232
## [259,]	2.583775181	3.1979344555	2.2426360513	3.587765875
## [260,]	0.524390922	-0.5887650902	0.1108433810	-0.476324682
## [261,]	-1.002442799	-0.0269445119	-0.3332800920	-0.138021415
## [262,]	-1.377777619	-0.0161264007	0.1815405869	-0.142967369
## [263,]	-1.155409443	1.5492542991	0.7543692296	1.622243681
## [264,]	-1.185152957	-0.6323981389	-0.3952761034	-0.658830391
## [265,]	-0.991111936	0.0545519263	-0.8674971838	-0.023275278
## [266,]	-0.857974302	2.7652100050	0.7253652477	2.415080137
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## [271,]	-1.280048930	-0.9915594328	-0.9010330379	-1.000150690
## [272,]	-0.013825046	-0.7744760002	-1.2463616975	-0.841830696
## [273,]	-0.121468240	2.7471798195	0.2449867973	2.968037815
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## [276,]	-0.573286382	0.8511255189	1.9507834834	0.571228416
## [277,]	-0.649769704	-0.6046316533	0.1144688787	-0.643497933
## [278,]	-1.818264899	-0.2772034858	-0.7048936103	-0.248810789
## [279,]	-1.076093405	-0.5307078931	-0.0921844924	-0.585135674
## [280,]	-0.604446254	-0.5671288676	-0.5390270887	-0.512430147
## [281,]	-0.084642937	0.8327347298	-0.3912880559	0.719607041
## [282,]	-0.236193223	0.5651667779	0.0927158923	0.420871408
## [283,]	-0.549208299	0.2370174029	-0.4019832742	0.018270737
## [284,]	0.572547088	-0.4250510064	-0.5430151362	-0.198856652
## [285,]	-0.488304913	-0.4106268580	0.3120585055	-0.256724316
## [286,]	-0.601613538	-0.4805839775	0.2413612995	-0.566341048
## [287,]	0.688688429	-0.4722900922	0.3138712543	0.164176386
## [288,]	-0.989695579	-0.9086205798	-1.3556704544	-0.866065872
## [289,]	-0.066230285	0.2936321852	1.2474369220	0.005411256
## [290,]	-0.459977757	-0.5033020112	1.3725165940	-0.451100315
## [291,]	1.292056856	1.7162138162	1.0027158247	0.738896262
## [292,]	-0.605862612	-0.4236085915	-0.4873637459	-0.343773109
## [293,]	0.269446516	-0.7059612955	-0.8216346374	-0.808692803
## [294,]	-0.799903632	-0.7146157845	0.0383334262	-0.807209017
## [295,]	-0.254605874	-0.8094545599	-0.9494334327	-0.754781902
## [296,]	-0.520881143	-0.6709827358	-0.9467143094	-0.686033139
## [297,]	-0.352334563	-0.8289271602	-0.3441565853	-0.790887368
## [298,]	0.010253037	0.8648284598	1.6099866959	0.629096080
## [299,]	-0.982613790	-0.6316769315	-0.9931206805	-0.596016773
## [300,]	0.391253288	-0.4268540249	-0.1338777164	-0.285410850
## [301,]	0.385587857	2.5452417426	-0.1012482368	2.161847283
## [302,]	-0.043568560	-0.1477467544	-0.3205908500	-0.141978178
## [303,]	1.684387970	2.4046062962	0.9537716052	2.442282885
## [304,]	0.453573032	-0.9255689541	0.6274768087	-0.905633505
## [305,]	-0.052066707	-0.2790065043	-0.2861486214	-0.193416102
## [306,]	-0.569037308	-0.5552289452	1.0317198066	-0.447638147
## [307,]	-0.546375583	-0.7748366039	-1.1662381975	-0.822046879
## [308,]	0.783584402	-0.8314513862	0.1597876004	-0.851722605
## [309,]	-1.338119600	-0.6518707392	-0.9615788502	-0.671195277
## [310,]	-1.078926121	-0.0276657193	-0.7039872358	-0.147913323
## [311,]	-0.214947855	-0.8837389239	0.3863812091	-0.869033444
## [312,]	-1.451428225	-0.3215577420	-0.5535290797	-0.451100315
## [313,]	-0.197951562	-0.2836943526	-1.0105230696	-0.257218911
## [314,]	-0.254605874	-0.9879533958	-1.5528975313	-0.870022635
## [315,]	1.528588611	-0.2465521706	2.8281539361	-0.318548743
## [316,]	-0.859390660	-0.8422694974	-0.9095529576	-0.899698360
## [317,]	-0.893383247	-0.6991098250	-1.1189254520	-0.706311551
## [318,]	-0.872137880	-0.0038658746	-1.2082939713	-0.157805232
## [319,]	3.488827823	0.2168235952	1.2583134153	0.446590370
## [320,]	-1.018022735	-0.0987046500	1.7821978385	-0.187480957
## [321,]	1.415279986	-0.1351256245	0.4607039127	-0.627670880

## [322,]	-1.676629117	0.6755115127	-0.9617601250	0.495555316
## [323,]	0.277944663	-0.6634100579	-0.3169653522	-0.619262758
## [324,]	0.552718079	0.5950968857	-0.3514075807	0.566777057
## [325,]	-0.213531498	-0.5325109116	-0.7424175119	-0.448627338
## [326,]	-0.418903380	-0.7037976733	-0.4828318738	-0.643003338
## [327,]	-0.821148999	-0.5339533264	-0.2462681463	-0.646960101
## [328,]	-0.297096609	-0.6190558017	-0.5567920276	-0.692462880
## [329,]	-0.003910542	0.1165757642	0.0274569330	0.199787256
## [330,]	0.357260701	0.5965393005	0.4353254286	0.046957272
## [331,]	-0.430234243	-0.2454703594	-0.8506386194	-0.117248408
## [332,]	0.368591563	-0.4932051073	-1.1791087145	-0.198362056
## [333,]	-0.356583637	-0.3879088244	1.3580146031	-0.448627338
## [334,]	-0.281516673	-0.6879311101	-0.4001705253	-0.661303368
## [335,]	-0.474141335	-0.7975546375	0.5712815938	-0.824519856
## [336,]	-0.295680251	1.4818214055	1.6534926688	1.587622001
## [337,]	1.357209316	-0.8069303340	-1.0435150991	-0.961572247
## [338,]	-0.278683957	0.8547315560	0.5676560960	0.743347621
## [339,]	0.072572780	-0.5166443484	1.4468392977	-0.538149109
## [340,]	-1.095922415	2.1774259597	-0.5299633444	1.778535833
## [341,]	0.187297763	-0.2021979144	-0.8089453954	-0.093507827
## [342,]	1.197160883	-0.7961122227	-0.4966087652	-0.710762910
## [343,]	0.888394880	-0.8837389239	-0.7148637290	-0.747362971
## [344,]	-0.799903632	0.7840532291	1.9344687435	1.141002338
## [345,]	0.320435397	-0.2184250813	-0.8653218852	-0.252767552
## [346,]	1.027197946	-0.2422249260	2.3423372391	-0.233478331
## [347,]	-0.328256480	-0.5847984494	-0.1175629766	-0.646465506
## [348,]	-0.520881143	-0.2249159480	-1.4841943492	-0.162751186
## [349,]	0.130643451	-0.8455149308	-0.8254414101	-0.878430757
## [350,]	0.426662233	-0.1592860730	-0.3024633613	-0.203308010
## [351,]	-0.777241907	-0.1866919549	-0.9869573343	-0.317064957
## [352,]	1.874179917	0.4155162387	0.1942298289	0.302168508
## [353,]	-0.224862360	2.1262202331	-0.6633816611	2.154428351
## [354,]	0.260948369	0.8951191714	0.5241501232	0.646901515
## [355,]	-0.278683957	0.0614033967	-0.7389732890	0.229462981
## [356,]	-0.135631818	-0.1621709027	0.4733931548	0.171100722
## [357,]	0.313353608	-0.3410303422	0.5314011186	-0.136537629
## [358,]	-0.561955519	-0.5440502303	0.2649270348	-0.558427521
## [359,]	0.483316546	0.4793430952	-0.0305510309	0.697844843
## [360,]	0.962045486	0.3704407751	0.0546481660	0.198303470
## [361,]	-1.550573272	-0.1949858402	-0.4788438262	-0.265627033
## [362,]	-0.826814430	-0.5159231410	0.5839708359	-0.414500254
## [363,]	-0.137048176	-0.6630494542	0.1235326230	-0.658335796
## [364,]	-0.972699285	-0.2389794927	0.4026959489	-0.258208102
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## [366,]	-1.023688166	0.6235845787	-0.5439215107	0.668663713
## [367,]	-0.367914499	2.0587873395	1.2238711867	2.107936382
## [368,]	-0.515215712	-0.5498198896	-0.7944434044	-0.490667949
## [369,]	-1.736116146	2.8914213030	-0.3006506124	2.407166610
## [370,]	-0.197951562	2.1738199226	-0.9371067404	2.322096198
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## [373,]	-0.137048176	-0.2299643999	0.1670385959	-0.227048591
## [374,]	-1.135580433	0.7519594990	-1.0139672925	0.619698767
## [375,]	-0.865056091	-0.8462361382	-1.2875110969	-0.738954849

## [376,]	0.413915013	-0.8318119899	-1.3194154770	-0.750330544
## [377,]	3.073834984	-1.0589923264	0.0256441841	-0.248810789
## [378,]	-0.731918457	-0.8520057976	-0.1211884743	-0.725106177
## [379,]	-0.542126510	-0.9554990620	-1.2238836116	-0.872990208
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## [381,]	0.898309385	-0.5371987598	-0.4188418387	-0.522816651
## [382,]	-0.047817634	-0.8689541719	-0.3369055898	-0.783963032
## [383,]	0.537138143	-1.0305046334	0.3936322046	-0.538149109
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## [385,]	-0.971282927	-0.8000788635	-1.2436425742	-0.630143857
## [386,]	-1.223394618	0.0379641557	0.7434927364	0.023711287
## [387,]	-0.178122552	-0.4996959741	-0.6996366385	-0.380373170
## [388,]	-1.141245865	-0.5447714377	-1.0786824271	-0.572276193
## [389,]	1.377038325	-0.2692702042	-0.2716466305	-0.146924133
## [390,]	-0.560539162	0.7411413877	2.9351061194	1.244867376
## [391,]	0.409665940	-0.7719517742	-1.2130071183	-0.750825139
## [392,]	1.158919222	0.4028951089	1.5628552253	0.148843928
## [393,]	0.657528557	0.8720405340	0.2069190710	0.894693819
## [394,]	0.462071179	0.7898228884	-0.5364892403	0.638987988
## [395,]	-0.168208048	-0.4365903251	0.7888114581	-0.493140926
## [396,]	-0.730502099	-0.9187174837	0.8486321708	-0.805725231
## [397,]	-0.284349388	-0.6908159397	0.2087318199	-0.669216895
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## [401,]	1.182997305	-0.0078325154	-0.8015131250	0.127081729
## [402,]	-1.046349891	-0.5516229082	-0.3115271056	-0.601951918
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## [405,]	-0.809818137	-0.0788714460	-0.5602362505	-0.130602484
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## [407,]	-0.573286382	-0.6003044088	-1.0507660945	-0.562384284
## [408,]	-0.234776865	0.3394288562	1.0534727930	-0.155332255
## [409,]	-0.298512966	0.1749935650	-0.6227760865	0.096416813
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## [418,]	0.827491494	3.4792053483	-0.0069852956	3.243527464
## [419,]	-0.006743257	-0.6486253058	-1.0353577291	-0.662292559
## [420,]	-0.192286131	-0.4279358361	0.8359429287	-0.444175979
## [421,]	-0.018074120	-0.4282964398	0.4045086978	-0.326462270
## [422,]	1.595157428	0.5085519956	0.5332138675	0.954045270
## [423,]	0.056992844	-0.5754227529	-0.8754732789	-0.593049200
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## [448,]	-0.557706446	-0.6662948875	-1.0781386025	-0.684549353
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## [457,]	-0.161126259	-0.3305728347	2.1918790830	-0.354159613
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## [460,]	-0.464226830	-0.8188302563	0.8522576685	-0.802757658
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## [469,]	1.542752189	1.8885823890	0.4498274195	1.451608261
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## [475,]	0.789249833	-0.9266507653	-1.2305907824	-0.774071124
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## [477,]	-0.383494435	-0.2068857626	-0.3604713251	-0.057896957
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## [491,]	-0.430234243	-0.6536737577	-0.1411287119	-0.637562788
## [492,]	-1.468424519	0.2820928665	-0.3097143567	0.146865546
## [493,]	-0.287182104	1.2607713320	0.1289708696	1.230029513
## [494,]	-0.377829004	-0.2797277118	0.4878951458	-0.374932621
## [495,]	-0.554873730	-0.2937912564	0.4643294105	-0.267110819
## [496,]	-0.753163824	-0.6233830462	0.7598074762	-0.628165475
## [497,]	0.813327916	-0.6230224425	-1.0580170900	-0.578705933
## [498,]	-0.331089196	-0.9086205798	-0.7900928071	-0.797811704
## [499,]	0.590959740	1.3959977228	-0.3115271056	0.981742613
## [500,]	-0.081810221	0.6679388349	-0.0015470489	0.662728568
## [501,]	0.834573283	-0.1196196651	-0.6789713014	-0.277991919
## [502,]	1.355792958	0.2521627587	0.5640305983	0.053387012
## [503,]	0.470569325	-0.5317897042	-0.2208896621	-0.643003338
## [504,]	-1.127082286	3.1943284184	-0.8549892166	3.347887097
## [505,]	4.563843402	0.0087552552	-0.2245151599	0.073170829
## [506,]	4.667237522	-0.4715688848	0.3138712543	-0.533697750
## [507,]	0.869982228	-0.8080121451	-0.7630828490	-0.930017059
## [508,]	2.402481381	-0.8195514638	-0.3387183386	-0.765663002
## [509,]	-0.882052385	-0.7156975956	-1.3527700562	-0.850733414
## [510,]	1.137673855	-0.2598945078	1.3489508587	0.035086981
## [511,]	0.677357566	-0.7672639260	-1.0426087246	-0.752308925
## [512,]	-1.319706949	-0.6742281691	-1.0761445787	-0.588103246
## [513,]	1.480432445	-0.0525473752	-0.5189055763	0.112243867
## [514,]	-0.906130468	0.0408489853	-1.0752382043	-0.150880896
## [515,]	-0.516632069	-0.0691351458	-0.0341765286	-0.116753812
## [516,]	-0.097390157	-0.5847984494	-0.3749733160	-0.680097994
## [517,]	-0.479806766	0.5038641474	-0.5335888421	0.174068294
## [518,]	-0.129966387	0.3704407751	-0.6220509869	0.389711897
## [519,]	1.378454683	0.1349665533	-0.0867462458	0.153295287
## [520,]	0.486149261	-0.0785108423	-0.3876625581	-0.183524194
## [521,]	2.005901194	-0.1852495401	-0.1574434517	-0.236445903
## [522,]	0.650446768	2.1143203107	-0.5736505921	2.069357939
## [523,]	0.089569074	-0.9764140771	-0.2426426486	-0.932242739
## [524,]	0.797747980	-0.3103790270	0.0582736637	-0.287883827
## [525,]	0.865733155	-0.5602773972	-0.0015470489	-0.440219216
## [526,]	1.198577241	-1.0041805626	-0.9744493671	-0.888817261
## [527,]	0.052743771	-0.7405792515	-1.1058736602	-0.703838574
## [528,]	-0.668182355	-1.0406015372	-1.3072700595	-1.036008858
## [529,]	0.251033865	0.5081913919	2.5707435967	0.605850095
## [530,]	0.464903894	-0.5548683415	-1.2922242439	-0.569803216
## [531,]	0.562632583	0.1198211975	1.2510624198	0.139941210
## [532,]	0.256699296	-0.7156975956	-0.6206007878	-0.728568345
## [533,]	-0.176706195	-0.7229096698	-1.3355489419	-0.738460254
## [534,]	-1.219145544	1.5449270545	0.9410823631	1.138529361
## [535,]	0.181632332	-0.9176356725	0.6637317861	-0.841336101
## [536,]	-0.040735845	1.0581120477	-0.4110470185	0.910026277
## [537,]	-0.421736096	-0.7301217440	1.1495484831	-0.480281445

## [538,]	1.593741070	-0.3947602948	1.3797675895	-0.350202850
## [539,]	1.423778133	-0.0990652537	0.4443891729	-0.185007980
## [540,]	2.083800873	-0.6691797172	0.4752059037	-0.702849383
## [541,]	0.711350154	-0.4571447365	0.9990903269	-0.612338422
## [542,]	0.086736358	-0.5444108340	-0.2498936440	-0.124172743
## [543,]	-0.849476155	-0.3680756204	0.3048075100	-0.340805537
## [544,]	-0.706424016	-0.6132861424	0.6891102703	-0.656357414
## [545,]	0.578212519	-0.4802233738	-0.3079016079	-0.390759674
## [546,]	-0.678096860	-0.2133766293	0.2159828154	-0.395705628
## [547,]	-0.111553735	-0.7023552584	-0.4529215174	-0.746868376
## [548,]	0.615037822	-1.0485348188	-0.3514075807	-0.928533273
## [549,]	-0.063397570	-0.3947602948	0.2649270348	-0.401640773
## [550,]	0.068323707	0.4126314091	1.2710026573	0.345198309
## [551,]	-0.469892262	-0.3204759308	0.1579748516	-0.371470453
## [552,]	0.385587857	-0.4513750771	0.4534529173	-0.431316498
## [553,]	-0.910379541	-0.5923711272	0.2721780303	-0.687022330
## [554,]	0.419580444	-0.3745664872	1.2003054514	-0.368502880
## [555,]	-0.809818137	-0.6980280139	0.2594887882	-0.674657445
## [556,]	-0.216364213	-0.6680979061	1.8528950444	-0.706806147
## [557,]	0.072572780	-0.5808318086	1.5827954629	-0.602446513
## [558,]	-0.312676545	0.4771794729	3.1000662665	0.371906462
## [559,]	-0.188037057	-0.6482647021	-0.1973239268	-0.317559552
## [560,]	0.411082297	-0.5999438051	3.0583730425	-0.460003033
## [561,]	-0.154044470	-0.1466649432	0.4987716390	0.010851806
## [562,]	-1.101587846	-0.3284092124	4.8566199209	-0.408070514
## [563,]	1.235402544	-0.5227746115	-0.0214872865	-0.249305384
## [564,]	0.848736861	2.0086634240	-0.3459693341	2.914126915
## [565,]	-0.930208551	2.7796341533	0.0709629058	2.377490885
## [566,]	-1.057680754	1.2993559289	2.2589507911	1.155840201
## [567,]	-0.894799605	0.1847298651	-0.2571446395	0.276449546
## [568,]	1.042777882	1.1569174639	0.6854847725	1.437264994
## [569,]	-0.560539162	-0.0702169570	0.3827557114	-0.157310636
##	worst area	worst smoothness	worst compactness	worst concavity
## [1,]	2.485390675	-0.213813513	1.315703888	0.7233896514
## [2,]	0.741749287	-0.604818670	-0.692317104	-0.4403925588
## [3,]	1.180297518	-0.296743909	0.814257044	0.2128891146
## [4,]	-0.288124629	0.689095329	2.741867851	0.8187979283
## [5,]	1.189310289	1.481763364	-0.048477225	0.8277424542
## [6,]	-0.289003924	0.156209255	0.445151962	0.1598845164
## [7,]	0.298364935	-0.908230679	-0.650995203	-0.3098687356
## [8,]	0.233516951	0.587513921	0.268696280	-0.2323495106
## [9,]	-0.352093318	-0.436293449	0.532821399	0.1204623464
## [10,]	-0.360446618	0.035976834	2.607292473	1.5085202629
## [11,]	0.003801211	-1.003151011	-0.905124890	-0.6918331217
## [12,]	0.303860527	-0.422971297	0.844969267	-0.1319720527
## [13,]	1.667646605	-1.299568890	3.210368855	1.8884969766
## [14,]	-0.082589493	0.908577781	0.322861473	0.6167178975
## [15,]	-0.464423216	-0.203821899	1.891976875	0.7657933300
## [16,]	-0.171178434	-0.477592120	0.944923593	0.5140214884
## [17,]	0.111294988	-0.440623149	-0.773844096	-0.3946760928
## [18,]	0.304300174	-0.004988783	-0.026141063	-0.0004543933
## [19,]	1.584113609	-0.182173403	-0.365650730	0.0667951907
## [20,]	-0.368799918	0.473276469	-0.607439687	-0.2658086633
## [21,]	-0.564223164	-0.980503353	-0.362858710	-0.4940597145

##	[22,]	-0.541581326	0.854290012	-0.623075000	-0.3989827164
##	[23,]	0.100523628	-0.083922533	1.552467208	1.0788517384
##	[24,]	1.179418224	-0.770346407	-0.719678902	-0.4884279759
##	[25,]	1.368686408	-0.330715396	-0.371793174	-0.1485359897
##	[26,]	1.562131241	0.329064175	0.698667404	0.1797612407
##	[27,]	-0.423975660	-0.862269255	0.283214785	-0.1684127140
##	[28,]	1.169526158	1.235303555	0.097266234	0.6266562596
##	[29,]	0.069528490	-0.602154240	0.284331593	0.1280817574
##	[30,]	0.456418157	-0.470931044	0.270929896	0.0720956505
##	[31,]	1.421444089	-0.264104637	0.461345680	0.6647533146
##	[32,]	0.014572571	-0.496243133	0.483681842	0.3364560843
##	[33,]	0.603260372	0.408664032	0.298291695	0.6137363888
##	[34,]	0.611833495	-0.674759968	0.430075052	0.1019107370
##	[35,]	-0.116662163	-0.951194619	0.178793226	-0.1740444526
##	[36,]	0.106678691	-0.421306028	-0.027257871	0.1675039274
##	[37,]	-0.339123722	-0.387334541	0.249710542	0.5385361151
##	[38,]	-0.575434172	-0.895574635	-1.149147463	-0.6116636668
##	[39,]	1.443426457	-0.052615476	-0.811815572	-0.4543062658
##	[40,]	-0.479591050	-0.557525031	-0.172442926	-0.0465021381
##	[41,]	-0.435406491	-1.252608304	-0.807348339	-0.5960935661
##	[42,]	-0.513663719	0.340721058	-0.437684853	-0.1969026856
##	[43,]	1.419245853	-0.164188498	4.194835209	2.1644521662
##	[44,]	-0.197996922	-0.655775901	-0.223816099	-0.3327269686
##	[45,]	-0.568179991	-1.229294539	-0.649878395	-0.5755542843
##	[46,]	0.686353721	-0.248784162	0.807556195	0.7833511032
##	[47,]	-0.706339170	0.641801690	-0.503576532	-0.5305003758
##	[48,]	-0.353632084	-0.169517358	-0.118277732	-0.0942062765
##	[49,]	-0.449914854	-0.517225522	-0.625867021	-0.2873417813
##	[50,]	-0.442660672	-0.861270094	-0.650995203	-0.3625420551
##	[51,]	-0.260866493	-0.394328671	-0.877818931	-0.6696374462
##	[52,]	-0.566861049	-0.853942910	-0.765468035	-0.6991212539
##	[53,]	-0.502672536	0.056293116	-0.954767011	-0.6222645865
##	[54,]	1.285373235	-1.046448004	0.230724804	-0.1233588055
##	[55,]	-0.229211884	-0.787998258	-0.847553432	-0.5255311947
##	[56,]	-0.370778331	0.428647260	-0.935669592	-0.6096759944
##	[57,]	1.366488171	-0.194163339	-0.135029854	-0.0809551269
##	[58,]	-0.005431383	-1.126380915	0.171533974	-0.2045220966
##	[59,]	-0.162165663	0.149881232	-0.942873004	-1.0336465246
##	[60,]	-0.703767233	1.022149125	-0.833593330	-0.4003078313
##	[61,]	-0.125674933	0.157541470	-0.809023552	-0.8026458599
##	[62,]	-0.482448757	1.628307035	0.342964019	-0.0627347963
##	[63,]	0.449383800	0.788012306	2.517389420	1.2289210072
##	[64,]	-0.369679212	0.565199316	0.776285568	0.3719029093
##	[65,]	-0.085227377	0.246466833	0.055944334	-0.0716793222
##	[66,]	-0.112045866	-0.112565159	-0.132796238	-0.2896607325
##	[67,]	-0.574554877	1.158701182	-0.442710490	-0.4887592547
##	[68,]	-0.487724526	0.746380582	-0.908084431	-0.3731429747
##	[69,]	-0.498276062	0.835305945	3.382915709	9.0076770783
##	[70,]	-0.483767699	0.306749570	-1.109333254	-0.5314942120
##	[71,]	1.224701901	-0.864933685	-0.500226108	-0.3048995545
##	[72,]	-0.327473067	3.386831080	3.808419601	0.8221107157
##	[73,]	0.640410573	-0.406651661	1.713287577	0.3520261850
##	[74,]	-0.373416215	-0.774009999	-0.269605232	-0.4738517114
##	[75,]	-0.454091503	-0.728381629	-0.406972630	-0.4516560359

##	[76,]	0.855398127	1.258617321	-0.192545472	0.1029045732
##	[77,]	-0.168980197	2.117896115	0.162599509	-0.6716251186
##	[78,]	2.076518640	0.299422387	1.837811682	0.4865253530
##	[79,]	1.672043079	1.112073650	2.393982123	2.5576800294
##	[80,]	-0.439363317	-0.582171012	-0.495200471	-0.3704927448
##	[81,]	-0.345498608	1.108743112	-0.470072289	-0.1998841942
##	[82,]	-0.601813013	-0.082257264	0.573584895	0.2619183680
##	[83,]	1.751179602	0.374692545	1.756284689	0.8416561613
##	[84,]	0.588312362	0.167866137	1.955634938	-0.3496221843
##	[85,]	-0.531469437	-0.357026646	-0.410881458	-0.3916945841
##	[86,]	0.885074323	-0.189833640	-0.501901320	-0.1270028716
##	[87,]	-0.032249872	0.775356262	0.243009693	0.7227270940
##	[88,]	0.380578990	-1.055440457	-0.394129337	0.1724731084
##	[89,]	-0.426173897	0.023653844	-0.030608295	-0.1611245818
##	[90,]	0.053261538	-0.510564446	1.040969091	0.4129814730
##	[91,]	-0.144579769	-0.723718876	-0.407531034	-0.6852075469
##	[92,]	-0.239543597	0.946212860	-0.057970094	0.4444529532
##	[93,]	-0.087645438	-0.852610695	-0.843086199	-0.6066944857
##	[94,]	-0.332309188	-0.385336218	-0.590129161	-0.4364172139
##	[95,]	0.149544307	0.735722860	0.651761463	0.5557626095
##	[96,]	1.044886135	-0.341373117	0.521653318	0.3454006102
##	[97,]	-0.349455434	-0.535543481	-0.764351227	-0.6818947595
##	[98,]	-0.445958027	1.361863998	-0.605764475	-0.8807282587
##	[99,]	-0.540482208	-0.295744747	-0.680032214	-0.4957161082
##	[100,]	-0.296477929	0.321070884	0.193870136	0.0436056790
##	[101,]	0.061614838	-0.389332864	-0.591804373	-0.1796761911
##	[102,]	-0.670551876	1.048793429	-0.817399612	-1.0565710133
##	[103,]	-0.564003341	-0.653111471	-1.082194817	-0.7024340413
##	[104,]	-0.627972030	0.671776531	-0.196454300	-0.0531277128
##	[105,]	-0.378252336	0.184518827	-0.183611007	-0.1024882450
##	[106,]	-0.124795639	0.040306533	0.203363005	0.7565175253
##	[107,]	-0.433428078	0.499254665	-0.132796238	-0.0809551269
##	[108,]	-0.683873191	-1.193657782	-0.838060563	-0.6699687249
##	[109,]	2.850297975	-0.175179273	3.417536761	2.3887278725
##	[110,]	-0.525973845	0.690094490	-0.553274493	-0.2472570539
##	[111,]	-0.383967751	2.267770324	0.214531086	-0.1548302857
##	[112,]	-0.436505609	1.954699755	0.834359590	0.6332818344
##	[113,]	-0.243720247	-0.580505743	2.735167002	3.6972788917
##	[114,]	-0.449035559	1.611654345	0.669630393	0.4643296775
##	[115,]	-0.689610589	0.406332656	-0.186961431	0.0230663971
##	[116,]	-0.341761606	0.253794017	-0.022790638	-0.4486745272
##	[117,]	-0.514323190	3.766512408	2.352101819	2.0120639463
##	[118,]	0.018529397	-0.018643989	0.008479989	-0.0590907301
##	[119,]	0.402121710	0.552210218	0.797504922	0.9145374839
##	[120,]	0.301222643	-1.004816280	-0.952421714	-0.2976114222
##	[121,]	-0.655889637	-0.333379826	-0.568909807	-0.5550150025
##	[122,]	1.102040290	0.353377102	-0.249502686	0.0505625325
##	[123,]	4.235187126	5.425120314	4.053000578	3.1771712714
##	[124,]	-0.354951026	-1.073425362	-0.710186033	-0.1018256875
##	[125,]	-0.564442988	-0.373679336	0.403271658	0.5855776960
##	[126,]	-0.493000294	-0.813310347	-0.910708930	-0.7542129082
##	[127,]	-0.450794148	-0.983500837	-0.767143247	-0.4615943981
##	[128,]	0.898923215	-0.870262546	0.102291870	0.2817950923
##	[129,]	-0.010926975	0.654457734	0.914211370	0.0048460665

## [130,]	0.505438837	-0.668764999	0.352456888	0.5186593907
## [131,]	-0.551693215	-0.089251393	-0.051269245	-0.7193292570
## [132,]	0.175263677	-0.266769067	-0.594037989	-0.1246839205
## [133,]	0.073485316	-0.720721391	-0.332704890	-0.3214634914
## [134,]	-0.272517148	-0.607483101	-0.576727464	-0.5006852893
## [135,]	0.615790322	-0.346368924	-0.628659041	-0.1107702134
## [136,]	-0.449914854	0.152545663	-0.751507934	-0.2840289939
## [137,]	-0.131170525	-0.154196884	-0.652112011	-0.1747070101
## [138,]	-0.608187899	-0.635459620	-0.573377039	-0.5815173016
## [139,]	1.353298750	0.985513208	0.520536510	1.1222492532
## [140,]	-0.307908760	1.408491529	0.530587783	-0.3320644111
## [141,]	-0.617200670	-0.333379826	-1.106876276	-1.0565710133
## [142,]	0.741749287	-0.090250555	-0.340522547	-0.0405391208
## [143,]	-0.416721479	-0.125554257	-0.453878571	-0.6729502336
## [144,]	-0.520917901	-0.571846344	-0.549924069	-0.5563401174
## [145,]	-0.496737296	-0.164521551	-0.428191984	-0.3880505180
## [146,]	-0.336485837	1.042132353	1.224684026	-0.1326346102
## [147,]	-0.343300371	-0.537541803	0.605972330	0.4835438444
## [148,]	-0.019939746	-0.416643275	1.300626978	0.7002001397
## [149,]	-0.420678305	-0.444619794	-0.140055491	-0.0249690200
## [150,]	-0.414743066	-1.399818082	-0.533730351	-0.4668948579
## [151,]	-0.122157755	-0.007986267	-0.785012177	-0.4109087510
## [152,]	-0.662264523	1.794833933	1.601606765	1.5118330503
## [153,]	0.209116523	1.308575390	3.930151685	12.0620670521
## [154,]	-0.546417447	0.658787433	-0.920983565	-0.6603616415
## [155,]	-0.385726341	0.513909032	-0.296408627	-0.0471646955
## [156,]	-0.523775608	-0.507233908	-0.550482473	-0.3960012077
## [157,]	1.177659634	0.664782402	1.343624091	0.6680661020
## [158,]	0.137893653	-1.198320536	0.063203586	-0.0441831869
## [159,]	-0.595438126	-0.503903370	-0.876087879	-0.7802182893
## [160,]	-0.479151402	-0.299408339	-1.080184562	-0.9957813647
## [161,]	-0.043900526	0.796671705	-0.079747852	0.3238674922
## [162,]	1.735791944	0.787679253	0.283214785	0.3825038290
## [163,]	1.247123915	-0.993825505	-0.151223572	0.2781510262
## [164,]	-0.412984476	1.431805295	0.350223272	-0.0213249539
## [165,]	1.264270162	-0.709730616	-0.002129688	-0.1217024118
## [166,]	-0.520917901	-1.134707260	-0.976935652	-0.7826697519
## [167,]	-0.634346917	0.255792340	-0.874580188	-0.6924956792
## [168,]	0.593588130	-0.305736361	-0.043451589	-0.1866330447
## [169,]	1.801739047	-0.288750618	0.606530734	0.4818874507
## [170,]	-0.352972613	-0.653111471	-0.657696052	-0.8155326028
## [171,]	-0.503551830	0.334393036	-0.763792823	-0.4990288956
## [172,]	0.067330253	-0.345702817	-0.829126098	-0.3439904457
## [173,]	0.094588389	-0.548865633	-0.126653793	0.3689214007
## [174,]	-0.467280924	2.637460038	-0.238334605	-0.5752230055
## [175,]	-0.403532058	0.556872972	-0.854812684	-1.0565710133
## [176,]	-0.636984801	0.709744664	-0.975595482	-1.0565710133
## [177,]	-0.282629037	1.938047065	3.495713329	2.9088354928
## [178,]	-0.192281507	-0.137877248	0.863396601	0.7227270940
## [179,]	-0.571477346	-1.206646880	-1.296957017	-1.0037320545
## [180,]	-0.407708708	0.497256342	-1.067341269	-0.8518407526
## [181,]	1.942426199	-0.802652625	-0.006038517	-0.0266254137
## [182,]	0.903979159	-0.928546960	1.234735299	0.2261402642
## [183,]	0.014132924	-0.804317894	-0.159041228	-0.4092523573

## [184,]	-0.386165988	0.104918970	0.659020716	0.9029427280
## [185,]	-0.457388859	-1.249277766	-0.643735951	-0.4688825303
## [186,]	-0.305710523	2.447619374	-0.752624742	-1.0036657987
## [187,]	-0.250974428	-1.390492576	-0.910038846	-0.5888054338
## [188,]	-0.494099412	-0.045288292	-0.936674719	-0.4013016676
## [189,]	-0.568619638	0.263119523	-0.932654210	-0.5411012954
## [190,]	-0.483987523	-0.348034193	-0.187519835	-0.3552539228
## [191,]	-0.189423799	0.309414001	6.138081329	2.8061390837
## [192,]	0.292649520	-0.822635853	-0.423166348	-0.3357084772
## [193,]	-0.409906945	-1.774503603	-1.046568638	-1.0565710133
## [194,]	-0.129631759	0.685098683	0.724353991	0.1900308816
## [195,]	-0.332748835	0.346382972	1.437435972	0.7823572670
## [196,]	-0.540482208	-0.580505743	-0.536522371	-0.2876730601
## [197,]	0.205819168	2.251117634	0.446827174	0.4888443042
## [198,]	0.791869085	-0.503237262	1.534598278	0.9675420821
## [199,]	0.305179469	-0.505235585	0.642268594	-0.1601307455
## [200,]	-0.459806919	-0.998155204	-0.530938331	-0.3940135353
## [201,]	-0.287904805	0.157541470	-0.429308792	-0.5924495000
## [202,]	0.008637332	-0.316727136	0.011830413	-0.1578117944
## [203,]	0.941349184	0.761368003	1.436877568	1.9110239309
## [204,]	0.272205918	0.593841943	0.312810200	-0.0256315775
## [205,]	-0.220858585	-0.029301710	-0.355599457	-0.1617871392
## [206,]	-0.305490699	-0.522554383	-0.351132224	-0.3810936645
## [207,]	-0.615661904	0.891925091	-0.725262943	-0.7923762190
## [208,]	0.618208382	-0.667099730	-0.583428312	-0.4119025872
## [209,]	-0.554990570	-0.596159272	0.275397128	-0.2197609186
## [210,]	-0.447057146	-0.915890916	-0.732522196	-0.4466868548
## [211,]	1.568725952	0.360704285	0.593687441	0.7618179852
## [212,]	-0.510366364	-0.507566962	-0.458345803	-0.3791059920
## [213,]	10.667227849	2.134548805	0.125186437	1.0599688503
## [214,]	0.399923473	8.022939926	3.354437102	3.7072172539
## [215,]	-0.205251104	1.278600548	0.648969443	0.1651849762
## [216,]	-0.387924577	-0.360024130	0.497083539	0.2383975775
## [217,]	-0.414962889	0.609828525	0.918120199	0.9132123689
## [218,]	-0.385726341	-0.786332989	0.319511049	0.8562324258
## [219,]	1.847902018	-0.078926726	0.346872848	0.0850155213
## [220,]	1.529157690	-0.500239778	0.053710717	-0.1740444526
## [221,]	-0.504211301	-0.968513416	-0.476214733	-0.5092985365
## [222,]	-0.424415307	-0.396326994	-0.130562622	-0.2137979013
## [223,]	-0.555869865	0.285767181	-0.640943930	-0.7738577375
## [224,]	-0.179092086	-0.757690363	-0.097058378	-0.2668024995
## [225,]	-0.344179666	-0.336044257	-0.806231531	-0.3082123419
## [226,]	0.174824030	0.015993606	-0.625308617	-0.2747531892
## [227,]	-0.625993617	-0.175845381	-0.972580100	-0.9632497926
## [228,]	-0.449695030	-0.973176169	0.368092202	0.1506087117
## [229,]	-0.479810873	-0.623469683	-0.141730703	-0.0574343364
## [230,]	-0.334287601	-0.019310096	0.731613243	0.4948073215
## [231,]	-0.183708383	-0.502571155	-0.301434263	-0.0445144656
## [232,]	-0.697546223	-1.128379238	-0.503018128	-0.5155928325
## [233,]	-0.546857094	-0.893243258	-1.042268926	-0.9497998758
## [234,]	0.652281052	-0.673094699	-0.271280444	0.0886595875
## [235,]	-0.608847370	1.531721434	-0.841969391	-0.6636744289
## [236,]	-0.401553645	0.115909745	-0.650436799	-0.8146712781
## [237,]	2.538148357	-0.204154953	0.176001206	0.4331894761

## [238,]	0.948823189	0.305750409	0.326770302	0.3536825787
## [239,]	-0.228112766	-0.244454463	0.165949933	0.2188521319
## [240,]	0.190431511	-0.726383306	0.132445690	-0.1945837344
## [241,]	-0.289003924	-0.190166694	-0.725821347	-0.4569564957
## [242,]	-0.688731294	-1.257604111	-1.147416410	-0.8414717281
## [243,]	-0.526413492	-0.125887311	1.879691986	1.8851841892
## [244,]	-0.008948562	-0.895907688	0.066554011	0.0601696159
## [245,]	0.441250324	1.188676023	0.393220384	0.2403852499
## [246,]	-0.376273923	2.997158138	-0.648761587	-0.4387361651
## [247,]	-0.588623592	-0.259774937	-0.546573644	-0.0365637759
## [248,]	-0.527292787	-0.512895822	1.699885879	1.6460009396
## [249,]	-0.520917901	0.049298986	-0.844761411	-0.6984586965
## [250,]	-0.477392813	-0.126220365	-0.736989428	-0.3592292677
## [251,]	2.144663979	-0.585501550	0.759533446	2.0965400247
## [252,]	-0.293400397	-0.220141535	-0.828009290	-0.6441289833
## [253,]	1.153479030	0.479937545	1.401139708	1.1961244120
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## [258,]	0.420367075	1.035471277	1.139248205	0.5941909432
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## [260,]	-0.380670396	-0.564852215	0.004571160	-0.0995067363
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## [263,]	1.112372003	-0.107902406	1.917105058	0.4798997783
## [264,]	-0.399135585	-1.403814728	-0.914282716	-0.6222645865
## [265,]	0.111734635	-0.848614049	-0.749274317	-0.3781121558
## [266,]	3.503174289	-0.829296929	-0.597388414	-0.3466406756
## [267,]	-0.290982337	0.142887103	0.576935319	0.0545378773
## [268,]	-0.298456342	-0.534544319	-0.079189448	-0.0299382011
## [269,]	-0.485746113	-0.507233908	-0.206505573	-0.1988903580
## [270,]	-0.430790194	0.619820139	1.249253805	1.3746836524
## [271,]	-0.649954398	-1.182000899	-1.215541706	-0.8966958939
## [272,]	-0.597196716	-0.189500586	-0.796180258	-0.6318716699
## [273,]	2.560130725	-0.450947816	1.346974515	1.0400921260
## [274,]	-0.526413492	2.251117634	-0.826892481	-0.7800526499
## [275,]	0.120307759	-0.531879889	-0.769935268	-0.5192368986
## [276,]	0.186914332	2.377678077	-0.591245969	-0.6368408510
## [277,]	-0.511025835	0.461286532	-0.920257640	-1.0073098649
## [278,]	-0.079072314	0.176525536	-0.800647491	-0.1876268809
## [279,]	-0.398256290	-1.099070504	-1.128207311	-0.7037591563
## [280,]	-0.460026743	-0.934541929	-0.562208958	-0.5738978906
## [281,]	0.644367399	0.116908907	-0.055178074	0.2645685979
## [282,]	-0.055111534	0.330729444	-0.617490960	-0.5550150025
## [283,]	0.281878160	-0.461938592	-0.194779088	-0.0431893507
## [284,]	-0.269219793	-0.825300284	0.521094914	0.2261402642
## [285,]	-0.374735157	-0.207485491	0.789128861	1.5694755509
## [286,]	-0.393200346	-0.219142374	-0.975539642	-0.9949531679
## [287,]	-0.405070824	-0.107236298	1.456421710	0.3967488148
## [288,]	-0.607968076	-0.769347246	-0.671656153	-0.5092985365
## [289,]	-0.124355991	2.897241999	3.190824713	1.6264554940
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## [291,]	0.808355861	0.240138811	4.518709562	2.2429652274

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## [294,]	-0.581589235	0.184518827	-0.585103524	-0.5888054338
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## [296,]	-0.496737296	-0.896906850	-0.967442783	-0.9150487360
## [297,]	-0.642920040	-1.186663653	-0.740898257	-0.7014402051
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## [301,]	2.036950379	-0.328050965	0.365858586	0.8111785173
## [302,]	-0.264163848	-0.170183466	0.458553659	0.5044144049
## [303,]	1.988589170	0.307415678	1.219658390	1.4773800615
## [304,]	-0.665122231	0.610827687	-0.899931732	-0.4576190532
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## [306,]	-0.486405584	-0.306069415	-0.117719328	-0.5278501459
## [307,]	-0.545977799	-1.135373368	-0.983748181	-1.0192027716
## [308,]	-0.671519100	0.115909745	-1.150040910	-0.9346273095
## [309,]	-0.438484022	-1.233291184	-1.213531451	-0.9893214293
## [310,]	-0.161066545	-0.963517609	-1.159645459	-0.9627860023
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## [315,]	-0.495198531	4.565841519	-0.639268718	-1.0565710133
## [316,]	-0.607748252	-0.703735648	-1.254518309	-0.9697759837
## [317,]	-0.538943442	-0.565518322	-1.100677991	-0.6845449894
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## [324,]	0.631397802	-0.518224683	-0.065229347	0.0002081641
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## [327,]	-0.360886265	-0.116228751	-0.790596218	-0.8671789582
## [328,]	-0.513663719	-0.770013353	-1.038136736	-1.0173476106
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## [331,]	-0.155351129	-0.400323639	0.389311556	0.1741295021
## [332,]	-0.432768607	-0.437625665	0.394895597	0.3987364872
## [333,]	-0.455410445	1.948038679	-0.806231531	-0.9473484131
## [334,]	-0.555430217	-0.474261582	-1.024902560	-1.0243144025
## [335,]	-0.595657950	0.279772213	-0.951863310	-0.8245433845
## [336,]	1.029498477	-0.195162501	-0.419257519	0.4348458698
## [337,]	-0.569938580	-0.602820348	-0.135588258	-0.0253002988
## [338,]	1.053239434	0.168865299	0.753391002	0.2400539712
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## [341,]	-0.180191205	-0.820637530	0.282097977	0.2155393445
## [342,]	-0.621377320	-0.362022453	0.515510873	0.6090984865
## [343,]	-0.649294927	0.124902198	-0.374585195	-0.1429042511
## [344,]	0.600622488	-0.761020901	0.457995255	0.3828351077
## [345,]	-0.347477021	0.830976246	-0.810140360	-0.5096298152

## [346,]	-0.461785332	3.436789149	1.185595742	-0.1916022257
## [347,]	-0.490582233	0.046301502	-0.810698764	-0.8633692527
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## [349,]	-0.617420494	0.716072686	-0.945385822	-0.7451689987
## [350,]	-0.300874402	-0.413312737	-0.073047004	-0.7976435509
## [351,]	-0.314503470	-0.152531615	-1.032329334	-0.8594932915
## [352,]	0.239232367	0.762034110	2.239862603	2.2406462762
## [353,]	2.478795965	-0.223805127	0.946598806	0.3566640873
## [354,]	0.506318132	1.158701182	-0.065229347	0.5706701528
## [355,]	-0.252733017	-0.499573671	0.468604932	0.4358397060
## [356,]	-0.282409213	0.936554300	0.942131573	1.0208779591
## [357,]	-0.412544829	0.255126232	0.772935144	0.6369259006
## [358,]	-0.430790194	-0.467267452	-0.979392630	-0.8825171639
## [359,]	-0.223276645	1.295253238	0.196103752	0.0081588539
## [360,]	-0.216681935	-0.068269004	-0.921151086	-0.2787285341
## [361,]	-0.264603496	-0.418974652	-1.160650587	-1.0303039221
## [362,]	-0.425514426	-0.513894984	-0.280773313	-0.4619256768
## [363,]	-0.507288833	-0.477259066	-0.503576532	-0.5500458214
## [364,]	-0.148536596	0.071946644	-0.414790287	-0.4496683635
## [365,]	-0.596097597	-0.881586375	-0.724704539	-0.5758855630
## [366,]	0.705698205	-0.277426788	-0.358391477	-0.2697840082
## [367,]	1.390668775	0.465616231	1.187270954	0.8992986619
## [368,]	-0.478491931	-0.402321962	-0.647644779	-0.3940135353
## [369,]	4.039544056	-0.490581218	-0.801764299	-0.3622107763
## [370,]	1.975399750	-1.020136754	0.152548236	0.1280817574
## [371,]	0.113493225	-0.468266614	0.764559083	0.9549534900
## [372,]	-0.497176944	-0.675759129	-0.593479585	-0.5427576891
## [373,]	-0.028073222	-0.870928654	0.071021243	0.0820340127
## [374,]	0.806377448	-0.276427627	-0.364533922	-0.1684127140
## [375,]	-0.578951351	-0.936207198	-0.536522371	-0.6693061674
## [376,]	-0.558947396	-0.842952135	-0.410881458	-0.4102461935
## [377,]	-0.727815943	0.485599459	2.845172601	4.0285576307
## [378,]	-0.558947396	-0.699072895	-0.750949530	-0.8078469361
## [379,]	-0.637204624	-0.609148370	0.243568097	-0.2472570539
## [380,]	-0.579171174	0.121238606	1.117470447	0.4633358413
## [381,]	-0.534107321	-0.135212817	-0.430425600	-0.3605543827
## [382,]	-0.629950443	-0.581171851	-0.360066689	-0.4857777460
## [383,]	-0.676794868	-0.665767515	1.123612892	0.3695839582
## [384,]	-0.464643040	-0.108568514	0.646177423	0.5193219482
## [385,]	-0.551253568	-0.922551992	-0.265137999	-0.1197147394
## [386,]	-0.161066545	0.423318399	-0.449969743	0.0661326332
## [387,]	-0.447936441	-0.877922783	0.267021067	0.3825038290
## [388,]	-0.378472159	-1.103400204	-0.632567869	-0.3979888802
## [389,]	-0.381769514	1.112073650	2.301287049	2.0813012027
## [390,]	0.654259465	1.398499915	0.865071813	1.4177498884
## [391,]	-0.625553970	-0.452613085	-0.660488072	-0.7751165967
## [392,]	-0.252513194	2.923886303	-0.324887234	-1.0565710133
## [393,]	0.584135712	0.075943290	0.212297470	0.5905468771
## [394,]	0.893647446	-0.608149208	0.657903908	0.5064020774
## [395,]	-0.398256290	0.368031469	-0.511952593	-0.4460242973
## [396,]	-0.608187899	-0.556192816	-0.711861246	-0.6815634808
## [397,]	-0.462664627	-0.532545996	-0.329912870	0.0379739404
## [398,]	-0.214703522	-0.539540126	0.518861297	0.4490908555
## [399,]	-0.601373365	-0.926548638	-0.602414050	-0.4138902596

## [400,]	-0.335826366	-0.525551867	-0.326004042	-0.3678425149
## [401,]	0.025783579	0.039307372	0.653436675	0.9857624128
## [402,]	-0.470138632	-0.288417564	-0.750391126	-0.6007314684
## [403,]	-0.442440849	-1.136372529	0.650644655	0.0870031938
## [404,]	-0.636984801	-1.383498446	-0.705160397	-0.5222184073
## [405,]	-0.223936116	0.220155583	-0.948568726	-0.6235897014
## [406,]	-0.319999062	0.825314331	-0.231075352	-0.3973263227
## [407,]	-0.406829413	-1.026797830	-0.726938155	-0.4499996422
## [408,]	0.019848339	-0.343038386	1.078940567	0.6577964611
## [409,]	0.208237229	0.063287245	0.125186437	-0.2253926571
## [410,]	-0.319779238	-0.384004003	-0.132796238	-0.3744680897
## [411,]	-0.586425356	-0.549531740	-0.867265095	-0.6712938399
## [412,]	-0.589063240	-0.627133275	-0.900322615	-0.7067406649
## [413,]	-0.737180432	-0.508233069	0.070462839	0.1807550769
## [414,]	-0.259987199	-0.863268416	0.145288983	0.0406241703
## [415,]	0.110855341	-0.069934273	-0.625867021	-0.2320182319
## [416,]	-0.457388859	0.950542559	0.282097977	-0.5165866687
## [417,]	-0.333408306	2.564188203	-0.487941218	-0.6037129771
## [418,]	2.993183364	0.385350266	0.748923769	0.4318643611
## [419,]	-0.504870772	-0.303071931	-0.717445286	-0.7559355577
## [420,]	-0.469259337	-0.044289131	-0.895464500	-0.8256034765
## [421,]	-0.440462436	0.078940774	-0.279656505	0.4166255391
## [422,]	0.200323576	0.977519916	1.505561267	0.6919181712
## [423,]	-0.537404676	-0.385336218	-0.303109475	-0.1849766510
## [424,]	-0.461125861	-1.019470647	0.145847387	0.1588906801
## [425,]	0.069968137	2.007988363	-0.370117962	-0.8571080846
## [426,]	-0.631709033	-0.438624826	-1.104586819	-0.9878638029
## [427,]	-0.430130723	0.107583400	0.736638880	0.6869489902
## [428,]	-0.442660672	-0.165853767	-0.223257695	-0.0657163049
## [429,]	-0.673387601	-0.385669272	-1.072981150	-0.7452352544
## [430,]	-0.375834275	0.098590948	-0.767143247	-0.8782436681
## [431,]	-0.354951026	-0.025305065	2.046654799	1.5691442722
## [432,]	-0.547516565	0.985513208	0.417231759	0.5537749371
## [433,]	0.267149973	0.681435091	0.093357406	0.7807008733
## [434,]	1.397263485	0.344051596	0.860046177	0.7061631570
## [435,]	-0.360886265	-0.427967104	-0.492966855	-0.5725727756
## [436,]	-0.472336868	-0.575176882	-0.422607944	-0.3327269686
## [437,]	-0.304171757	-0.376676820	-0.662163284	-0.8224894564
## [438,]	-0.167001784	0.311412324	-0.699576356	-0.5238748010
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## [440,]	-0.463543921	-0.734043543	-0.906855942	-0.7513639111
## [441,]	-0.487724526	0.507581010	1.167168408	1.0735512786
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## [443,]	-0.241741834	-0.202822738	-0.777194520	-0.8001281415
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## [445,]	-0.100395211	-0.973842277	-0.551599281	-0.0710167647
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## [447,]	0.079420555	-0.775009160	-0.297525435	0.1681664848
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## [451,]	-0.441781378	-0.062274036	1.962894190	1.1507392248
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## [453,]	-0.582028882	0.097591786	0.022998494	-0.0822802419

## [454,]	-0.321757651	-0.302405823	-0.724146135	-0.5215558498
## [455,]	-0.593899361	-1.050111596	-0.945888386	-0.6404849171
## [456,]	-0.250754604	-0.399657532	-0.726938155	-0.7936682061
## [457,]	-0.378252336	0.939218730	-0.072488600	0.3596455960
## [458,]	-0.500254475	-0.423970459	-0.971407452	-0.5563401174
## [459,]	-0.420898129	-0.328717073	-0.921597810	-0.8683715617
## [460,]	-0.638743390	-0.150866346	-0.713536458	-0.5762168418
## [461,]	1.645664238	0.565865424	-0.183611007	-0.1551615645
## [462,]	11.032135149	0.202836786	1.578153795	1.6118792295
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## [464,]	-0.603351779	-0.110566836	-0.472864309	-0.3675112362
## [465,]	-0.515422309	-0.357026646	-0.589012353	-0.5384510655
## [466,]	-0.363304326	-0.703402594	2.294586201	1.4886435386
## [467,]	-0.284387627	0.084602688	0.352456888	0.4722803673
## [468,]	-0.425514426	1.311905928	-0.720795710	-0.7150226334
## [469,]	1.419245853	-0.091582770	2.500078894	1.1268871556
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## [472,]	0.101622746	0.158540631	-0.554391301	-0.5712476607
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## [474,]	-0.114463926	0.099257055	-0.962417146	-1.0565710133
## [475,]	-0.675739715	-0.854942071	0.304434139	-0.1432355299
## [476,]	-0.630829738	-0.722053607	-0.419815923	0.0426118427
## [477,]	-0.205031280	-0.977172815	0.413322931	-0.1220336906
## [478,]	-0.578291880	-1.243282798	-0.688408275	-0.7284725502
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## [481,]	-0.529271200	-0.709397562	-0.492408451	-0.5937746149
## [482,]	-0.262185435	-1.111393495	-0.818516421	-0.5497145426
## [483,]	-0.604450897	-0.862935363	-0.611906919	-0.6146451755
## [484,]	-0.432988431	-1.264265187	-0.965600049	-0.4940597145
## [485,]	-0.581809058	-0.335378149	-0.406414226	0.0485748600
## [486,]	-0.223056821	0.064286407	2.749127104	2.6338741394
## [487,]	-0.447496793	-1.167679586	-0.644852759	-0.4602692831
## [488,]	0.602381077	-0.583836281	0.159249084	0.3533513000
## [489,]	-0.280650624	0.651793304	-0.701251568	-0.4433740674
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## [491,]	-0.490142586	-0.647782610	-0.749832721	-0.7448377199
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## [494,]	-0.345938255	1.115404188	-0.862630341	-0.8440225744
## [495,]	-0.313624175	0.253460963	-0.276864485	-0.6123262243
## [496,]	-0.406609590	-0.541538449	-0.657137648	-0.3433278882
## [497,]	-0.482228934	-0.445285902	-0.025582659	-0.1833202573
## [498,]	-0.624894499	-1.080752545	-0.657137648	-0.6106698306
## [499,]	1.218546838	0.310746216	0.372001030	0.4126501942
## [500,]	0.763951478	-0.124888150	0.135796114	0.2890832246
## [501,]	-0.120838813	-0.971843954	-0.406972630	-0.3953386503
## [502,]	-0.028293045	0.878935993	0.730496435	0.0949538835
## [503,]	-0.480250521	0.886263176	-0.547690453	-0.2035282604
## [504,]	3.074518123	-0.428966266	0.451294406	0.2605932530
## [505,]	-0.446177851	0.913240534	1.113561619	0.0972728347
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## [508,]	-0.616321375	1.858114154	-0.111576884	-0.4602692831
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## [510,]	-0.171837905	0.831642354	1.335806434	0.9373957169
## [511,]	-0.600054423	-0.019643150	0.764000679	0.2741756813
## [512,]	-0.431229841	-0.110899890	-0.754299954	-0.5656159221
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## [514,]	-0.070938838	-0.695409303	-0.410881458	-0.0511400404
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## [518,]	0.425642843	-0.650113987	-0.136705066	-0.0455083018
## [519,]	-0.131170525	-0.588832088	-0.122186561	-0.5911243850
## [520,]	-0.257569138	0.155876201	-0.551040877	-0.4016329463
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## [523,]	-0.681565042	-0.946198812	-1.093251217	-0.9291612103
## [524,]	-0.305270876	-0.100575223	-0.165742077	-0.3655235637
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## [526,]	-0.727244402	0.285101074	-0.438801661	-0.4599380044
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## [537,]	-0.479151402	-0.309066899	0.019648070	0.4822187294
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## [544,]	-0.494319236	-0.688748227	-0.656579244	-0.5603154623
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## [546,]	-0.199975335	-0.390665079	-0.250619494	-0.3870566818
## [547,]	-0.601593189	0.014994445	-1.018034190	-0.7213169295
## [548,]	-0.725661671	1.075437733	0.299408503	-0.1909396683
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## [550,]	-0.161286369	0.406998763	-0.378494023	-0.6335280636
## [551,]	-0.432328960	0.845297559	-0.806231531	-1.0565710133
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## [553,]	-0.474315281	0.597505535	-0.734197408	-0.6166328479
## [554,]	-0.494099412	1.298583776	-0.398596569	0.2672188278
## [555,]	-0.516741251	0.456623779	-0.220465675	0.2347535114
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## [560,]	-0.513663719	0.386016374	0.242451289	0.8443063912
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##	[562,]	-0.385286693	0.184185773	-0.926958489	-1.0565710133
##	[563,]	-0.388803872	-0.804650948	1.282199644	1.3813092272
##	[564,]	1.724800761	-0.213813513	0.984011877	1.5423106943
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##	[567,]	0.180539446	-0.379008196	0.660695928	0.5103774223
##	[568,]	1.008615228	-0.172847896	2.015942576	1.3011397723
##	[569,]	-0.465742158	0.049298986	-1.162493320	-1.0565710133
##		worst concave points	worst symmetry	worst fractal dimension	sd radius
##	[1,]	0.660239001	1.147746771	0.9062856452	1.885030997
##	[2,]	0.259933354	-0.804742292	-0.0993563172	1.804339810
##	[3,]	1.423574870	0.236827152	0.2933013299	1.510541129
##	[4,]	1.114026779	4.728519774	2.0457108675	-0.281217023
##	[5,]	1.143198850	-0.360774828	0.4988891643	1.297433635
##	[6,]	-0.069062786	0.134000901	0.4864178435	-0.165352754
##	[7,]	-0.227888508	-0.828936704	-0.6106804717	1.367779798
##	[8,]	0.434965782	-0.687399393	0.6111310519	0.163619008
##	[9,]	0.075176901	0.107387047	-0.0173479347	-0.161214745
##	[10,]	0.409035052	-0.320854048	2.3752560728	-0.243974936
##	[11,]	-0.681514219	-0.718852129	-0.2845365358	0.604317029
##	[12,]	0.165934457	-0.055925235	0.1319299966	0.858804619
##	[13,]	4.716777604	2.939342993	3.4183119977	0.970530878
##	[14,]	1.316610608	1.121132918	-0.2996532883	0.118100903
##	[15,]	0.726686497	-0.112782104	1.6243313906	-0.256388965
##	[16,]	-0.145234306	-0.238593047	0.6315386678	0.246379200
##	[17,]	-0.114441564	-0.779338159	-0.6462048402	0.579488972
##	[18,]	0.190244516	-0.441826109	0.1311741589	0.970530878
##	[19,]	0.553274739	-0.844663072	-0.6794616958	2.286417927
##	[20,]	0.219416588	-0.089797412	-0.5649522953	-0.239836927
##	[21,]	-0.859950056	-0.455133036	-0.5177124436	-0.366046219
##	[22,]	0.391207675	-0.032940543	-0.3125025280	-1.249511266
##	[23,]	1.737984974	1.957049858	0.2264096999	0.372588492
##	[24,]	-0.229509179	-1.174916798	-0.6832408839	2.669183813
##	[25,]	-0.080407480	-0.709174364	-0.3756149698	2.108483514
##	[26,]	1.972982216	0.306990948	1.3790620807	1.237432496
##	[27,]	0.279381401	-0.726110452	-0.0317088496	0.279483277
##	[28,]	1.185336287	0.288845139	0.1595180699	1.042946046
##	[29,]	-0.156579000	-0.346258181	-0.3128804468	0.827769547
##	[30,]	0.282622742	-0.156332045	-0.0199933664	0.773975422
##	[31,]	-0.035028702	-0.057134956	0.2891442229	1.423642928
##	[32,]	-0.219785155	0.264650726	0.7075003493	0.113962893
##	[33,]	-0.109579552	0.057788502	0.0223335407	0.953978839
##	[34,]	-0.348952605	-0.618445318	0.0382061308	1.630543407
##	[35,]	-0.442303233	-0.424890021	0.0083505446	0.815355518
##	[36,]	0.024936111	0.888866561	-0.4270119284	0.773975422
##	[37,]	-0.030166690	-0.031730823	0.0858239013	-0.078454553
##	[38,]	-0.025304679	0.746119529	-0.7626038347	-0.614326794
##	[39,]	1.195060311	-1.531542434	-0.7712959674	-0.264664984
##	[40,]	0.133521044	-0.819258939	-0.2297383079	-0.152938725
##	[41,]	-0.796581834	-0.815629777	-0.9481619721	-0.070178534
##	[42,]	-0.231129849	-0.843453351	-0.2852923734	-0.709501015
##	[43,]	2.363563840	3.966395790	1.4554016810	1.618129378
##	[44,]	-0.362404171	-0.405534491	-0.1806088621	0.229827162
##	[45,]	-0.801119712	-1.138625180	-0.7841452071	-0.008108390

## [46,]	0.835271429	0.452157421	-0.0982225608	1.355365769
## [47,]	-0.952814484	0.628776630	-0.4583791899	-1.511447273
## [48,]	0.057349524	-0.376501196	-0.0574073289	-0.123972658
## [49,]	-0.994465719	-0.757563188	-0.5188462001	-0.519152574
## [50,]	0.007108734	-0.499892698	-0.6949563671	-0.231560908
## [51,]	-0.705176010	0.425543567	-0.7656271852	-0.680534948
## [52,]	-0.622359741	-0.882164411	-0.4700946732	-0.330873138
## [53,]	-0.615228790	-0.070441882	-0.4383494928	-0.655706890
## [54,]	-0.417182838	0.749748691	0.5030462713	0.896046705
## [55,]	-0.417182838	-1.160400151	-0.8185358191	0.378795507
## [56,]	-0.506643858	1.407836702	-0.5324512774	-0.709501015
## [57,]	0.580826140	-0.244641650	-0.4485533008	2.042275361
## [58,]	0.149727750	-0.286981871	0.1183249193	0.331208396
## [59,]	-1.236431845	0.164243916	-0.3041883141	-0.421909349
## [60,]	-0.981176220	0.043271855	-0.3805279144	-1.399100313
## [61,]	-0.583949847	2.575217090	0.8155851300	-1.086059887
## [62,]	-0.411186357	1.575987867	-0.2822690229	-1.387100085
## [63,]	1.083234037	-0.428519183	0.8760521401	0.289828301
## [64,]	0.616480894	2.586104575	0.7660777654	-1.295029372
## [65,]	0.178899822	-0.507151022	-0.0732799191	0.169826022
## [66,]	0.007108734	-0.186575061	-0.2157553118	0.215344128
## [67,]	-0.399193394	0.271909050	0.1670764462	-1.212269180
## [68,]	-0.458834074	0.155775872	-0.6231517926	-0.815020259
## [69,]	3.472102555	2.592153178	2.1783603710	-1.232959228
## [70,]	-0.507940394	0.241666035	-0.7138523078	-0.581222718
## [71,]	0.308553473	-0.808371454	-0.7924594210	1.777442747
## [72,]	0.950339045	0.588855850	6.8535940113	-1.352340804
## [73,]	-0.085269492	-0.637800848	0.9463450394	1.458816009
## [74,]	-0.419775911	-1.009185075	-0.2512796802	0.062237773
## [75,]	-0.620901137	-0.808371454	-0.5634406201	-0.446737406
## [76,]	1.020027882	-0.609977274	-0.6979797176	0.724319307
## [77,]	-0.576494762	0.626357189	0.8953259996	-0.452944420
## [78,]	1.443022918	0.649341880	1.2316737435	1.266398563
## [79,]	2.290633661	7.065699990	0.8284343696	1.469161033
## [80,]	-0.586218786	-0.370452593	-0.3578527856	-0.419840344
## [81,]	-0.695289920	-0.288191592	0.0684396359	-0.653637885
## [82,]	0.329622191	0.096499562	0.3053947319	-0.152938725
## [83,]	1.378196093	-1.196691769	0.7929100012	2.840911211
## [84,]	1.075130684	1.211861963	2.4924109049	0.840183576
## [85,]	-0.772920043	-0.099475177	-0.4489312196	-0.537773617
## [86,]	0.389587004	0.381993626	-0.0150804219	1.378124822
## [87,]	0.942235691	0.440060215	-0.2063073414	-0.012246399
## [88,]	0.033039464	-0.109152942	-0.1730504858	1.715372604
## [89,]	0.183761834	-0.116411265	0.2524860980	-0.504669540
## [90,]	0.718583144	0.450947700	0.3953394095	0.014650663
## [91,]	-0.517016150	0.037223252	-0.3408464390	-0.032936447
## [92,]	0.945477033	0.504175407	-0.6246634678	0.033271706
## [93,]	-0.158199671	-1.191852887	-0.8918520689	0.018788673
## [94,]	-0.393845181	-0.205930590	-0.7474870822	-0.241905931
## [95,]	1.088096049	-0.672882746	-0.1038913430	0.405692569
## [96,]	0.144865738	0.729183440	0.2328343197	1.645026441
## [97,]	0.550033397	0.200535534	-0.1462182501	-0.711570020
## [98,]	-1.060913215	-0.306337401	0.7075003493	-1.106749935
## [99,]	-0.796095633	-0.487795492	-0.4700946732	-0.663982909

## [100,]	0.396069687	-0.716432687	0.2483289911	0.012581658
## [101,]	-0.303897961	-0.712803526	-0.5441667606	0.149135975
## [102,]	-1.911765299	0.731602882	0.1153015688	-1.725382369
## [103,]	-0.810195468	-0.734578497	-0.8551939440	-0.606050775
## [104,]	-0.109579552	-0.538603758	-0.0849954023	-1.139854012
## [105,]	-0.515719613	0.793298633	-0.1299677411	-0.978471638
## [106,]	0.337725544	-0.613606436	1.2483021713	0.008443649
## [107,]	0.353932250	-0.591831465	0.0170426773	-0.647430871
## [108,]	-1.010996560	-0.971683736	-0.9217076551	-0.616395799
## [109,]	2.107497879	1.279606317	0.4694114969	2.509870444
## [110,]	-0.868863745	-0.588202303	-0.4980606654	-0.674327933
## [111,]	-0.253819238	1.484049100	0.3186218904	-1.079852873
## [112,]	1.807673811	0.108596768	0.7921541635	-0.608119780
## [113,]	1.802811799	0.619098865	3.4712206316	-0.200525836
## [114,]	0.590550163	0.281586815	1.1326590144	-1.057093820
## [115,]	-0.226267837	-0.418841418	0.0041934376	-1.374065355
## [116,]	-0.662066172	-0.939021279	0.0230893783	-0.537773617
## [117,]	1.828742530	0.398929714	0.6825577076	-1.418342057
## [118,]	0.148107080	-0.547071802	0.0336711051	0.525694847
## [119,]	0.340966885	-0.676511907	0.8273006132	0.811217509
## [120,]	-0.313297851	3.580494916	-0.7153639830	0.891908696
## [121,]	-0.864812068	-0.859179719	-0.6004766638	-0.713639025
## [122,]	0.682928390	-0.011165572	0.2929234111	1.237432496
## [123,]	1.041096600	3.015555391	2.2977827160	2.017447303
## [124,]	0.033039464	-0.174477855	-0.3729695381	-0.117765644
## [125,]	-0.229509179	-1.023701722	0.1062315172	-0.415702334
## [126,]	-1.131574455	-0.976522618	-0.7886802328	-0.161214745
## [127,]	-0.796743901	-0.884583852	-0.8805145045	0.128445927
## [128,]	0.293967437	-0.029311382	-0.4190756334	1.251915530
## [129,]	1.747708998	0.361428375	0.4493817998	-0.032936447
## [130,]	-0.221405826	-0.576105097	-0.2157553118	1.316054678
## [131,]	-0.599994486	0.735232043	0.1315520777	-0.606050775
## [132,]	-0.140372294	-0.795064527	-0.5041073664	0.618800063
## [133,]	-0.415724235	-0.628123083	-0.5373642219	0.662249163
## [134,]	0.143245068	-0.466020521	-0.5536147309	0.254655219
## [135,]	-0.429175801	-0.680141069	-0.6008545826	1.293295626
## [136,]	-0.466127092	-0.492634375	-0.4406170057	-0.368115224
## [137,]	0.206451223	-0.841033910	-0.0332205249	-0.608119780
## [138,]	-0.517664418	-0.669253584	-0.8340304904	-0.817089264
## [139,]	2.627733153	3.766791889	-0.3431139519	0.471900722
## [140,]	1.272852501	-0.573685656	-0.1333690104	-0.899849456
## [141,]	-1.911765299	0.269489609	-0.2172669870	-1.168820079
## [142,]	-0.020442667	-0.216818076	-0.1363923609	0.755354379
## [143,]	-0.368886853	0.275538212	-0.1016238301	-0.721915044
## [144,]	-0.682648689	0.060207944	-0.7210327652	-0.370184229
## [145,]	-1.002244938	-0.463601080	-0.5422771665	-0.893642442
## [146,]	-0.112820893	1.297752127	2.1269634124	-0.645361866
## [147,]	1.075130684	4.323263370	0.3174881339	-0.523290583
## [148,]	0.562998763	1.574778146	2.1061778776	-0.003970380
## [149,]	0.507895961	-0.726110452	-0.4787868059	-0.086730572
## [150,]	-0.967238452	-0.877325528	-0.6873979909	-0.192249816
## [151,]	-0.043132056	1.084841299	-0.8548160252	-0.436392382
## [152,]	-0.255439909	0.308200668	3.0177180555	-1.484964012
## [153,]	6.643754990	1.804625062	9.8429318251	-1.086059887

## [154,]	-0.577629231	0.403768596	-0.8223150072	-0.885366422
## [155,]	-0.366293780	0.864672149	-0.1193860143	-0.310183090
## [156,]	-0.627545887	-0.308756842	-0.4954152337	-0.554325655
## [157,]	1.071889342	-0.334160975	0.4433350988	0.869149643
## [158,]	-0.112820893	-0.646268892	-0.8601068886	0.403623564
## [159,]	-0.895118609	-0.712803526	-0.4780309682	-0.647430871
## [160,]	-1.332213480	-0.504731581	-0.4270119284	-0.808813245
## [161,]	-0.044752726	1.624376691	0.4784815484	-0.610188785
## [162,]	2.616388459	1.332834024	-0.1636025155	1.191914391
## [163,]	0.199968540	-0.445455271	-0.1802309433	2.164346644
## [164,]	-0.072304127	-0.211979193	0.5869442479	-0.556394660
## [165,]	0.718583144	-0.118830707	-0.0207492041	2.429179257
## [166,]	-0.869349946	-0.157541766	-0.8597289698	-0.059833510
## [167,]	-1.045678911	-0.968054574	-0.6329776817	-0.966057609
## [168,]	0.687790402	0.044481576	-0.1140951510	0.782251441
## [169,]	0.631066929	-1.099914120	0.5007787584	1.421573923
## [170,]	-0.441330831	-0.851921396	-0.7894360705	-0.032936447
## [171,]	0.099486961	-0.157541766	-0.5846040736	-0.572946699
## [172,]	-0.381366017	-0.646268892	-0.7282132227	0.353967449
## [173,]	0.227519941	-0.317224886	0.1410000481	0.521556837
## [174,]	0.653756318	1.243314699	0.3741759559	-1.017782729
## [175,]	-1.911765299	1.464693571	-1.0554909151	-0.978471638
## [176,]	-1.911765299	0.794508353	-0.1496195194	-1.449790930
## [177,]	2.075084466	2.208671745	5.3381395697	-1.036403772
## [178,]	1.026510565	-0.450294153	0.2982142744	0.314656358
## [179,]	-1.611617097	-0.533764875	-1.0682267791	-0.469496459
## [180,]	-0.711172492	-1.196691769	-0.1677596225	-0.546049636
## [181,]	0.099486961	-0.579734259	-0.3960225857	3.486440707
## [182,]	0.627825588	-0.309966563	0.5669145508	2.154001620
## [183,]	-0.328856289	-0.613606436	-0.5158228496	0.794665470
## [184,]	0.862822830	0.134000901	0.3737980371	-0.806744240
## [185,]	-0.937904314	-1.067251664	-0.4610246216	0.316725363
## [186,]	-1.522156078	0.585226688	-0.1227872837	-0.910194480
## [187,]	-0.822836699	-1.191852887	-1.0233678159	1.042946046
## [188,]	0.008729405	-0.190204223	-0.8026632289	-0.674327933
## [189,]	-0.900466822	1.312268774	-0.6828629651	-0.765364144
## [190,]	-0.869187879	-0.035359985	-0.7822556130	-0.603981770
## [191,]	0.788271981	3.700257256	2.9950429267	-0.109489625
## [192,]	0.935753009	1.595343396	0.7861074625	-0.521221579
## [193,]	-1.911765299	2.110684376	-0.7962386091	-1.303719192
## [194,]	0.229140611	-0.213188914	0.7093899434	-0.128110668
## [195,]	1.139957509	0.593694732	0.1954203572	-0.039143462
## [196,]	-0.547160623	-0.243431929	-0.6208842797	-0.494324516
## [197,]	1.426816212	0.767894500	0.1931528443	0.024995687
## [198,]	0.428483100	0.103757885	0.4690335781	0.722250302
## [199,]	0.067073548	-0.773289556	-0.1503753570	1.467092028
## [200,]	-0.728189534	-0.643849451	-0.6873979909	0.426382617
## [201,]	-0.064200774	-0.666834142	-0.1726725670	-0.378460248
## [202,]	0.268036707	-0.556749567	-0.4296573601	0.858804619
## [203,]	2.083187819	0.146098107	0.5291226694	1.831236872
## [204,]	0.180520492	-0.068022441	0.2687366070	0.606386034
## [205,]	-0.231129849	-0.329322093	-0.0789487013	-0.268802994
## [206,]	-0.573091354	-0.642639730	-0.3453814647	0.310518348
## [207,]	-0.691886511	0.188438328	-0.4999502594	-1.210200175

## [208,]	-0.091752175	0.290054859	-0.4587571088	0.730526322
## [209,]	-0.565960403	0.556193393	0.1523376125	-0.355701195
## [210,]	-0.716844839	-1.331091729	-0.8083320111	0.229827162
## [211,]	2.569389010	1.357028436	-0.5403875724	1.442263971
## [212,]	-0.818298821	-0.530135714	-0.3298867934	-0.614326794
## [213,]	0.368518286	3.301049455	0.2573990426	2.449869305
## [214,]	4.452608290	0.146098107	3.3124947300	0.372588492
## [215,]	0.725065826	2.957488802	0.3677513361	0.122238912
## [216,]	0.413897064	-0.139395957	0.2891442229	-0.107420620
## [217,]	0.981131787	0.893705443	0.3703967678	-0.624671818
## [218,]	-0.176027048	0.611840541	0.3091739200	-0.990885667
## [219,]	0.862822830	-0.190204223	0.0945160340	1.957446164
## [220,]	-0.164682353	-0.873696367	-0.5815807231	2.406420205
## [221,]	-0.832560723	-0.826517263	-0.4005576115	-0.192249816
## [222,]	-0.556884647	-0.256738856	-0.3313984686	-0.266733989
## [223,]	-0.676166006	0.704989028	-0.0128129090	-1.055024815
## [224,]	-0.511992071	-0.341419299	-0.2509017614	0.680870207
## [225,]	-0.443923904	-0.692238275	-0.4908802079	-0.233629912
## [226,]	0.172417139	0.256182682	-0.8812703421	0.103617869
## [227,]	-1.105967859	-0.668043863	-0.3680565935	-0.982609648
## [228,]	-0.039890714	-0.802322851	-0.2233136880	0.029133697
## [229,]	-0.503240449	-0.833775587	-0.1465961689	-0.428116363
## [230,]	0.517619985	-0.452713595	0.6886044087	-0.221215884
## [231,]	0.331242862	-1.061203061	-0.5513472180	0.687077221
## [232,]	-1.405305725	-0.625703642	-0.6586761610	-0.866745379
## [233,]	-1.357658009	-0.167219531	-0.4765192930	-0.808813245
## [234,]	-0.143613635	-0.914826867	-0.3431139519	1.696751560
## [235,]	-0.352193946	0.397719993	-0.0963329667	-1.191579132
## [236,]	-0.284611980	-0.957167089	-0.3287530370	-0.194318821
## [237,]	0.869305513	-0.561588449	-0.2803794288	3.049880696
## [238,]	0.796375334	-0.862808881	0.0521891269	1.645026441
## [239,]	-0.273267286	-1.051525296	-0.0778149449	-0.107420620
## [240,]	0.315036155	-1.001926751	-0.3914875600	1.291226621
## [241,]	-0.124165587	-0.188994502	-0.7561792149	-0.293631051
## [242,]	-1.302069006	-0.404324770	-0.9201959799	-0.635016842
## [243,]	0.217795917	-0.071651603	1.8442801400	-0.763295140
## [244,]	-0.279749968	0.657809924	-0.2675301892	-0.260526975
## [245,]	0.614860223	0.159405033	0.0582358279	1.113292208
## [246,]	-0.075545468	1.719944619	-0.0887745905	-0.990885667
## [247,]	-1.039358296	-0.111572383	-0.5838482360	-0.481910488
## [248,]	0.308553473	-0.953537927	1.4210110690	-0.388805272
## [249,]	-0.899656487	0.125532856	-0.4443961939	-0.831572298
## [250,]	-0.281370639	-0.456342757	-0.3820395896	-0.748812106
## [251,]	1.109164767	0.419494964	0.4561843384	1.926411092
## [252,]	-0.424799990	0.287635418	-0.8820261797	-0.682603953
## [253,]	1.282576525	-0.710384084	1.3091471002	1.864340949
## [254,]	0.133521044	-0.798693689	-0.5415213289	0.740871346
## [255,]	0.141624397	-0.833775587	-0.4704725920	1.951239150
## [256,]	0.263174695	0.009399678	-0.4156743640	0.024995687
## [257,]	0.550033397	-0.143025119	0.5506640418	1.816753838
## [258,]	1.535401144	-0.205930590	1.8386113578	0.302242329
## [259,]	2.989142705	0.342072845	0.8492199043	0.740871346
## [260,]	-0.255439909	-1.281735128	-0.1647362719	0.459486694
## [261,]	-0.585246383	-0.934182397	-0.7191431712	1.667785493

## [262,]	-0.668548854	-0.778128439	-0.8378096786	0.740871346
## [263,]	1.571055898	0.838058295	0.7721244664	0.852597604
## [264,]	-1.073230312	-1.259597241	-0.9258647621	0.339484416
## [265,]	-0.311353046	-1.101123841	-0.8389434350	1.098809175
## [266,]	-0.407782949	-0.831356145	-0.5653302141	3.356093405
## [267,]	0.300450119	1.753816796	-0.1802309433	-0.908125475
## [268,]	-0.304222095	-0.029311382	-0.2970078566	-0.303976075
## [269,]	-0.885880786	0.651761321	-0.6189946856	-0.490186507
## [270,]	0.916304961	0.815073604	1.4417966037	-0.947436566
## [271,]	-1.327027334	-0.626913362	-0.9122596848	-0.281217023
## [272,]	-0.477633853	-0.438196947	-0.3695682688	-0.817089264
## [273,]	0.618101564	-0.157541766	0.3095518388	2.466421344
## [274,]	-0.376666072	-0.309966563	0.1761464977	-1.141923017
## [275,]	-0.530629783	-0.768450674	-0.3945109105	0.962254859
## [276,]	1.183715616	0.754587573	-0.2996532883	-0.800537226
## [277,]	-1.371595776	0.367476978	-0.8192916567	-0.841917322
## [278,]	0.459275842	-0.150283442	-0.8007736349	0.763630398
## [279,]	-1.094299030	-0.859179719	-1.0086289822	-0.159145740
## [280,]	-0.221405826	-0.636591127	-0.8321408964	-0.266733989
## [281,]	0.183761834	-0.749095144	-0.1318573352	1.541576201
## [282,]	1.079992696	1.048549681	-0.6764383453	-0.612257790
## [283,]	0.517619985	-0.521667669	-0.6915550978	1.556059234
## [284,]	0.047625500	-0.805952013	0.1081211113	0.471900722
## [285,]	0.963304410	-0.213188914	-0.0373776318	-0.490186507
## [286,]	-1.437881205	0.625147468	-0.6726591571	-0.572946699
## [287,]	0.734789850	-0.220447238	1.5948537231	-0.626740823
## [288,]	-0.954759288	-0.526506552	-0.6477165154	-0.548118641
## [289,]	3.739513210	1.649780824	1.0287313407	-0.912263485
## [290,]	-0.303411760	1.647361383	-0.3276192805	-0.808813245
## [291,]	2.579113034	2.446986705	7.2050585078	-0.103282610
## [292,]	0.011970746	-0.643849451	-0.3703241064	-0.003970380
## [293,]	-0.077166139	0.690472381	0.3518787459	-0.523290583
## [294,]	-0.521554027	-0.317224886	-0.7599584030	-0.663982909
## [295,]	-0.618794266	-0.822888101	-0.9081025779	-0.572946699
## [296,]	-0.805009322	0.120693974	-0.7531558644	-0.330873138
## [297,]	-0.389469371	1.072744093	-0.1386598738	-1.013644720
## [298,]	0.409035052	-0.579734259	-0.3918654788	-0.601912766
## [299,]	-1.062047685	-1.150722386	-0.6945784483	-0.010177395
## [300,]	0.114072996	1.705427972	-0.6892875849	-1.104680931
## [301,]	0.896856914	-0.205930590	0.3749317936	1.998826260
## [302,]	0.362035604	0.829590251	0.3235348349	-0.581222718
## [303,]	1.225853053	0.824751368	0.8061371596	1.533300182
## [304,]	-0.420910381	-0.318434607	-0.1806088621	-1.077783868
## [305,]	-0.646507733	-0.193833384	-0.1450844937	-0.742605092
## [306,]	-0.777619988	1.378803407	-0.5985870697	-0.792261207
## [307,]	-1.525073285	-0.859179719	-0.4572454335	-0.384667262
## [308,]	-1.349068454	0.782411147	-0.6205063609	-1.359375421
## [309,]	-1.386343879	-0.694657716	-0.8423447043	-0.268802994
## [310,]	-1.130439985	-0.764821512	-0.5751561033	-0.318459109
## [311,]	-0.673897067	0.356589493	-0.8064424171	-0.757088125
## [312,]	-0.886853189	-0.677721628	-1.0960038119	0.039478720
## [313,]	-0.179268389	-0.391017844	0.2256538623	-0.430185368
## [314,]	-1.155398313	-0.026891940	-0.6964680424	-0.812951255
## [315,]	-1.911765299	4.949898645	1.1432407412	-1.513930079

## [316,]	-1.362357954	-0.799903409	-0.9262426809	-0.606050775
## [317,]	-1.093164561	-0.093426574	-1.0750671096	-0.707432010
## [318,]	-0.059338762	-0.942650441	-0.5078865545	1.152603300
## [319,]	2.183669399	1.738090428	1.5427009269	-1.284684348
## [320,]	-0.096614187	1.667926633	-0.3151479597	-0.697086986
## [321,]	0.590550163	0.804186118	1.4365057403	-1.032265763
## [322,]	0.109210984	0.167873078	-0.8336525716	1.405021884
## [323,]	-0.273267286	-1.030960046	-0.2599718129	-0.461220440
## [324,]	0.306932802	0.863462428	-0.1700271353	1.868478958
## [325,]	-1.078416458	-0.872486646	-0.6537632165	-0.521221579
## [326,]	-0.862543129	0.003351075	-0.7599584030	-0.529497598
## [327,]	-0.837746869	-0.772079835	-0.4984385842	-0.152938725
## [328,]	-1.271438331	-0.711593805	-0.7709180486	-0.661913905
## [329,]	0.344208227	-0.626913362	-0.3801499956	0.622938072
## [330,]	0.010350075	2.380452072	0.9040181323	0.302242329
## [331,]	0.452793159	-0.710384084	-0.2546809495	0.515349823
## [332,]	-0.321725338	0.842897178	0.2936792487	-0.382598257
## [333,]	-1.106778194	2.647800326	-0.6923109354	-0.887435427
## [334,]	-1.435126065	-0.428519183	-0.6669903749	-0.726053053
## [335,]	-0.854763910	-0.139395957	-0.5944299628	-0.603981770
## [336,]	0.914684291	-0.272465224	-0.0233946357	0.976737892
## [337,]	-0.726244729	-0.502312140	0.7203495890	-0.527428593
## [338,]	1.031372576	0.137630063	0.2456835594	1.711234594
## [339,]	-0.608259906	0.055369061	-0.3843071025	-1.057093820
## [340,]	0.196727199	-0.695867437	-0.2558147060	2.979534533
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## [342,]	-0.532574588	-0.368033152	0.0892251706	-1.141923017
## [343,]	-0.274887957	0.355379772	-0.3317763875	-0.899849456
## [344,]	1.015165870	2.058666390	-0.1915685077	1.340882736
## [345,]	-0.094993516	-0.122459868	-0.2626172446	-0.663982909
## [346,]	0.188623846	1.960679020	1.1198097747	-1.115025955
## [347,]	-1.021206785	-0.087377971	-0.3933771540	-0.543980631
## [348,]	-0.182509730	0.131581459	-0.3824175085	0.207068109
## [349,]	-0.888311792	-0.361984549	-0.2970078566	-0.777778173
## [350,]	-0.206819790	0.822331927	-0.2573263812	-0.715708029
## [351,]	-0.891391067	0.195696652	-0.4258781720	-0.618464804
## [352,]	1.788225764	4.220437118	1.3359793360	0.225689152
## [353,]	0.532206020	0.339653404	-0.1549103828	3.488509712
## [354,]	0.917925632	0.079563473	0.4051652986	0.463624703
## [355,]	0.472241207	1.267509111	0.3915602213	-0.858469360
## [356,]	1.274473172	0.707408469	0.1549830442	-0.599843761
## [357,]	1.128612815	0.974756724	0.7256404524	-0.430185368
## [358,]	-0.932556101	-0.617235598	-0.6454490026	-0.239836927
## [359,]	0.528964679	0.946933150	0.1432675610	-1.301029485
## [360,]	-0.847795026	-0.135766795	-0.4088718254	-0.879159408
## [361,]	-1.325568730	-0.013585014	-1.0343274615	-0.527428593
## [362,]	-0.874698159	-0.272465224	-0.6949563671	-0.428116363
## [363,]	-0.291580864	-0.175687575	-0.6280647372	-0.521221579
## [364,]	-0.237612532	-0.435777506	-0.6779500205	0.385002521
## [365,]	-1.022989522	-0.923294912	-0.6503619471	-0.318459109
## [366,]	0.456034501	-0.736997938	-0.7138523078	1.663647484
## [367,]	2.198255435	2.003019242	0.1855944681	1.638819426
## [368,]	-0.763196019	-0.140605678	-0.6934446919	-0.409495320
## [369,]	0.028177452	-0.957167089	-0.7527779456	2.996086571

## [370,]	0.472241207	-0.648688333	0.0004142495	2.356764090
## [371,]	0.772065275	1.382432569	0.1096327866	0.643628120
## [372,]	-0.428041332	-0.492634375	-0.7663830228	-0.014315404
## [373,]	0.264795365	-0.458762198	0.2162058919	1.328468707
## [374,]	0.084900925	-0.941440721	-0.7875464764	1.882961992
## [375,]	-0.885070451	-0.134557075	-0.6114363094	-0.295700056
## [376,]	0.026556782	-0.145444560	-0.0373776318	0.144997965
## [377,]	2.818972288	-0.528925993	3.1764439571	-1.121232969
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## [381,]	-0.025304679	0.065046826	-0.0279296615	-0.709501015
## [382,]	-0.783778536	-0.210769473	-0.7006251493	-0.864676374
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## [384,]	-0.208440460	-0.261577738	1.1674275452	-0.432254373
## [385,]	-0.539381404	-0.717642408	-0.4466637067	-0.419840344
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## [387,]	-0.150096318	-0.161170928	0.3125751893	-0.649499876
## [388,]	-0.774702781	-0.492634375	-0.6896655037	-0.157076735
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## [390,]	3.664962361	0.863462428	0.9240478294	0.941564811
## [391,]	-0.788802615	-0.140605678	-0.5381200596	-1.011575715
## [392,]	-1.911765299	-0.228915282	1.1114955608	-1.261925295
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## [394,]	0.175658480	-0.011165572	0.0880914142	2.060896404
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## [397,]	0.303691461	-0.894261617	-0.5029736099	-0.303976075
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## [399,]	-1.236269778	-1.044266972	-0.0974667232	-0.742605092
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## [402,]	-0.966590184	-1.159190430	-0.7323703296	-0.510876555
## [403,]	-0.185751072	0.698940425	-0.0339763625	-0.442599397
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## [405,]	-0.242474544	1.107825991	-0.8929858253	-0.639154852
## [406,]	-0.039890714	0.029964928	-0.4134068512	-0.800537226
## [407,]	-0.494164694	-0.670463304	-0.8215591696	0.298104320
## [408,]	0.261554024	0.743700088	1.4875247801	-0.386736267
## [409,]	0.486827243	-0.774499277	-0.1734284047	0.995358935
## [410,]	-0.169544365	0.633615512	-0.5774236162	-0.448806411
## [411,]	-0.960107502	-0.862808881	-0.8071982547	-0.666051914
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## [413,]	-1.080685397	-0.124879310	0.2978363556	-1.304339893
## [414,]	0.026556782	-0.179316737	0.0831784696	0.101548864
## [415,]	-0.439061892	1.326785421	-0.7728076427	0.204999104
## [416,]	-0.408431217	0.246504917	-0.5755340221	-0.666051914
## [417,]	-0.505995589	1.197345316	0.3004817873	-1.121232969
## [418,]	1.554849192	-0.043828029	0.7634323337	1.427780937
## [419,]	-0.577791298	-0.412792815	0.2339680762	-0.541911627
## [420,]	-0.913594254	0.773943103	-0.6556528105	-0.808813245
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## [425,]	-0.206819790	1.265089670	0.1625414204	-1.046748796
## [426,]	-1.340154766	-0.736997938	-0.5230033070	-1.067438844
## [427,]	0.136762385	0.129162018	0.3911823025	-0.856400355
## [428,]	-0.218164484	-0.254319415	-0.4175639581	-0.726053053
## [429,]	-0.908732242	-0.054715514	-0.5358525467	-0.949505571
## [430,]	-1.000462200	-0.138186236	-0.9882213663	-0.506738545
## [431,]	1.708812902	-0.671673025	0.7517168505	0.016719668
## [432,]	-0.020442667	0.160614754	0.8352369083	-0.701224996
## [433,]	1.183715616	0.479980995	0.0793992815	1.191914391
## [434,]	1.060544648	0.397719993	0.2721378763	1.322261693
## [435,]	-0.569687945	-0.608767553	-0.6148375787	0.008443649
## [436,]	-0.361269701	-1.005555913	-0.3586086232	0.159480999
## [437,]	-0.858005252	0.204164696	-0.5354746279	-0.376391243
## [438,]	-0.445706642	-0.059554397	-0.3646553242	-0.126041663
## [439,]	-0.770975238	-0.048666911	-0.5551264062	-0.132248677
## [440,]	-0.167923695	-1.068461385	-0.6390243827	-0.281217023
## [441,]	0.953580386	-0.651107775	0.4463584493	-0.808813245
## [442,]	0.911442949	-0.556749567	-0.0211271229	0.850528600
## [443,]	-0.994951920	-1.000717031	-0.4651817286	-0.206732850
## [444,]	-0.111200222	-0.407953932	0.2634457436	-1.102611926
## [445,]	-0.331125228	-0.918456029	-0.5135553367	0.850528600
## [446,]	0.786651310	0.048110737	-0.0456918457	-0.680534948
## [447,]	-0.112820893	-0.988619824	-0.4682050791	1.088464151
## [448,]	-0.793988761	0.241666035	-0.6892875849	0.033271706
## [449,]	-0.477795920	-0.798693689	-0.5547484874	0.006374644
## [450,]	0.464137854	-1.240241711	-0.6008545826	1.947101140
## [451,]	0.605136199	0.362638096	1.9387598433	-0.719846039
## [452,]	0.517619985	-0.499892698	-0.7512662703	1.069843108
## [453,]	-0.426258594	-0.374081755	0.1916411691	-0.657775895
## [454,]	-0.070683456	0.184809166	-0.0876408340	-0.097075596
## [455,]	-0.795447365	-0.539813479	-0.8151345498	-0.399150296
## [456,]	-0.432417142	-0.593041185	-0.3060779082	-0.252250955
## [457,]	-0.418317308	0.509014289	-0.6299543312	-0.651568881
## [458,]	-0.866270672	-0.855550557	-0.7433299752	-0.397081291
## [459,]	-0.884908384	-1.015233678	-0.8619964826	-0.399150296
## [460,]	-0.922507943	-0.580943979	-0.5770456973	-1.158475055
## [461,]	0.451172489	-0.011165572	0.2351018326	1.384331836
## [462,]	2.298737014	-0.432148344	0.2883883853	4.090590107
## [463,]	-0.879398104	-0.774499277	-0.7191431712	-0.179835788
## [464,]	-0.696748523	0.084402356	-0.3880862907	-0.723984048
## [465,]	-0.540191740	-1.162819592	-0.8022853101	-0.283286027
## [466,]	0.305312131	-0.639010568	1.6394481431	-0.171559769
## [467,]	0.120555679	-0.596670347	-0.2134877989	-0.303976075
## [468,]	-1.274193471	0.772733382	-0.3083454210	-1.059162825
## [469,]	1.833604541	-0.461181639	2.8363170251	1.096740170
## [470,]	1.071889342	-0.299079077	0.5374368833	-0.601912766
## [471,]	-0.310056509	-0.113991824	0.0446307507	-1.061231830
## [472,]	0.005488064	-0.008746131	0.0332931863	-0.552256651
## [473,]	-0.550564032	-1.096284959	-0.5350967091	0.188447066
## [474,]	-1.911765299	1.314688215	-0.2490121673	-0.583291722
## [475,]	-0.827374576	-1.018862840	0.3314711300	-0.895711446
## [476,]	-0.556884647	-0.548281523	-0.5691094023	-0.450875416
## [477,]	0.276140060	-0.537394037	-0.3971563422	0.037409716

## [478,]	-1.128657248	-0.893051896	-0.6424256521	-0.233629912
## [479,]	-0.609070242	-0.255529135	-0.0683669745	-0.800537226
## [480,]	1.678020160	0.869511031	0.9675084929	0.231896166
## [481,]	-1.075337184	-0.726110452	-0.7319924108	-0.606050775
## [482,]	-0.894794475	-1.200320931	-0.5963195568	0.029133697
## [483,]	-0.487033743	-0.430938624	-0.3809058332	-0.297769061
## [484,]	-0.415400101	-0.930553235	-0.8582172945	-0.270871999
## [485,]	-0.182509730	-1.063622502	-0.5815807231	0.153273984
## [486,]	2.498079502	1.424772790	2.2040588503	-0.515014564
## [487,]	-0.916025260	-0.968054574	-0.9734825326	0.039478720
## [488,]	-0.005856631	-0.407953932	-0.2206682563	1.591232316
## [489,]	-0.020442667	0.118274533	-0.2199124187	-0.610188785
## [490,]	-1.053296063	-0.112782104	-0.7917035833	0.602248024
## [491,]	-1.174198093	-0.539813479	-0.5275383328	-0.434323377
## [492,]	-0.706472547	-0.909987985	-0.9394698394	0.734664331
## [493,]	0.759099910	1.019516386	0.0759980122	1.088464151
## [494,]	-0.632245832	0.985644209	0.3877810332	-0.637085847
## [495,]	-1.010024157	0.038432973	-0.4134068512	-0.366046219
## [496,]	0.005488064	-0.115201545	-0.7493766762	-0.053626495
## [497,]	-0.239233203	-0.357145667	-0.0876408340	-0.390874277
## [498,]	-0.761737416	-0.626913362	-0.8517926747	-0.457082430
## [499,]	0.637549612	-0.528925993	0.5517977982	1.340882736
## [500,]	0.485206572	-1.133786298	-0.0256621486	1.570542268
## [501,]	-0.284611980	-1.208788975	-0.2108423672	0.101548864
## [502,]	0.707238449	0.459415745	1.2093765335	-0.053626495
## [503,]	-0.030166690	0.903383208	-0.4969269089	-0.558463665
## [504,]	1.582400593	0.177550843	-0.3393347637	3.004362590
## [505,]	1.634262053	0.590065570	2.5150860337	-1.239166242
## [506,]	0.438207124	0.698940425	2.9043424115	-1.172958089
## [507,]	-0.308273772	-0.888213014	0.0068388693	-0.643292862
## [508,]	0.170796469	0.200535534	1.7074735295	-0.947436566
## [509,]	0.407414381	-0.859179719	-0.4632921345	0.217413133
## [510,]	1.396023470	0.061417664	0.8333473142	0.173964032
## [511,]	0.564619433	0.248924359	1.1439965788	-0.790192202
## [512,]	-0.200337107	-0.573685656	-0.7618479971	-0.136386687
## [513,]	0.250209330	-0.422470579	0.0793992815	0.029133697
## [514,]	-0.510209333	-0.623284201	-0.5721327528	0.101548864
## [515,]	-0.384607359	-1.091446076	-0.7849010447	0.271207257
## [516,]	-0.117682905	0.459415745	-0.9742383702	-0.786054192
## [517,]	-0.216543814	-0.599089788	-0.4043367996	1.156741309
## [518,]	-0.002615290	-1.206369534	-0.1526428699	1.543645206
## [519,]	0.104348972	-0.288191592	-0.1874114008	-0.252250955
## [520,]	-0.420910381	-0.301498519	-0.0626981923	-0.376391243
## [521,]	0.648894306	2.350209056	0.0400957249	-1.179165103
## [522,]	0.676445707	-0.211979193	0.3647279856	2.824359173
## [523,]	-1.252962686	-0.686189672	-0.5668418894	-0.897780451
## [524,]	0.279381401	-0.222866679	-0.0181037724	-0.239836927
## [525,]	-0.809547199	-0.278513827	-0.1137172321	-1.040541782
## [526,]	-0.725434393	-0.559169008	0.0491657764	-1.406134929
## [527,]	-0.717493107	-0.847082513	-0.7588246466	-0.190180812
## [528,]	-0.981500354	-1.119269650	-0.9179284670	-0.550187646
## [529,]	2.712008026	-0.410373373	0.6549696343	-0.341218162
## [530,]	-0.617497729	-0.111572383	-0.5898949370	-0.583291722
## [531,]	0.221037258	-0.265206900	-0.5581497567	-0.572946699

## [532,]	-1.053133996	-0.732159055	-0.4247444156	-0.603981770	
## [533,]	-1.088950817	-1.209998696	-0.7928373398	-0.086730572	
## [534,]	0.841754112	0.921529017	-0.4066043125	1.440194966	
## [535,]	0.269657377	-0.885793573	-0.0986004796	-0.961919600	
## [536,]	0.391207675	-0.128508471	-0.4179418769	1.661578479	
## [537,]	0.156210433	-0.729739614	-0.0147025030	-0.202594840	
## [538,]	0.950339045	-0.597880068	0.7668336030	-0.680534948	
## [539,]	-1.911765299	1.001370577	1.1628925195	-1.488067519	
## [540,]	0.298829449	0.061417664	1.4194993937	-1.570620810	
## [541,]	0.509516632	-0.259158297	0.6489229333	-0.829503293	
## [542,]	-0.028546020	0.016658002	0.8752963025	-0.010177395	
## [543,]	0.144865738	-0.222866679	-0.4417507622	0.049823744	
## [544,]	-0.434199880	-0.399485888	-0.9266205997	-0.392943281	
## [545,]	-0.443275636	-0.682560510	-0.0740357567	-0.252250955	
## [546,]	-0.442789434	0.039642693	-0.4580012711	-0.190180812	
## [547,]	-1.021206785	-0.597880068	-0.4493091384	-1.038472777	
## [548,]	-0.133889611	0.269489609	0.7917762447	-1.125370978	
## [549,]	-0.353490482	0.179970284	0.1357091847	-1.104680931	
## [550,]	-0.952814484	0.498126804	-0.3091012587	-0.670189924	
## [551,]	-1.911765299	1.148956492	-0.5921624499	-0.953643581	
## [552,]	-0.252198568	1.037662196	0.3507449895	-0.879159408	
## [553,]	-0.403731272	-0.190204223	-0.7818776942	-0.496393521	
## [554,]	0.165934457	2.062295551	0.3129531082	-1.329167951	
## [555,]	-0.676814274	-0.434567786	-0.3756149698	-0.492255512	
## [556,]	0.877408866	-0.255529135	0.4319975344	-1.123301974	
## [557,]	-0.764006355	0.626357189	-0.5732665092	-1.162613065	
## [558,]	-1.911765299	1.148956492	-0.1779634304	-1.195717141	
## [559,]	0.691031743	-0.503521860	0.2309447257	-0.163283749	
## [560,]	0.141624397	-0.684979952	0.3564137717	-0.783985187	
## [561,]	0.723445156	0.031174649	0.5703158201	-0.200525836	
## [562,]	-1.911765299	-0.078909927	-0.7641155100	-0.899849456	
## [563,]	0.694273084	0.100128724	0.8870117857	0.258793229	
## [564,]	2.340874451	0.003351075	0.9138440214	1.659509474	
## [565,]	2.065360442	-1.137415459	0.1678322838	1.899514030	
## [566,]	0.807720029	-0.188994502	-0.4901243703	1.535369186	
## [567,]	0.611618882	-0.890632455	0.0366944556	0.560867929	
## [568,]	0.785030640	0.326346478	0.9032622947	1.959515169	
## [569,]	-1.911765299	0.752168132	-0.3824175085	-1.409652237	
##	sd texture	sd perimeter	sd area	sd smoothness	sd compactness
## [1,]	-1.358098490	2.301575480	1.9994781593	1.306536657	2.614364658
## [2,]	-0.368878647	1.533776431	1.8888270199	-0.375281748	-0.430065806
## [3,]	-0.023953307	1.346290616	1.4550042985	0.526943750	1.081980138
## [4,]	0.133866306	-0.249719577	-0.5495376931	3.391290721	3.889974667
## [5,]	-1.465480907	1.337362721	1.2196510812	0.220362270	-0.313118999
## [6,]	-0.313560433	-0.114908349	-0.2441054208	2.046711944	1.720102933
## [7,]	0.322599039	1.367122374	1.2740984673	0.518184279	0.021196330
## [8,]	0.400695342	0.099361153	0.0288340565	1.446688190	0.724148332
## [9,]	0.822089979	-0.031581320	-0.2481450656	1.661295226	1.816711165
## [10,]	2.440961268	-0.286026354	-0.2971477131	2.318255541	5.108382433
## [11,]	1.334596970	0.492188574	0.4731949817	-0.624926668	-0.630273655
## [12,]	0.260772799	0.870136167	0.7348937083	0.316716450	1.948911903
## [13,]	0.693556480	1.322482894	0.7928538290	-1.255608570	0.864611617
## [14,]	0.322599039	0.141024667	-0.0071714730	-0.843913439	-0.393202139
## [15,]	1.030346788	0.045793778	-0.3212099450	1.433548984	3.293800184

##	[16,]	1.863374024	0.501116469	0.1099782255	1.551801841	2.564153801
##	[17,]	0.846495074	0.480284712	0.4521185741	0.614538459	-0.426887904
##	[18,]	0.944115453	0.879064063	0.7629955850	2.037952473	1.074353173
##	[19,]	0.846495074	2.367046717	2.6651413634	0.824765759	0.386019521
##	[20,]	-1.044086270	-0.225019065	-0.2974989865	0.509424808	-0.489174790
##	[21,]	-0.843964493	-0.332451413	-0.4392378271	-0.051181327	0.148312424
##	[22,]	-1.629808546	-1.253810272	-0.9935473448	0.001375499	-0.886412586
##	[23,]	-1.073372384	0.530876123	0.1761932724	0.290438037	2.168187166
##	[24,]	1.612815051	2.402758300	3.0462730660	0.338615127	0.036450261
##	[25,]	0.957131504	2.075402117	2.3437261488	2.108028240	0.658047963
##	[26,]	-0.695906918	1.343314651	1.0194252098	0.969297028	0.893848319
##	[27,]	1.225587547	0.450525059	0.0286584198	0.881702320	2.606102112
##	[28,]	0.257518786	0.971318988	0.9175559068	0.062691795	-0.270535107
##	[29,]	1.795039759	1.251059727	0.6822026895	1.389751630	2.267337720
##	[30,]	-1.001784106	0.822520723	0.6084352632	-0.300826246	0.171193321
##	[31,]	1.355748052	1.584367841	1.3865059741	0.732791315	1.089607104
##	[32,]	0.397441330	0.361246100	0.0142562080	1.372232688	2.054418262
##	[33,]	1.043362839	0.858232306	0.8139302365	1.359093482	0.645971934
##	[34,]	0.849749087	1.611151529	1.6376664970	0.811626553	2.572416347
##	[35,]	0.257518786	0.757049486	0.6681517512	0.535703221	2.072850095
##	[36,]	0.543871898	0.780857208	0.6119479978	1.048132266	0.821392145
##	[37,]	0.761890745	0.266015210	-0.1422361178	0.535703221	1.077531075
##	[38,]	-0.466499027	-0.678556178	-0.5878265001	-1.548612870	-1.322484348
##	[39,]	-0.077644516	-0.348819222	-0.3192779410	-1.686136562	-1.289942628
##	[40,]	0.055770002	0.001154298	-0.2462130616	1.253979831	1.069268529
##	[41,]	0.743993676	-0.141692037	-0.1627856152	-1.005963650	-0.317568062
##	[42,]	1.572139893	-0.596419536	-0.6438546167	2.563520725	0.098737147
##	[43,]	1.219079521	2.087305978	1.3531349955	-0.335864129	3.115202071
##	[44,]	0.377917254	0.173760286	0.0467490029	0.903600997	0.750842712
##	[45,]	0.685421448	-0.052413078	-0.2456861514	0.785348140	0.865247197
##	[46,]	-0.708922968	1.289747276	1.2056001429	1.556181576	1.619045638
##	[47,]	-0.604794564	-1.488018741	-1.1212352470	-0.116877358	-0.753576267
##	[48,]	0.369782222	-0.132764141	-0.2128420830	2.024813267	1.031769281
##	[49,]	-0.809797360	-0.517258858	-0.5233678204	0.745930522	-0.245747469
##	[50,]	0.999433668	-0.245850822	-0.3192779410	-0.708141641	-0.528580779
##	[51,]	0.006959813	-0.722600465	-0.6394636985	-1.045381269	-1.068506391
##	[52,]	-0.404672786	-0.332749009	-0.3932210040	-1.027862327	-0.610570660
##	[53,]	-0.707295962	-0.702066304	-0.6206705685	-0.786976879	-1.050010999
##	[54,]	-0.251734192	0.828472653	0.7735337888	-0.191332860	-0.156130622
##	[55,]	0.978282586	0.310654690	0.2624309065	0.286058302	-0.308669936
##	[56,]	-0.521817242	-0.757419259	-0.6575542816	-0.327104659	-1.061832796
##	[57,]	0.400695342	1.870060510	2.2207804383	1.315296127	0.615464072
##	[58,]	0.817208960	0.251135384	0.1842725620	0.194083858	1.110581260
##	[59,]	-0.557611381	-0.506545383	-0.4504785778	-1.325684336	-1.222571098
##	[60,]	-1.672110710	-1.409453257	-1.0638020365	1.792687289	-0.828574763
##	[61,]	-1.338574414	-1.113047113	-0.8992304211	-0.213231537	-0.988995274
##	[62,]	0.221724647	-1.345172407	-1.0655584038	1.380992159	-0.536843326
##	[63,]	0.623595208	0.352318204	0.1386070123	1.385371894	2.354412245
##	[64,]	-1.048967289	-1.240120831	-1.0019779078	-1.489486441	-0.549554935
##	[65,]	1.267889711	0.135072737	0.0135536611	2.309496070	0.965033332
##	[66,]	1.254873660	0.218399765	0.0781879775	1.420409778	0.555083927
##	[67,]	0.957131504	-1.197266931	-0.9657967415	0.982436235	-0.558453062
##	[68,]	-0.298917376	-0.870803537	-0.7269307897	-0.147535506	-1.034629952
##	[69,]	-0.492531128	-1.242799200	-0.9763349453	0.693373696	1.158249795

##	[70,]	-0.962735954	-0.642546998	-0.5720191944	-0.121257093	-1.167275596
##	[71,]	0.146882356	1.745069968	1.7307539635	-0.572369843	-0.131342984
##	[72,]	-1.628181539	-1.330292580	-1.0471165472	-0.511053547	-0.067784937
##	[73,]	1.324834932	1.319506929	1.4058260143	1.144486446	3.083423047
##	[74,]	-0.783765259	0.090433257	-0.1197546165	0.382412481	0.635167066
##	[75,]	-0.401418774	-0.522317999	-0.4733113526	-0.646825345	-0.444684157
##	[76,]	-0.181772921	0.640986839	0.6014097941	0.772208934	-0.316296902
##	[77,]	-2.145569549	-0.473214572	-0.4831470094	0.557601898	-0.739593496
##	[78,]	-1.101031492	1.274867449	1.2811239365	0.675854755	1.964801415
##	[79,]	0.983163605	1.876012441	1.3039567113	1.380992159	2.301659066
##	[80,]	-0.139470756	-0.457739552	-0.4539913123	-0.151915241	-0.255281176
##	[81,]	1.054751883	-0.676473002	-0.6243589398	1.021853854	-0.550826096
##	[82,]	-0.404672786	-0.315488411	-0.4666371569	0.929879410	1.429007077
##	[83,]	1.292294806	3.108062078	2.9531855995	1.091929621	2.245727984
##	[84,]	1.145864237	1.012982502	0.7331373410	0.299197508	0.174371223
##	[85,]	-0.126454706	-0.579754130	-0.5491864196	0.233501476	-0.342991281
##	[86,]	0.325853051	1.337362721	1.2688293654	0.325475920	-0.288331361
##	[87,]	0.581293044	0.033889916	-0.1259019020	-0.077459739	-0.360151954
##	[88,]	0.770025777	1.358194478	1.3039567113	-0.327104659	0.421612027
##	[89,]	0.784668834	-0.470238607	-0.5372431220	-0.086219210	-0.050624264
##	[90,]	-1.210040915	0.063649569	-0.1352106487	-0.204472067	0.347249112
##	[91,]	0.558514955	-0.129788176	-0.1350350119	-0.913989206	-0.493623853
##	[92,]	0.026483888	0.007106229	-0.0872618216	-0.292066775	-0.346804764
##	[93,]	-0.541341317	-0.082172731	-0.0877887317	-1.391380368	-0.829210343
##	[94,]	0.042753952	-0.287811933	-0.3182241206	0.067071530	-0.503157560
##	[95,]	-0.235464129	0.483260678	0.2536490700	0.995575441	1.055285759
##	[96,]	0.962012523	1.453425367	1.5270153575	-0.585509049	0.633260325
##	[97,]	-0.774003221	-0.747598573	-0.6768743218	-0.804495820	-1.021282762
##	[98,]	0.099699173	-1.144294748	-0.9036213394	-0.033662385	-1.013973586
##	[99,]	-1.385757598	-0.723195658	-0.6464891677	0.470007190	-0.439599513
##	[100,]	0.843241061	0.066625535	-0.0951654744	0.470007190	0.307207542
##	[101,]	1.560750849	0.039841847	0.0455195458	-0.257028892	-0.381126110
##	[102,]	-0.998530093	-1.691872365	-1.2213481827	1.140106710	-0.852091240
##	[103,]	1.165388313	-0.674985020	-0.5844894022	-0.878951323	-1.052807553
##	[104,]	0.187557514	-1.042814332	-0.9122275391	1.030613325	-0.152952720
##	[105,]	-0.385148711	-0.983295025	-0.8391626597	-0.458496721	-0.671586386
##	[106,]	-0.533206286	-0.025629390	-0.0937603805	2.357673160	0.989185389
##	[107,]	0.582920050	-0.647308542	-0.6303305886	1.595599195	0.074585089
##	[108,]	0.294939931	-0.645820560	-0.5909879612	-0.611787461	-0.368414500
##	[109,]	0.379544260	2.962239778	2.5983994063	1.652535755	2.831097599
##	[110,]	0.565022981	-0.692543215	-0.6366535108	1.643776285	-0.220324250
##	[111,]	-0.684517874	-1.058884544	-0.9020406088	0.627677665	-0.494259434
##	[112,]	-0.033715345	-0.543447353	-0.6203192950	-0.160674712	-0.186002905
##	[113,]	-0.316814445	-0.007773598	-0.3013629946	-1.877968974	1.048929954
##	[114,]	-0.476261065	-1.030910470	-0.8890434908	-0.103738152	-0.313754580
##	[115,]	-0.985514043	-1.273154046	-1.0471165472	1.753269670	-0.113546731
##	[116,]	0.076921084	-0.586896447	-0.5226652735	0.772208934	-0.091301414
##	[117,]	-1.400400655	-1.307080051	-1.0724082362	-0.633686139	-0.421803260
##	[118,]	0.275415856	0.590395429	0.3765947805	2.427748926	1.231341550
##	[119,]	0.784668834	0.685626318	0.6874717914	2.327015012	1.514174860
##	[120,]	0.350258146	0.652890700	0.6681517512	-1.102317830	-0.852091240
##	[121,]	-1.579371350	-0.699983128	-0.6500019023	0.982436235	-0.097021638
##	[122,]	-0.126454706	1.134997080	1.1739855316	0.785348140	-0.159944105
##	[123,]	-0.274512281	2.191464763	2.0943219932	1.630637078	1.081344558

## [124,]	-1.577744344	-0.132764141	-0.2372555884	-0.046801591	-0.479641083
## [125,]	-0.476261065	-0.454465990	-0.4364276394	-1.308165395	-0.007404792
## [126,]	-0.341219540	-0.207163273	-0.2716803873	-0.730040318	-0.758025330
## [127,]	1.620950083	0.176736251	-0.0559984837	0.645196607	0.216955115
## [128,]	0.008586819	1.218324108	1.1546654914	-1.325684336	-0.177104778
## [129,]	-1.195397858	-0.040509216	-0.2072217077	0.272919095	0.216319535
## [130,]	1.285786781	1.233203935	1.2442402233	-0.213231537	0.837917237
## [131,]	-1.280002187	-0.472619379	-0.5890559572	0.452488248	0.026916554
## [132,]	0.052515990	0.524924192	0.4837331854	0.973676764	-0.094479317
## [133,]	0.976655580	0.667770527	0.5171041640	0.312336714	0.325639376
## [134,]	-1.045713277	0.209471869	0.0741483327	-0.440977780	-0.377312627
## [135,]	0.929472396	1.140949010	1.2459965906	0.618918194	-0.170113393
## [136,]	1.251619648	-0.452978008	-0.3986657426	0.417450364	-0.648069908
## [137,]	-0.032088339	-0.627964768	-0.5864214062	-0.230750479	-0.962682242
## [138,]	-0.595032526	-0.813367406	-0.7351857160	-0.585509049	-0.568622349
## [139,]	-0.691025899	0.420765406	0.1595077831	0.382412481	-0.240662825
## [140,]	-1.611911476	-0.914550227	-0.7843640002	0.189704122	-0.458031347
## [141,]	-1.883621532	-1.212146757	-0.9443690606	-0.392800690	-1.158186796
## [142,]	-0.066255472	0.646938769	0.6189734670	-0.042421856	-0.194901031
## [143,]	0.176168470	-0.732123554	-0.6631746569	0.391171952	-0.477098761
## [144,]	-0.627572652	-0.300310988	-0.4158781421	-0.051181327	0.003400076
## [145,]	-0.806543348	-0.877053064	-0.7717181557	-1.084798888	-0.838744050
## [146,]	-1.491513009	-0.624988803	-0.6515826328	0.439349042	-0.016302918
## [147,]	0.114342230	-0.456251570	-0.5073848781	0.268539360	0.984736326
## [148,]	-0.033715345	-0.004797633	-0.1244968082	-1.430797987	-0.013760596
## [149,]	-0.948092897	0.039841847	-0.1996693283	-0.033662385	0.122253625
## [150,]	-0.523444248	-0.299715795	-0.2716803873	-1.544233134	-0.456760186
## [151,]	-0.254988205	-0.489284785	-0.4634756957	-0.116877358	-0.913742546
## [152,]	0.657762341	-1.463615826	-1.1078868556	1.341574540	1.123292869
## [153,]	-1.006665125	-1.077930722	-0.8783296503	-0.138776035	0.145770102
## [154,]	-1.525680141	-0.922882930	-0.7724207026	0.075831001	-1.045879726
## [155,]	-0.842337487	-0.285431161	-0.3570398378	0.675854755	-0.182189422
## [156,]	-0.074390503	-0.614870520	-0.5556849786	-0.467256192	-0.479641083
## [157,]	-0.092287573	0.763001416	0.7401628102	0.413070629	0.607201525
## [158,]	0.389306298	0.388029788	0.2659436411	-1.954614344	-0.529216360
## [159,]	-1.182381808	-0.689864846	-0.6108349116	-0.213231537	-0.833023826
## [160,]	-1.216548940	-0.868720361	-0.7211347776	-0.668724022	-1.088908524
## [161,]	0.086683122	-0.546125722	-0.5913392347	0.150286503	-0.413540714
## [162,]	-1.280002187	1.170708663	1.0791416978	-0.874571587	-0.334728735
## [163,]	0.115969236	2.012906845	2.3735843928	0.500665338	0.828383530
## [164,]	0.488553683	-0.592253184	-0.5753562923	0.561981634	-0.130071823
## [165,]	0.413711393	2.289671619	2.6739231999	-0.419079103	0.661225866
## [166,]	0.023229876	-0.147643967	-0.1729725455	-1.220570686	-0.980796286
## [167,]	-2.222038846	-0.999960431	-0.8194913460	0.491905867	-0.817134314
## [168,]	0.101326179	0.697530180	0.6663953839	-0.681863228	-0.269263947
## [169,]	1.082410990	1.429617645	1.3689423011	0.229121741	0.818214242
## [170,]	-0.435585907	-0.079196765	-0.1525986849	-0.471635928	-0.575613734
## [171,]	-1.633062558	-0.603859449	-0.5822061247	0.268539360	-0.811414090
## [172,]	0.682167436	0.277919071	0.1984991370	0.338615127	-0.633451557
## [173,]	-1.405281674	0.527900157	0.3888893516	0.907980732	0.661225866
## [174,]	-1.441075813	-1.049063859	-0.8502277737	-0.471635928	-1.092340659
## [175,]	-1.053848308	-1.013649872	-0.8295026396	-1.084798888	-1.184436269
## [176,]	-1.405281674	-1.455283123	-1.0913770030	-0.708141641	-1.167529829
## [177,]	-0.209432028	-1.017518626	-0.8612928876	-0.099358416	0.258903426

## [178,]	0.451132538	0.483260678	0.1767201826	0.399931423	1.350195098
## [179,]	0.543871898	-0.567850269	-0.4773509973	-2.238859173	-1.397927750
## [180,]	-1.550085236	-0.611894555	-0.5442685912	-0.708141641	-1.269985401
## [181,]	1.167015319	3.378874921	4.1018498091	0.649576342	0.947872659
## [182,]	1.269516717	2.060522290	2.1224238698	0.732791315	3.204183337
## [183,]	1.162134300	0.655866665	0.6822026895	0.395551687	0.638344969
## [184,]	-1.297899256	-0.837472725	-0.7260526060	-0.887710794	-0.592774407
## [185,]	0.382798273	0.194592043	0.1624936075	-0.099358416	0.480721011
## [186,]	-0.731701057	-0.948476231	-0.7790948983	0.864183378	-0.968402467
## [187,]	0.111088217	0.950487231	0.9298504779	-0.392800690	-0.062064712
## [188,]	-0.697533924	-0.679746564	-0.6306818620	-0.003004237	-0.955055277
## [189,]	0.130612293	-0.824080882	-0.6868856154	0.141527032	-0.980478495
## [190,]	-0.990395062	-0.613382538	-0.5864214062	-0.997204179	-0.567351188
## [191,]	1.871509056	-0.025629390	-0.2075729811	0.916740203	4.312000101
## [192,]	-0.354235591	-0.542256967	-0.5288125590	-1.685698589	-1.045434820
## [193,]	-0.788646278	-1.339518073	-1.0130430217	-2.680336505	-1.442609057
## [194,]	2.222942421	-0.165499759	-0.1961565937	2.020433532	1.374982736
## [195,]	0.342123115	0.337438378	-0.1684059905	-0.033662385	1.338119069
## [196,]	-0.598286539	-0.489582381	-0.4917532091	-0.992824444	-0.658874776
## [197,]	1.355748052	0.129120806	-0.1294146366	1.810206231	0.368223268
## [198,]	-0.158994832	0.649914735	0.6101916305	-1.933591614	-0.368414500
## [199,]	1.038481820	1.759949794	1.4181205853	-0.007383972	1.943827259
## [200,]	0.722842594	0.316606620	0.2870200486	0.999955176	0.961855429
## [201,]	0.436489481	-0.449704446	-0.4253625255	0.461247719	-0.318203643
## [202,]	0.026483888	0.959415127	0.6295116707	0.251020418	0.557626248
## [203,]	1.139356212	2.075402117	1.9415180387	0.929879410	1.032404862
## [204,]	2.631321008	0.632058943	0.4784640835	3.951896856	1.694679714
## [205,]	-0.168756870	-0.333641799	-0.3559860174	0.448108512	-0.104648604
## [206,]	-0.884639651	0.310654690	0.1912980311	0.732791315	0.501059586
## [207,]	-0.399791767	-1.195778948	-0.9642160110	0.399931423	-0.823490119
## [208,]	-0.102049611	0.676698423	0.5785770192	-0.931508148	-0.671586386
## [209,]	0.566649987	-0.231566189	-0.4237817949	0.110868884	1.181766273
## [210,]	-1.587506381	0.191616078	0.0915363689	-0.445357515	-0.226680055
## [211,]	0.351885153	1.518896604	1.3619168319	-0.638065874	0.239836012
## [212,]	-0.111811649	-0.655938842	-0.5871239532	-0.191332860	-0.421167680
## [213,]	-1.172619770	2.417638127	2.8425344600	-0.795736349	-0.652518972
## [214,]	0.389306298	0.391005753	0.2466236009	-0.353383071	-0.476463181
## [215,]	1.492416583	0.230303627	-0.1216866205	1.030613325	0.963762171
## [216,]	0.203827577	-0.085148696	-0.2291762988	0.597019517	1.161427698
## [217,]	0.231486685	-0.626774382	-0.6136450993	0.356134068	0.319919152
## [218,]	-0.196415977	-0.948178635	-0.8375819291	-1.624820266	-0.728153048
## [219,]	0.482045658	1.876012441	1.9819144864	0.128387826	0.440043861
## [220,]	3.210535258	2.170633006	2.8038943795	0.369273275	0.987914229
## [221,]	-1.517545110	-0.224721469	-0.3062808230	-0.055561062	-0.043632879
## [222,]	-1.390638617	-0.183355551	-0.3407056219	0.229121741	0.098737147
## [223,]	-0.461618008	-1.051147034	-0.8869358501	0.360513804	-0.700823087
## [224,]	0.750501701	0.554683845	0.3643002095	0.999955176	1.231341550
## [225,]	-0.337965527	-0.250612367	-0.3017142680	-0.208851802	-0.782812968
## [226,]	-1.428059762	0.093409222	-0.0129674851	-0.116877358	-0.646798747
## [227,]	-0.956227929	-1.005614765	-0.8521597777	0.075831001	-0.883234683
## [228,]	-1.035951239	0.206495904	-0.1271313591	-0.822014762	0.689191406
## [229,]	0.916456346	-0.493748732	-0.4506542145	-0.423458838	0.579235985
## [230,]	0.727723613	-0.058365008	-0.3066320965	1.985395648	1.779847498
## [231,]	-0.128081712	0.780857208	0.5416933061	1.661295226	0.884314611

## [232,]	1.313445888	-0.816640968	-0.7522224787	-1.766723694	-0.705907731
## [233,]	2.619931963	-0.857709289	-0.7195540471	-1.420286622	-1.178461813
## [234,]	1.904049182	1.649839078	1.7412921672	-0.440977780	0.138778717
## [235,]	-1.060356334	-1.235656883	-0.9566636316	0.789727876	-1.011304148
## [236,]	0.748874695	-0.267575369	-0.2899466072	-0.160674712	-0.654425713
## [237,]	1.437098368	2.938432055	3.6241179054	0.688993961	1.006346062
## [238,]	0.080175097	1.620079425	1.5270153575	-0.419079103	-0.147232496
## [239,]	2.418183180	-0.141692037	-0.2047627935	-1.062900211	-0.074140741
## [240,]	3.122676916	1.010006537	0.9263377433	0.180944651	0.757834097
## [241,]	-1.078253403	-0.391375526	-0.3463259973	-0.200092331	-0.795524578
## [242,]	-0.863488569	-0.696709566	-0.5922174183	-1.255608570	-1.121831593
## [243,]	0.371409228	-0.598205115	-0.7160413125	0.102109414	1.465235164
## [244,]	0.107834204	-0.275610476	-0.3066320965	-1.694458060	-0.699551927
## [245,]	0.789549853	1.120117253	0.9421450489	0.610158723	0.270343875
## [246,]	0.615460176	-0.999365238	-0.8391626597	0.837904966	-0.963953403
## [247,]	0.348631140	-0.564874303	-0.4892942949	-0.975305502	-0.657603615
## [248,]	-1.297899256	-0.067292904	-0.4241330684	-0.305205981	2.101451217
## [249,]	1.547734798	-0.871398730	-0.7462508299	0.767829199	-0.727517467
## [250,]	-0.730074051	-0.785095736	-0.6828459706	0.286058302	-0.610570660
## [251,]	0.215216622	1.727214176	1.9836708537	-0.493534605	0.400002291
## [252,]	-0.523444248	-0.718434113	-0.6528120899	-0.616167197	-0.949335052
## [253,]	-0.014191270	1.563536084	1.8484305721	1.691953374	2.168822747
## [254,]	-0.095541586	0.703482110	0.5996534268	0.404311158	-0.087487931
## [255,]	-0.180145914	1.661742939	1.9169288966	0.759069728	0.393010906
## [256,]	-0.586897494	0.024962020	-0.0958680213	0.824765759	0.457204534
## [257,]	1.723451481	2.123017561	1.8361360011	-0.186953125	1.770949371
## [258,]	-0.490904121	0.373149961	0.0846865364	1.932838823	1.245959900
## [259,]	0.970147554	1.084405670	0.6066788959	0.789727876	1.671163237
## [260,]	3.882488868	0.566587706	0.2712127430	2.449647604	1.920310782
## [261,]	2.193656307	1.637935216	1.6921138830	0.868563114	0.255725524
## [262,]	0.942488447	0.623131047	0.5926279576	-0.366522277	-0.671586386
## [263,]	0.254264773	0.911799682	0.7278682392	-0.830774233	0.206150247
## [264,]	0.975028573	0.257087314	0.1897173006	-1.049761004	-0.467565054
## [265,]	0.594309094	0.989174780	0.9755160275	1.026233589	0.015476105
## [266,]	3.495261364	3.176509280	4.4812251444	0.338615127	0.064415802
## [267,]	-0.445347944	-0.862470834	-0.8005225793	-0.484775134	-0.017574079
## [268,]	0.709826543	-0.285728758	-0.3847904410	-1.395760103	-0.516504750
## [269,]	-0.331457502	-0.535412247	-0.4971979478	-0.296446511	-0.466929474
## [270,]	-0.076017510	-0.915145420	-0.8258142683	0.049552588	0.004671237
## [271,]	-0.817932392	-0.381554840	-0.3442183565	-2.045274867	-1.295980642
## [272,]	-1.545204217	-0.862768430	-0.7430893688	0.150286503	-0.658239196
## [273,]	0.407203367	2.637859559	2.6405522213	-0.226370744	1.387058765
## [274,]	-0.780511246	-1.165721699	-0.9227657429	0.627677665	-1.020520065
## [275,]	1.466384482	0.828472653	0.7717774215	-0.038042120	-0.468200634
## [276,]	-1.088015441	-0.827354443	-0.7169194961	0.154666239	-1.084205229
## [277,]	-1.088015441	-0.889552118	-0.7422111851	-0.283307304	-1.149034437
## [278,]	-0.224075085	0.646938769	0.6242425689	-0.353383071	-0.878785620
## [279,]	0.068786053	-0.248529191	-0.2484963391	-1.198672009	-1.131619532
## [280,]	-0.640588703	-0.264599404	-0.3698613190	-0.607407726	-0.520318233
## [281,]	1.663252247	1.563536084	1.4813498079	2.007294325	0.825205628
## [282,]	-1.206786902	-0.671413861	-0.6092541811	-1.259988305	-1.075815566
## [283,]	0.483672665	1.343314651	1.3127385477	0.851044172	0.766732224
## [284,]	-0.095541586	0.584443498	0.2641872738	0.180944651	1.374982736
## [285,]	-0.974124999	-0.450597236	-0.5005350456	-1.450068823	-0.143419013

## [286,]	-0.422569856	-0.645820560	-0.5558606153	-1.251228834	-1.195050463
## [287,]	0.262399805	-0.448216463	-0.5874752266	-0.909609471	0.171828902
## [288,]	-1.649332622	-0.591062798	-0.5332034773	-1.585840621	-0.887048166
## [289,]	-0.544595330	-0.862768430	-0.7780410779	-1.295026188	-0.444684157
## [290,]	0.075294078	-0.832413584	-0.7399279077	-0.900850000	-0.999037445
## [291,]	-0.577135456	-0.165499759	-0.1989667814	-1.425104331	-0.044904040
## [292,]	0.083429110	0.054721673	-0.1243211714	-0.046801591	0.309749864
## [293,]	-0.935076847	-0.549101687	-0.5184499920	0.697753432	-0.301678551
## [294,]	0.011840832	-0.681829740	-0.6371804210	0.198463593	-0.498708497
## [295,]	-1.333693395	-0.557136794	-0.5741268352	-0.112497623	-0.680484512
## [296,]	-1.423178743	-0.389589947	-0.3854929879	-0.673103757	-0.934716701
## [297,]	-1.766477077	-1.036862401	-0.8577801530	-1.718984578	-1.138992265
## [298,]	-0.372132660	-0.659510000	-0.5743024719	-0.817635027	-1.109247099
## [299,]	-0.067882478	-0.043485182	-0.1069331352	-1.660734097	-0.238756084
## [300,]	-0.237091135	-1.105904796	-0.9095929882	-0.791356614	-1.068569949
## [301,]	0.091564141	1.899820163	2.0591946473	0.750310257	0.999990258
## [302,]	-0.424196862	-0.569338251	-0.5783421167	-1.198672009	-0.244476308
## [303,]	0.610579158	1.533776431	1.4321715237	0.102109414	0.539194415
## [304,]	-0.185026933	-1.086263425	-0.8872871235	0.391171952	-0.952512955
## [305,]	-0.661739785	-0.731230764	-0.6863587052	-0.786976879	-0.479005503
## [306,]	0.966893542	-0.769918313	-0.7097183902	-1.616936742	-0.751033945
## [307,]	-0.850472518	-0.454168394	-0.4279970764	-0.857052645	-0.760567652
## [308,]	-0.912298758	-1.379693604	-1.0451845432	-1.478537103	-1.283523265
## [309,]	-1.421551737	-0.350604801	-0.3194535777	-1.845558932	-1.245134204
## [310,]	-1.346709446	-0.395839474	-0.3656460375	-1.347583014	-1.244435066
## [311,]	0.142001337	-0.783905350	-0.6981263661	-0.440977780	-0.925182994
## [312,]	-0.638961697	-0.105980453	-0.0698737854	-1.369481691	-1.165623087
## [313,]	-1.509410078	-0.452978008	-0.4597873244	-0.567990107	-0.212697284
## [314,]	-2.083743309	-0.774679857	-0.7249987857	-1.014723121	-0.582605120
## [315,]	-0.526698260	-1.506172130	-1.1249236183	0.102109414	-1.122403615
## [316,]	-0.970870986	-0.677960985	-0.5908123245	-0.962166296	-1.301255960
## [317,]	-1.498021034	-0.763668786	-0.6454353473	-1.413279045	-1.277167460
## [318,]	-0.110184642	1.001078641	1.0615780248	0.483146396	0.140049878
## [319,]	-0.370505654	-1.149949083	-1.0248106826	-0.449737251	0.766096644
## [320,]	-0.889520670	-0.758907241	-0.6405175189	-2.114474687	-1.316573449
## [321,]	-0.824440417	-1.063348492	-0.8609416141	0.342994862	-0.116089053
## [322,]	-0.430704888	1.277843414	1.3636731992	-1.181153067	-0.639171782
## [323,]	-0.747971120	-0.430360671	-0.4936852132	0.978056499	-0.198078934
## [324,]	1.005941693	1.899820163	1.8572124086	1.175144594	1.238968515
## [325,]	-0.699160930	-0.480654485	-0.5224896368	-0.296446511	-0.390659817
## [326,]	-0.744717107	-0.552375249	-0.5377700322	0.264159624	-0.845735435
## [327,]	-1.249089067	-0.263706614	-0.2295275723	-0.186953125	-0.911200224
## [328,]	-0.557611381	-0.729742781	-0.6273447642	-1.360722220	-1.146364999
## [329,]	0.765144758	0.670746492	0.4222603302	1.166385123	0.256996685
## [330,]	-0.076017510	0.191616078	0.1661819788	0.448108512	-0.271170688
## [331,]	-0.601540551	0.507068400	0.3326855982	0.487526131	1.230070389
## [332,]	-0.606421570	-0.239601295	-0.4325636314	-0.156294977	0.451484310
## [333,]	0.016721851	-0.903241559	-0.7806756289	0.439349042	-1.001516209
## [334,]	-0.588524501	-0.749384152	-0.6812652401	-0.690622699	-0.993571453
## [335,]	0.452759544	-0.676473002	-0.5906366877	-0.445357515	-1.040731524
## [336,]	1.215825509	1.069525843	0.8455448478	0.548842427	-0.311212258
## [337,]	-1.426432756	-0.591657991	-0.5349598446	-0.795736349	-0.360787535
## [338,]	1.414320280	1.602223633	1.7430485345	0.763449463	1.451887974
## [339,]	0.189184521	-1.049956648	-0.8721823648	0.342994862	-0.724975145

## [340,]	0.822089979	2.831297304	3.5573759483	0.837904966	0.086025538
## [341,]	-0.678009848	0.123168876	-0.0324631620	-0.130016564	0.526482805
## [342,]	-0.424196862	-1.071681195	-0.9255759305	-0.397180425	0.555083927
## [343,]	-0.939957866	-0.818426547	-0.7738257964	0.413070629	-0.211426124
## [344,]	1.461503463	1.498064847	1.1581782260	-0.462876457	0.581778306
## [345,]	-1.223056966	-0.687484074	-0.6396393352	0.597019517	-0.907386741
## [346,]	-1.008292131	-1.082394670	-0.9194286450	0.159045974	-0.576249315
## [347,]	0.224978660	-0.616656100	-0.5584951663	-0.151915241	-0.756754169
## [348,]	-1.260478111	0.206495904	0.0003809064	-0.454116986	-0.339177799
## [349,]	-0.795154303	-0.821104916	-0.7109478473	0.907980732	-0.904208839
## [350,]	-1.294645244	-0.719326903	-0.6751179545	-0.134396300	-0.417989777
## [351,]	-0.965989967	-0.703851883	-0.5937981489	-1.436053669	-1.204457054
## [352,]	-0.245226167	0.361246100	0.0609755780	0.991195706	1.591080097
## [353,]	-0.341219540	3.631831972	4.1334644204	0.903600997	2.157382298
## [354,]	1.227214553	0.414813476	0.2975582524	1.472966603	-0.118631374
## [355,]	-1.603776445	-0.822592899	-0.7501148380	-1.915196725	-0.817769894
## [356,]	-0.528325267	-0.542852160	-0.5851919491	-0.997204179	-0.343626862
## [357,]	-0.134589737	-0.388101964	-0.5082630617	0.084590472	0.073313928
## [358,]	-0.015818276	-0.313107638	-0.3270059571	-0.747559260	-0.975393852
## [359,]	-1.297899256	-1.249643920	-1.0162044829	-1.351962749	-0.822854538
## [360,]	-0.106930630	-0.936572370	-0.7745283433	0.040793117	-0.949335052
## [361,]	-0.764241183	-0.608323397	-0.5179230818	-1.727306075	-1.341043298
## [362,]	0.573158012	-0.426194320	-0.4555720429	-0.804495820	-0.556546320
## [363,]	0.050888983	-0.578563744	-0.5284612856	-0.112497623	-0.447226479
## [364,]	-0.036969358	0.295774863	0.2255471933	0.062691795	-0.548919355
## [365,]	-0.647096728	-0.401791405	-0.3812777064	-0.484775134	-0.550826096
## [366,]	0.112715223	1.605199598	1.5797063763	0.014514705	-0.105919765
## [367,]	1.323207926	1.569488014	1.3882623413	-0.200092331	0.555083927
## [368,]	-0.266377249	-0.399113036	-0.4496003941	0.194083858	-0.236849342
## [369,]	0.124104268	2.744994310	3.9736349967	0.172185180	-0.581333959
## [370,]	0.019975863	2.611075872	2.3648025563	-0.130016564	0.853171168
## [371,]	0.870900169	0.655866665	0.4995404911	0.399931423	1.348923937
## [372,]	-1.618419502	-0.082172731	-0.1079869556	-0.865812116	-0.512055687
## [373,]	-0.624318640	1.334386755	1.1493963895	-0.576749578	0.188989574
## [374,]	-0.407926799	1.771853655	1.8712633469	1.043752531	0.325639376
## [375,]	-0.889520670	-0.241089278	-0.3688074986	-0.957786560	-0.283882297
## [376,]	-1.063610346	0.173760286	-0.0335169824	-0.388420955	0.004671237
## [377,]	-0.464872020	-0.915145420	-0.9285617549	-0.791356614	0.684106763
## [378,]	1.619323076	-0.302096567	-0.3512438257	-0.944647354	-0.690018219
## [379,]	-0.982260030	-0.276800862	-0.3926940938	-0.213231537	0.356782819
## [380,]	1.162134300	-0.461310711	-0.6542171838	3.767947968	4.345050285
## [381,]	-0.837456468	-0.664569141	-0.7104209371	1.253979831	-0.072234000
## [382,]	-0.788646278	-0.819319337	-0.7613555886	-1.001583915	-0.356338471
## [383,]	0.493434702	-0.592253184	-0.6888176194	-1.943665005	0.426696671
## [384,]	-0.414434824	-0.358044715	-0.4919288459	0.452488248	0.667581670
## [385,]	-1.351590465	-0.317571586	-0.4511811247	-0.690622699	0.090474601
## [386,]	0.981536599	-0.150619933	-0.2149497238	-0.051181327	-0.611206241
## [387,]	-1.039205251	-0.583622885	-0.6159283768	-1.303785659	-0.070962839
## [388,]	-0.928568822	-0.226209451	-0.2376068618	-2.081626672	-0.832388245
## [389,]	-1.097777479	-0.819319337	-0.7562621235	-0.970925767	0.169286580
## [390,]	0.774906796	1.033814259	0.7594828504	-0.318345188	-0.081767707
## [391,]	-1.631435552	-1.012757082	-0.8537405082	0.084590472	-0.567351188
## [392,]	-0.468126033	-1.287141083	-0.9898589735	0.597019517	-0.783448549
## [393,]	0.607325145	1.036790225	0.8402757459	1.564941047	0.870967422

## [394,]	0.498315721	1.926603851	2.1083729315	0.780968405	2.017554594
## [395,]	0.019975863	-0.563386321	-0.5639399049	0.474386925	-0.489174790
## [396,]	-0.054866428	-0.322630727	-0.3443939933	-1.128596242	-0.833659406
## [397,]	0.247756748	-0.295549443	-0.3607282091	0.456867983	0.017382847
## [398,]	-0.751225133	-0.492260750	-0.5086143352	-1.621754451	-0.464387152
## [399,]	-0.866742581	-0.787774105	-0.6742397709	-0.892090529	-0.421803260
## [400,]	-0.193161965	-0.632726312	-0.5595489867	-0.349003336	-0.519047072
## [401,]	0.342123115	1.259987623	0.7436755448	2.405850249	2.144670689
## [402,]	-0.900909714	-0.583920481	-0.5112488861	0.220362270	-0.615019724
## [403,]	-0.173637889	-0.325904289	-0.4543425858	-1.711539028	-0.142783432
## [404,]	-0.432331894	-0.522913192	-0.5263536448	-0.664344287	-0.371592403
## [405,]	-1.436194794	-0.688972057	-0.6103080014	-1.207431480	-1.187423497
## [406,]	-0.015818276	-0.729147588	-0.7169194961	0.172185180	-0.571164671
## [407,]	-0.992022068	0.257087314	0.1182331517	-0.515433282	-0.521589394
## [408,]	0.216843628	-0.465179466	-0.4123654075	-1.679566959	-0.385575173
## [409,]	-0.043477383	0.917751612	0.8227120730	0.693373696	0.757834097
## [410,]	0.521093810	-0.543447353	-0.4748920831	-0.366522277	-0.475192020
## [411,]	1.731586512	-0.660402790	-0.6310331355	0.566361369	-0.585147441
## [412,]	0.124104268	-0.813367406	-0.7186758634	0.198463593	-0.674128708
## [413,]	0.376290247	-1.209765985	-1.0179608501	-1.041001534	-0.416718616
## [414,]	0.955504497	0.087457292	-0.0236813255	-1.080419152	0.509957713
## [415,]	1.827579885	0.084481327	0.0892530914	-0.769457937	-0.988995274
## [416,]	0.249383754	-0.659807597	-0.6282229478	0.448108512	-0.226044474
## [417,]	0.905067302	-1.146675521	-0.9153890002	0.886082055	-0.857811464
## [418,]	0.320972032	1.483185021	1.5235026229	0.846664436	0.927534084
## [419,]	-1.424805749	-0.569635848	-0.5509427869	-0.042421856	-0.594681149
## [420,]	0.527601835	-0.833306374	-0.7422111851	-0.182573389	-0.911835804
## [421,]	0.211962609	-0.619929661	-0.6324382293	-0.327104659	-0.384939593
## [422,]	-1.193770852	0.203519939	-0.1253749918	-0.051181327	0.694276050
## [423,]	-0.977379011	-0.753848100	-0.7111234840	0.399931423	-0.236849342
## [424,]	-0.028834326	-0.174427655	-0.3017142680	-0.773837672	0.396824389
## [425,]	-0.407926799	-1.055610982	-0.8776271034	0.325475920	-0.756754169
## [426,]	0.530855848	-1.111261534	-0.8857063930	-0.865812116	-1.165178181
## [427,]	-0.668247810	-0.769323120	-0.7731232495	0.014514705	0.288140128
## [428,]	1.035227807	-0.701471111	-0.6868856154	-0.090598945	-0.538114486
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## [430,]	-0.767495196	-0.547316108	-0.5159910778	-1.119836771	-1.005583924
## [431,]	0.307955982	0.539804018	-0.0841003604	0.417450364	2.890206583
## [432,]	-0.450228963	-0.525293965	-0.6406931556	0.553222163	0.054882095
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## [434,]	0.854630106	1.132021114	1.2688293654	0.290438037	0.584956209
## [435,]	-0.835829461	-0.147643967	-0.1810518350	-0.462876457	-0.630909235
## [436,]	0.833479023	0.197568008	-0.0198173175	1.267119038	0.651692158
## [437,]	-0.211059034	-0.360723083	-0.4455607493	-0.480395399	-0.566080027
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## [439,]	0.379544260	-0.189307482	-0.2309326661	-0.900850000	-0.890861649
## [440,]	-1.035951239	-0.319357166	-0.3366659772	-1.268747776	-0.969673627
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## [460,]	1.829206892	-1.167507278	-0.9320744895	-0.935887883	-0.911200224
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## [462,]	0.926218384	4.283568374	5.9249590593	0.145906768	1.088971524
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## [464,]	-0.269631262	-0.731528360	-0.6770499586	0.080210736	-0.466929474
## [465,]	-0.290782344	-0.361913469	-0.3389492547	-0.182573389	-0.367143339
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## [468,]	-0.172010883	-1.075847546	-0.8788565605	0.281678566	-0.818405475
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## [475,]	-1.029443214	-0.788071701	-0.7859447307	0.036413382	0.861433714
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## [477,]	0.257518786	0.144000633	-0.0914771031	-0.747559260	0.563346473
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## [480,]	-0.427450875	0.441597163	0.1038309399	0.233501476	1.219901101
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## [505,]	-1.512664091	-1.137152432	-1.0193659440	2.532862577	0.570973438
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##	[45,]	0.482261779	0.701234369	1.2805506994	0.677371224
##	[46,]	2.216000531	1.874172965	1.4518846883	0.437631680
##	[47,]	-0.974903222	-1.353461958	0.3301319681	-0.545688020
##	[48,]	1.094846276	1.432989836	1.6151368853	1.879944642
##	[49,]	0.157275637	-0.747367617	-0.2485243342	-0.051813486
##	[50,]	-0.210850258	0.206805000	-0.0480958943	-0.818094154
##	[51,]	-1.039948383	-1.178357895	-0.7560608675	-1.014093688
##	[52,]	-0.801385547	-0.437322369	-0.8966840471	-0.204626683
##	[53,]	-0.863554725	-0.785704910	-0.1871027155	-0.546241691
##	[54,]	-0.047398526	0.272221809	0.1943578637	-0.225666181
##	[55,]	-0.004738103	0.584092642	-0.3649021380	-0.288784675
##	[56,]	-0.869642224	-0.782662267	0.6550200038	-0.752207304
##	[57,]	0.554640698	1.437553799	1.0283987911	-0.055689183
##	[58,]	0.414676165	1.046574267	1.2886324913	1.409324290
##	[59,]	-1.295839020	-1.574509919	-0.7463627172	-1.165799543
##	[60,]	-0.744824537	-1.071256887	0.5160131826	0.349597991
##	[61,]	-1.200763797	-1.351179976	1.0607259588	-0.207395038
##	[62,]	-0.873237203	-1.320905685	0.1151563028	-0.377925707
##	[63,]	2.013243691	0.972029531	-0.0917375707	1.619719271
##	[64,]	-0.635057832	-0.969632635	0.6162274025	0.052830333
##	[65,]	0.624143634	0.867058373	0.7794795996	1.060511558
##	[66,]	0.144813042	0.711883617	0.6792653796	0.285925826
##	[67,]	-0.853536716	-0.752083713	-0.0367813856	0.452027126
##	[68,]	-0.612529294	-0.684537054	-0.8094006943	-0.970907350
##	[69,]	4.696536347	0.918783291	2.1453024361	1.857797802
##	[70,]	-0.806658184	-0.848687605	-0.8368787868	-1.098805352
##	[71,]	-0.016721368	0.978114816	-0.5653305779	-0.999698242
##	[72,]	-0.617322600	-1.015424401	-1.0453890187	1.353957189
##	[73,]	1.842602000	1.145460141	0.6663345125	2.765818246
##	[74,]	0.027377047	0.360458434	-0.5039089592	1.054974848
##	[75,]	-0.488382671	-0.426064593	-0.4570345660	-0.434953820
##	[76,]	0.051343576	0.568879430	-0.4053110977	-1.111539785
##	[77,]	-0.895382276	-0.616686131	-0.3083295945	-0.666388299
##	[78,]	0.510062953	1.454288331	1.3742994858	1.486838230
##	[79,]	2.377055610	2.071944713	4.1043288004	0.868941391
##	[80,]	-0.474961414	-0.537729564	-0.1968008658	-0.263869480
##	[81,]	-0.463457480	-0.811415237	-0.2242789584	0.252705565
##	[82,]	1.023905348	0.854887804	1.0122352072	0.977460908
##	[83,]	1.799462247	2.618099001	-0.8821368217	1.171245759
##	[84,]	-0.138950669	1.057223515	-0.9532565907	0.447597758
##	[85,]	-0.218040216	-0.582456406	0.7730141660	-0.260547454
##	[86,]	0.208564011	0.754480609	1.2837834161	0.102107053
##	[87,]	0.300595484	0.120089695	0.1927415053	-0.857404795
##	[88,]	1.453864890	1.232175446	1.7056529550	0.494659793
##	[89,]	-0.137992008	0.089663272	0.1151563028	0.479710676
##	[90,]	-0.056505807	0.381756930	0.4044844539	0.043417926
##	[91,]	-0.864225788	-0.686819036	-0.6122049711	-0.635936394
##	[92,]	0.059492196	0.501941300	-0.5572487860	-0.867370873
##	[93,]	-0.657586370	-0.220686240	-1.4123023724	-1.211754236

## [94,]	-0.642727121	-0.540011546	-0.3600530628	-0.991946848
## [95,]	1.189274403	1.474065506	-0.1079011546	-0.088909443
## [96,]	0.659614098	0.649509450	1.2740852658	-0.014717529
## [97,]	-1.065976034	-0.848687605	-1.0890306951	-0.563959163
## [98,]	-1.206467831	-1.381302134	-1.5626237023	0.328558493
## [99,]	-0.383409271	-0.458164469	-0.2081153745	0.200106820
## [100,]	0.226299243	0.637338881	-0.2953987274	0.530648409
## [101,]	0.214315978	0.057715528	-0.4036947393	-0.552332072
## [102,]	-1.304682669	-1.743528697	0.0505019673	0.546704868
## [103,]	-0.755849141	-0.613034961	-0.3341913287	-0.839687323
## [104,]	-0.037332583	-0.260392721	-0.4505691325	0.052830333
## [105,]	-0.921841325	-1.256249537	-0.1208320217	-0.466513067
## [106,]	1.751529187	1.277815080	0.3980190204	3.131241108
## [107,]	0.072434122	0.109440447	-0.1531591894	0.388908632
## [108,]	-0.376219312	-0.459229394	0.1329362450	-0.669710325
## [109,]	3.300725662	2.683515810	1.8656724352	0.772048966
## [110,]	0.190828779	-0.484179060	-0.1159829465	0.242185816
## [111,]	-0.681552899	-0.932056003	-0.5944250289	0.040649571
## [112,]	-0.242486077	-0.062468841	-1.0906470535	0.050615649
## [113,]	1.946616739	0.546059613	-0.8126334110	1.342883769
## [114,]	-0.683949552	-0.810046048	-0.8368787868	0.349597991
## [115,]	-0.127446735	-0.146141504	0.0408038170	0.982997618
## [116,]	-0.584248789	-0.641027270	-0.7479790756	0.081067555
## [117,]	-0.564596235	-1.158428588	-2.0184367673	-0.372388996
## [118,]	0.950088437	1.340189246	1.1059839936	1.248759699
## [119,]	2.221273168	1.350838494	0.6032965355	2.284124474
## [120,]	-0.226668167	0.059236849	3.2024008208	-1.264352982
## [121,]	-0.297129764	-0.380729223	0.1862760718	0.071101477
## [122,]	0.263687029	0.803162885	-0.0109196514	0.034005519
## [123,]	1.476872758	1.676401217	0.5192458994	-0.213485419
## [124,]	-0.077596353	0.114004410	-0.0190014433	-0.215146432
## [125,]	0.280942930	-0.377686581	-1.3783588463	-0.424434071
## [126,]	-0.915705894	-0.967046389	-0.8675895962	-0.671371338
## [127,]	0.514856259	0.278307093	0.9201027792	-0.273835558
## [128,]	0.232530540	0.109440447	-0.0965866459	-1.026274450
## [129,]	-0.365194708	0.421311280	-0.5022926009	-0.340829749
## [130,]	1.414559781	0.891399511	0.6534036454	0.038988558
## [131,]	-0.829426387	-0.498023083	0.9184864208	0.468637256
## [132,]	0.512459606	0.559751503	-0.1030520794	-0.207948709
## [133,]	0.129474463	0.252444634	0.9362663631	-0.429417110
## [134,]	-0.485506687	0.346766544	-0.2873169355	-0.732828819
## [135,]	0.595863129	0.354373150	0.3365974017	-0.434400149
## [136,]	-0.261179970	-0.323983945	-0.1159829465	-0.181372501
## [137,]	-0.803302870	-0.683472129	-1.9214552641	-0.582230306
## [138,]	-0.634099171	-0.454056902	-0.3632857796	-0.902252146
## [139,]	0.303471468	0.792513637	0.8295867095	-0.690749823
## [140,]	-0.889150979	-0.433519067	-1.2910754934	-0.891732397
## [141,]	-1.304682669	-1.743528697	0.3301319681	-0.134864136
## [142,]	0.038401650	0.106397805	-0.1757882068	-0.130988439
## [143,]	-0.935214649	-0.769578906	-0.5119907512	-0.165316042
## [144,]	-0.302881731	-0.203951707	1.0477950917	-0.153135279
## [145,]	-0.837095676	-1.224301793	-0.9710365329	-0.900037462
## [146,]	-0.853153251	-0.824346466	-0.2808515019	1.088195108
## [147,]	0.854222319	1.093735222	4.6441925015	1.054974848

## [148,]	-0.106356188	-0.464858282	-0.0788067036	0.455902823
## [149,]	0.182680159	0.689063800	-0.3390404038	-0.393982166
## [150,]	-0.555009623	-0.827845505	-0.8902186136	-0.764388066
## [151,]	-0.915849693	-0.785704910	0.4772205813	-1.084963577
## [152,]	1.274595248	-0.544879774	0.6808817380	3.579714620
## [153,]	2.633497477	0.646466808	0.3349810433	2.322881444
## [154,]	-0.963591020	-0.905889279	-0.0674921949	-0.898376449
## [155,]	0.137623083	-0.264500288	1.5327026076	0.132005287
## [156,]	-0.373343328	-0.494371912	0.3430628352	-0.145383885
## [157,]	0.412758843	0.561272824	-0.7075701159	-0.363530260
## [158,]	-0.402582495	-0.460142186	-0.6041231792	-1.341313251
## [159,]	-0.890732770	-0.674800599	-0.6251358382	-0.274942900
## [160,]	-1.215814777	-1.142150452	-0.2630715597	-0.392874824
## [161,]	-0.367112031	-0.540315810	0.4319625465	-0.225666181
## [162,]	-0.219478208	0.959858962	-0.7398972837	-1.186839041
## [163,]	1.959558665	1.674879896	1.1997327800	0.458671178
## [164,]	-0.495572629	-0.496958158	-1.0227600013	0.380603567
## [165,]	0.587714509	1.825490689	1.1124494272	0.438739022
## [166,]	-0.933201460	-0.868160515	-0.4117765312	-1.278748428
## [167,]	-0.802823539	-1.043264578	-1.3104717941	-0.385123430
## [168,]	-0.193594356	0.498898658	-0.1466937558	-0.645902472
## [169,]	1.039243927	0.874664979	-1.1973267070	0.501303846
## [170,]	-0.986023692	-0.452687713	-0.8029352607	-1.088839274
## [171,]	-0.709354074	-0.314856018	-0.1192156633	-0.898930120
## [172,]	-0.037332583	0.021203821	-0.0270832352	-0.566727518
## [173,]	1.489814684	1.035925019	0.5095477490	0.944240648
## [174,]	-1.115922282	-1.088447816	-1.6143471707	-0.598840437
## [175,]	-1.304682669	-1.743528697	-0.3083295945	-1.235008418
## [176,]	-1.304682669	-1.743528697	-0.4990598841	-0.302626450
## [177,]	0.366263776	-0.235899451	-0.4634999996	1.785820571
## [178,]	1.505153263	1.352359815	0.2476976904	0.622557795
## [179,]	-1.266446468	-1.602669573	-0.9791183248	-1.412736810
## [180,]	-1.172483292	-1.136521564	-1.8066938187	-0.595518410
## [181,]	1.254942694	2.345782517	-0.0723412701	-0.173067436
## [182,]	1.945178747	2.672866562	1.9351758458	2.461299194
## [183,]	0.086334710	0.600827174	0.8667629524	0.130897945
## [184,]	-0.514745853	-0.785704910	-1.7614357839	-0.535721942
## [185,]	0.435287381	0.121611016	0.4432770552	0.762636559
## [186,]	-1.271512992	-1.585007035	0.0521183257	-0.386230772
## [187,]	0.391188966	0.646466808	0.4933841652	-0.806467062
## [188,]	-0.575620838	-0.071596768	-0.5313870518	-0.718433373
## [189,]	-0.882727949	-1.088447816	0.4836860149	-1.006895965
## [190,]	-0.622595236	-1.011012570	-0.6768593066	-1.156387136
## [191,]	2.763875398	0.952252356	3.6614466025	3.358246219
## [192,]	-1.056101824	-1.017402119	-1.1666158977	-0.843563020
## [193,]	-1.304682669	-1.743528697	-1.6030326620	-1.016308372
## [194,]	0.816355202	0.476078841	0.5079313907	2.023899102
## [195,]	0.894965419	0.883792905	0.1604143376	0.169654915
## [196,]	-0.459143504	-0.490720741	0.1992069388	-0.800376681
## [197,]	0.521087557	0.801641564	0.2897230085	0.519574989
## [198,]	-0.089579618	-0.346803762	-0.8595078042	-1.016862043
## [199,]	0.547930070	0.854887804	0.4723715062	0.457563836
## [200,]	1.076152383	1.052659551	2.9938905889	0.960850778
## [201,]	-0.644644444	-0.100501870	-0.3762166467	-0.122129703

## [202,]	0.376809049	1.206312986	0.0440365338	-0.292106701
## [203,]	2.478194365	2.414241968	0.4804532981	0.203428846
## [204,]	0.922287263	1.318890750	2.4750395469	1.365030609
## [205,]	-0.024390657	-0.199387744	0.1830433550	0.196784794
## [206,]	0.290050211	0.161165366	0.8312030679	0.744919087
## [207,]	-1.006874572	-0.893414446	0.1426343953	-0.561744479
## [208,]	-0.378615965	-0.076160732	0.6049128938	-1.066138763
## [209,]	0.210960664	-0.030521098	1.9836665974	1.309663509
## [210,]	0.115094545	-0.168961321	-0.9387093652	-0.509699405
## [211,]	0.546012748	1.177407885	0.0133257244	-1.400556048
## [212,]	-0.599587368	-0.691839396	-0.5911923121	-0.222344155
## [213,]	0.229654557	0.682978515	-2.0249022008	-1.588804189
## [214,]	0.038880981	-0.071596768	-2.0976383282	-0.872907583
## [215,]	0.489931068	0.952252356	2.9470161957	1.032828008
## [216,]	0.917493957	0.772736462	1.1787201210	1.215539439
## [217,]	0.040318973	-0.012265244	0.8021086170	-0.016378542
## [218,]	-0.381971279	-1.200264919	-0.0529449694	-0.324219619
## [219,]	0.429056083	1.025275771	0.2735594246	-0.077282352
## [220,]	0.610243047	0.728618149	-0.3034805193	-0.457654331
## [221,]	-0.461540157	-0.517952390	-0.8417278620	0.179067322
## [222,]	-0.069447733	-0.360647784	0.2654776327	-0.120468690
## [223,]	-0.989666604	-0.895392164	0.2493140488	0.222807331
## [224,]	0.601135766	0.506505264	1.7654582153	1.243222989
## [225,]	-0.448598231	-0.271194101	-0.6380667053	-0.427202426
## [226,]	-0.522415143	-0.089852622	0.2606285575	-1.285946151
## [227,]	-1.178187326	-1.064410942	-0.4618836412	-0.069530958
## [228,]	0.326000006	0.354373150	0.0860618518	-0.018039555
## [229,]	0.569979277	0.051630244	-0.1208320217	0.659100081
## [230,]	1.706472112	1.264123189	0.8182722008	2.234294083
## [231,]	1.100598243	2.125190953	0.3365974017	0.368976476
## [232,]	-0.782691654	-1.426637504	-0.0836557788	-0.723970083
## [233,]	-1.193573838	-1.286828092	0.0165584412	-0.602162463
## [234,]	0.682621967	0.634296239	-0.7495954340	-0.036864369
## [235,]	-0.961434032	-0.644830572	-0.2323607503	-0.119915019
## [236,]	-1.006011777	-0.532100676	-1.0906470535	-0.430524452
## [237,]	1.485021378	2.201257009	0.3268992514	0.156366811
## [238,]	0.209043341	0.454780345	-1.0712507529	-0.701823243
## [239,]	0.163986266	-0.493154855	-1.6337434713	-0.331417342
## [240,]	0.248827780	1.399520771	-0.0771903452	0.056152359
## [241,]	-0.569868871	-0.340566345	-0.5992741040	-1.043438252
## [242,]	-1.005436581	-1.127089373	0.0003948573	-0.892286068
## [243,]	2.259619615	0.109440447	0.6582527206	2.533276425
## [244,]	-0.653272394	-0.814610011	-0.3842984386	-1.148082071
## [245,]	0.352842519	0.635817560	0.0311056667	-0.432185465
## [246,]	-0.738593240	-0.718766780	-0.0286995936	-0.357993550
## [247,]	-0.203180968	-0.987432092	-0.2161971665	-0.662512602
## [248,]	2.399104818	0.631253597	-0.4230910399	1.874407932
## [249,]	-0.765435753	-0.810046048	0.8215049176	-0.137078820
## [250,]	-0.439970281	-0.281843349	-0.3826820803	-0.324219619
## [251,]	2.046317502	1.458852295	0.3640754943	-0.302072779
## [252,]	-0.916185224	-0.747976145	-0.2598388429	-1.056172685
## [253,]	2.764354729	2.070423392	-0.2452916174	2.533276425
## [254,]	0.314496072	1.081564653	0.3834717949	-0.155903634
## [255,]	0.764587498	1.297592254	0.7730141660	0.307518995

## [256,]	0.233489202	0.346766544	0.2703267079	-0.242276311
## [257,]	0.732951679	1.209355629	-0.1337628888	0.916557097
## [258,]	0.818272524	1.647496116	0.5774348013	1.946385162
## [259,]	1.658059722	2.001963941	0.6081456106	0.994071038
## [260,]	1.428939699	1.320412071	0.9879898314	2.018362392
## [261,]	0.511021614	0.838153271	0.4044844539	-0.219022129
## [262,]	-0.724213322	-0.490720741	-0.7253500582	-1.040669897
## [263,]	-0.203180968	0.581049999	0.2687103495	-0.504162695
## [264,]	-0.221395531	-0.440060747	-0.3519712709	-0.866817202
## [265,]	0.559434004	1.274772437	0.5095477490	-0.456546989
## [266,]	0.345173230	0.780343068	-0.0529449694	-0.097768179
## [267,]	-0.386285254	-0.537729564	0.0634328344	-0.447134582
## [268,]	-0.608215319	-0.801983046	-0.7350482085	-0.758851356
## [269,]	-0.349856129	-0.864205080	1.1366948030	-0.737811858
## [270,]	-0.090058948	-0.435192520	-0.4780472251	0.169654915
## [271,]	-1.119373462	-1.236472362	-0.7156519079	-1.259369943
## [272,]	-0.693536164	-0.412372703	-0.2711533516	-0.206287696
## [273,]	1.478790080	1.057223515	-0.1095175130	0.256581262
## [274,]	-1.110889311	-0.958679123	-0.0965866459	-0.121576032
## [275,]	-0.307675037	-0.015307886	-0.6412994221	-0.247259350
## [276,]	-0.961817497	-0.382554809	-1.1003452038	-1.307539320
## [277,]	-1.280931838	-1.574509919	-0.2307443919	-1.112093456
## [278,]	-0.245362060	0.225060853	-0.5394688437	-1.471425936
## [279,]	-0.796592241	-0.954723688	-0.9144639894	-1.180194989
## [280,]	-0.606777327	-0.223272486	0.0876782102	-0.820862509
## [281,]	1.453385559	1.104384470	0.5774348013	0.733845667
## [282,]	-0.981853515	-0.482353475	0.3236665346	-0.944884813
## [283,]	0.764108167	1.682486502	1.1140657856	-0.336400381
## [284,]	1.104432888	0.891399511	-0.2113480913	1.237686279
## [285,]	0.298198831	-0.196345101	-1.4575604072	-0.701823243
## [286,]	-1.277940816	-1.610078407	-0.6396830637	-1.087178261
## [287,]	-0.171065818	0.013597215	-0.7043373992	0.878353798
## [288,]	-0.736196587	-0.927187775	-0.9564893074	-0.819201496
## [289,]	-0.563637574	-0.326570191	0.0876782102	-0.767156421
## [290,]	-0.943794666	-0.799853196	0.5919820267	-0.775461486
## [291,]	-0.240568754	-0.190259817	-1.0162945677	0.223914673
## [292,]	-0.439970281	0.521718475	0.0989927189	0.042864255
## [293,]	-0.230502812	-0.137013577	0.7746305244	0.658546410
## [294,]	-0.673883610	-0.353041178	0.3236665346	-0.768263763
## [295,]	-1.053848970	-0.778554700	-0.8595078042	-0.815325799
## [296,]	-1.125796492	-0.860858174	-0.1256810968	-0.886195687
## [297,]	-1.166443726	-1.257618726	-1.2248047996	-0.969800008
## [298,]	-1.011332347	-0.654262763	-1.4915039333	-0.819201496
## [299,]	-0.554530292	-0.597973881	-0.4279401151	-0.397857863
## [300,]	-1.105377009	-1.268115842	-1.0890306951	-0.895608094
## [301,]	1.629299886	1.268687153	0.1086908692	0.849562906
## [302,]	-0.392037221	-0.583521331	-0.3487385541	-0.349134814
## [303,]	1.059375812	1.181971848	0.6356237032	0.594874245
## [304,]	-0.900942511	-0.750410259	-1.1116597125	-0.305948476
## [305,]	-0.717023363	-0.905432883	-1.1197415045	-0.418897361
## [306,]	-0.957695254	-1.011012570	0.5548057839	-0.913325566
## [307,]	-1.250997643	-1.363198413	-0.4036947393	-0.005305122
## [308,]	-1.234125206	-1.532217191	0.1458671121	-0.326987974
## [309,]	-1.238582981	-1.407316726	-1.0243763597	-1.219505630

## [310,]	-1.217252769	-1.206198072	-1.2829937015	-1.004681281
## [311,]	-0.925292505	-0.870138233	0.9475808718	-0.795393642
## [312,]	-1.077192370	-0.859184721	-0.5992741040	-1.494680119
## [313,]	-0.456746851	-0.463945489	-0.5443179189	-0.078389694
## [314,]	-0.905687884	-1.026225781	0.7907941083	-0.531846245
## [315,]	-1.304682669	-1.743528697	0.3899372284	-0.154242621
## [316,]	-1.211788401	-1.319992893	-1.5901017949	-1.229471708
## [317,]	-1.107390197	-1.461780023	-0.9823510416	-1.305324636
## [318,]	0.542178103	0.958337641	-0.1434610391	-0.108841599
## [319,]	0.904552031	-0.001615996	0.3786227197	1.193392599
## [320,]	-1.197456416	-1.312690551	-1.6159635291	-1.363460091
## [321,]	-0.394913205	-0.261153382	-0.4731981499	0.723879589
## [322,]	-0.055547146	0.424353922	0.2493140488	-1.362906420
## [323,]	-0.446201578	0.013597215	-0.8384951452	0.087711607
## [324,]	1.256860016	2.341218554	4.2950590900	1.021754588
## [325,]	-0.745303868	-0.898282674	-0.3875311554	-0.240061627
## [326,]	-0.815765465	-0.891284596	-0.3438894790	-0.834150613
## [327,]	-1.050301924	-0.847470548	-1.2943082102	-0.726184767
## [328,]	-1.267620828	-1.318167307	-1.1795467648	-0.751653633
## [329,]	0.419469471	0.664722662	0.3268992514	-0.107734257
## [330,]	0.298198831	-0.150705467	-0.2663042764	-0.244490995
## [331,]	1.070879746	1.270208474	0.1911251469	0.403857749
## [332,]	0.343735238	-0.243810321	1.1237639359	0.427111931
## [333,]	-1.240692035	-1.435917563	0.6323909864	-1.036794200
## [334,]	-1.278233207	-1.489924464	-0.1386119639	-0.540704981
## [335,]	-1.131212928	-1.137586488	-0.5604815028	-0.657529563
## [336,]	0.574293253	1.035925019	-0.4489527741	-0.440490530
## [337,]	-0.609653310	-0.853555832	-0.7576772259	0.938703938
## [338,]	0.916535296	1.372136990	1.2579216819	0.816896317
## [339,]	-0.798988894	-0.754822091	-0.0109196514	-0.404501915
## [340,]	1.005211455	1.434511157	-0.4974435257	-0.363530260
## [341,]	0.495203705	0.407619390	0.2460813321	0.204536188
## [342,]	0.776091432	-0.508215934	0.1313198866	0.791981122
## [343,]	-0.202701638	-0.108108476	0.6469382119	0.379496225
## [344,]	0.964468355	1.687050465	1.8495088514	-0.263869480
## [345,]	-0.783650315	-0.547161755	-0.2194298832	-0.325880632
## [346,]	-0.961386099	-1.123742466	-0.7544445091	0.051722991
## [347,]	-1.088696304	-0.968719842	-0.0335486688	-0.172513765
## [348,]	-0.273642565	0.159644044	0.3365974017	-0.114931980
## [349,]	-0.833836228	-0.747367617	-0.0804230620	0.203982517
## [350,]	-1.155035658	-1.018467043	0.3608427775	-0.445473569
## [351,]	-1.158678571	-1.095141629	-0.2743860684	-0.869031886
## [352,]	1.989277161	1.504491929	2.1727805287	1.165709049
## [353,]	1.787478982	2.449232355	1.2757016242	0.232773409
## [354,]	0.626540287	0.578007357	-0.3988456641	0.577710444
## [355,]	-0.729006628	-1.146866547	-0.5249216183	-0.762173382
## [356,]	-0.160041215	-0.334024664	-1.2603646841	-0.668049312
## [357,]	-0.071365055	0.170293292	0.3430628352	-0.042954750
## [358,]	-1.051356451	-0.898282674	-0.8708223130	-0.709574637
## [359,]	-0.852146657	-1.019075572	-0.7544445091	-0.533507258
## [360,]	-0.756328472	-0.974957259	-0.7221173414	-0.143169201
## [361,]	-1.287517841	-1.494792691	-1.0793325448	-1.591018873
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## [363,]	-0.703122776	-0.479006569	-0.2533734093	-0.640365762

## [364,]	-0.507555894	-0.355627424	-0.8190988446	-1.066138763
## [365,]	-0.650875741	-0.680581619	-0.2582224845	-0.449902937
## [366,]	-0.009531409	0.941603109	-0.4715817915	-0.918862276
## [367,]	0.470278514	1.530354388	0.5984474603	-0.422219387
## [368,]	-0.148057950	-0.400354266	0.5127804658	-0.511914089
## [369,]	0.066682155	1.025275771	-0.6316012718	-1.052296988
## [370,]	0.975013628	1.956324306	-0.2582224845	0.099338698
## [371,]	2.092333239	1.676401217	3.1086520344	0.675156540
## [372,]	-0.651834403	-0.499392272	-0.6687775147	-0.901698475
## [373,]	0.624143634	1.247388657	-0.2760024268	0.150276430
## [374,]	0.688853264	1.469501543	-0.3422731206	-0.741687555
## [375,]	-0.659503692	-0.680581619	0.6824980964	-0.384016088
## [376,]	-0.291377797	0.159644044	0.4077171707	0.313055705
## [377,]	1.585680803	0.485206768	-0.4909780921	1.996215552
## [378,]	-0.924381777	-0.864052948	-0.3341913287	-0.738365529
## [379,]	-0.073282378	-0.140056219	0.7859450331	0.688444644
## [380,]	2.722652968	2.096285851	2.0256919155	3.120167687
## [381,]	-0.227626828	0.261572561	0.7148252641	0.454241810
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## [384,]	0.232051210	-0.252025455	-0.1321465304	1.542205330
## [385,]	0.069078808	-0.348020819	-0.2663042764	-0.594964739
## [386,]	-0.022473335	0.323946727	-0.6849410985	-0.862941505
## [387,]	0.169738233	-0.353041178	-0.3616694212	0.237756448
## [388,]	-0.781732993	-1.053305297	-0.5798778034	-0.980873428
## [389,]	0.143375050	-0.485091853	-1.2021757822	1.126952078
## [390,]	0.530674169	1.032882377	-0.5249216183	-0.438829517
## [391,]	-0.891739364	-0.724852064	0.0585837592	-0.372388996
## [392,]	-1.304682669	-1.743528697	-0.7366645669	0.260456959
## [393,]	1.357040110	1.483193433	0.4626733558	0.994071038
## [394,]	2.076035998	1.941111095	1.4987590815	0.927630517
## [395,]	-0.536315730	-0.790268873	0.2396158985	-0.727292109
## [396,]	-0.899168988	-0.540011546	-0.6105886127	-0.988624822
## [397,]	0.343255907	0.466950914	-0.3794493635	-0.392321153
## [398,]	-0.393475213	-0.481440682	-1.4753403495	-0.742794897
## [399,]	-0.308154368	-0.898282674	-0.5022926009	0.422682563
## [400,]	-0.610132641	-0.928709096	-0.1968008658	-0.151474266
## [401,]	3.025589902	1.244346015	0.5564221422	1.985142132
## [402,]	-0.578976152	-0.697468284	-0.7124191911	-0.627077658
## [403,]	-0.535836399	-0.738239690	0.4950005236	-0.635382723
## [404,]	-0.437094297	-0.467444528	0.6404727783	-0.310377844
## [405,]	-1.068804085	-1.014359476	-0.9742692497	-1.340205909
## [406,]	-0.627867873	-0.543662717	-1.0502380938	-0.366852286
## [407,]	-0.197429001	-0.025957134	-0.1984172242	-0.765495408
## [408,]	-0.423673041	-0.891436729	-0.6671611563	-0.134864136
## [409,]	0.277587616	1.259559226	0.2573958408	0.060028056
## [410,]	-0.644644444	-0.293405390	0.8958574034	-0.829721245
## [411,]	-0.436614967	-0.420283573	0.1167726611	-0.359654563
## [412,]	-0.793236927	-0.613034961	0.1571816208	-0.284355307
## [413,]	-0.409293123	-1.353461958	-0.8481932955	0.449258771
## [414,]	0.187952795	0.246359349	0.4238807546	0.474173966
## [415,]	-0.563158243	-0.743260050	0.5370258416	-1.234454747
## [416,]	-0.746741860	-0.486461042	0.2816412166	-0.577800938
## [417,]	-1.010325753	-1.169990629	-0.0464795359	-0.050152473

## [418,]	0.713778455	1.502970608	0.1652634127	1.154635628
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## [420,]	-1.132986451	-1.088447816	0.1216217363	-0.704037927
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## [423,]	-0.201263646	-0.062468841	-0.1838699987	-0.535721942
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## [426,]	-1.245485341	-1.351179976	-0.8918349720	-0.184694527
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## [429,]	-1.085149258	-1.128306430	-0.8368787868	-0.726184767
## [430,]	-1.138402887	-1.194027503	-1.1892449151	-1.311968688
## [431,]	3.018399943	2.021741115	-0.0561776862	1.747063601
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## [433,]	1.239604115	1.562302132	0.2121378059	-0.176943133
## [434,]	0.570458608	0.854887804	0.1717288463	-0.044615763
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## [437,]	-0.963063756	-0.772317284	0.6663345125	-0.365191273
## [438,]	-0.769270398	-0.609688054	-0.2840842187	-0.642580446
## [439,]	-1.076856839	-0.847470548	-0.6267521966	-0.821416180
## [440,]	-1.004621718	-0.493611251	-1.2361193083	-0.932704051
## [441,]	0.986038231	0.622125670	-0.5831105202	0.629755518
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## [443,]	-1.136102100	-1.239667137	-1.6838505813	-0.877336951
## [444,]	-1.113621495	-1.405491141	-0.3261095367	-0.918308605
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## [447,]	1.762553791	1.253473941	0.1151563028	0.376727870
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## [450,]	0.803892606	1.725083494	-1.0227600013	-0.536829284
## [451,]	0.237803177	-0.412372703	-0.9629547410	0.862297339
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## [454,]	-0.646561766	-0.117236402	-0.4764308667	-0.323665948
## [455,]	-0.400185842	-0.244875246	0.5968311019	-0.589428029
## [456,]	-0.969007455	-0.562527099	-1.1391378051	-0.398411534
## [457,]	0.096400652	-0.703705700	-0.0270832352	-0.650331840
## [458,]	-0.638413146	-0.829975355	-0.7382809253	-0.889517713
## [459,]	-1.090805359	-0.842754452	-0.9613383826	-1.164692201
## [460,]	-0.960043973	-1.003253832	-0.9370930068	-0.655314879
## [461,]	-0.039729236	0.622125670	0.1765779214	0.368422805
## [462,]	1.970583268	2.249939286	-0.4198583231	-0.535721942
## [463,]	-0.599108037	-0.897217749	-0.8983004055	-1.068907118
## [464,]	-0.413127768	-0.482657739	0.4998495987	-0.294321385
## [465,]	-0.405458478	-0.153748110	-1.0760998280	-0.813664785
## [466,]	1.837808694	0.320904085	-0.0901212123	2.267514344
## [467,]	0.854222319	0.053151565	-0.5459342773	-0.122129703
## [468,]	-0.997479693	-1.363198413	0.2525467656	-0.287677333
## [469,]	1.171059840	1.293028291	-0.9694201745	2.129096593
## [470,]	0.222464598	0.410662032	-0.3891475138	0.484693715
## [471,]	-0.692577503	-0.745542032	0.4416606968	0.071655148

## [472,]	-1.039900450	-0.899651863	-0.8029352607	-0.972014692
## [473,]	0.205688027	0.001426646	-0.3438894790	-0.067316274
## [474,]	-1.304682669	-1.743528697	-0.7948534688	-0.914432908
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## [476,]	0.361470470	-0.255220229	0.1701124879	-0.328095316
## [477,]	-0.100604221	0.293520304	-0.5928086705	-0.297089740
## [478,]	-0.641289130	-0.796962686	-0.3584367045	-0.385677101
## [479,]	-0.060340452	-0.613034961	0.0650491928	0.434863325
## [480,]	1.521929834	0.956816320	0.6744163045	0.410501801
## [481,]	-0.528167110	-0.877896970	-0.7997025440	-0.368513299
## [482,]	-0.502762588	-0.503651971	-0.8805204633	-0.438275846
## [483,]	-0.418879735	0.287435020	0.5273276913	0.515699292
## [484,]	-0.469688777	-0.362625502	-0.6186704047	-0.794286300
## [485,]	0.614557022	0.465429593	-0.5556324276	-0.118254006
## [486,]	1.042119911	0.298084268	0.5337931248	1.077121688
## [487,]	-0.136554016	-0.552638512	-0.7205009830	-0.995822545
## [488,]	1.540623727	1.390392844	0.5903656684	0.340185584
## [489,]	-0.590480087	-0.250352002	-0.1563919062	-0.205180354
## [490,]	-0.117380792	-0.414350420	2.8710473516	-0.427202426
## [491,]	-0.715106041	-0.779771757	0.3220501762	-0.106073244
## [492,]	-0.802344209	-0.474594737	-1.8066938187	-1.397234022
## [493,]	-0.085265642	0.521718475	0.5661202926	-0.426095084
## [494,]	-1.224442728	-1.335814633	-1.0033637006	-0.756636672
## [495,]	-0.935693979	-1.105334480	-0.3455058374	-0.534614600
## [496,]	-0.489820662	-0.196345101	-0.8595078042	-0.994161532
## [497,]	0.412279512	0.396970142	0.5321767664	1.071584978
## [498,]	-0.332600228	-0.141577541	0.2169868811	-0.406162928
## [499,]	0.388792313	0.786428353	-0.6316012718	0.581586141
## [500,]	1.177770468	1.471022864	-0.6800920234	0.334648874
## [501,]	-0.415045090	-0.194823780	-1.1698486144	0.085496923
## [502,]	0.315934063	0.570400751	1.2126636471	1.902091482
## [503,]	-0.399227180	-0.463945489	0.4109498875	-0.474264461
## [504,]	0.513897598	1.700742356	0.0117093660	-0.618772593
## [505,]	-0.298567756	-0.184174532	0.2218359563	2.284124474
## [506,]	0.091607346	-0.108108476	-0.0852721372	2.904235996
## [507,]	0.389750975	-0.513084162	-0.3099459529	0.246061513
## [508,]	-0.702643445	-0.296143768	-0.1951845074	1.819040832
## [509,]	-0.371426006	0.320904085	-0.9710365329	-0.644795130
## [510,]	1.528640462	1.547088921	0.1814269966	1.259833119
## [511,]	-0.015283376	-0.137013577	-0.4796635834	0.821879356
## [512,]	-0.776939687	-0.533317733	-0.9160803478	-1.247189180
## [513,]	1.142779335	1.376700954	1.1059839936	1.492374940
## [514,]	-0.110190833	-0.346043101	-0.4441036989	-0.745563252
## [515,]	0.069078808	-0.039649024	-1.0001309839	-0.797608326
## [516,]	-0.526249787	-0.444016182	0.2573958408	-0.892286068
## [517,]	0.497121027	0.553666219	0.2800248582	-0.294321385
## [518,]	0.701315860	0.710362296	-0.5685632947	0.410501801
## [519,]	-0.707436751	-0.076160732	-0.5152234679	0.275959747
## [520,]	-0.622595236	-0.519625843	0.2751757830	0.089926291
## [521,]	-0.825543809	-0.638745288	1.2611543987	0.325236467
## [522,]	0.928039230	2.021741115	0.4141826042	0.706715787
## [523,]	-1.183268230	-1.312690551	-0.5556324276	-0.432739136
## [524,]	-0.377177973	0.209847642	-0.0836557788	0.352366346
## [525,]	-0.900415248	-0.750410259	-0.6445321389	0.450919784

```

## [526,] -0.463936810      -0.448580146  0.1329362450      1.160172339
## [527,] -0.032539277       0.427396564  0.9976879817       0.149722759
## [528,] -0.446201578      -0.115715081  0.3382137601      -0.444366227
## [529,] -0.557406276      -0.199387744 -1.1973267070      -0.632060697
## [530,] -0.527208449      -0.618815981 -0.1935681490      -0.189677566
## [531,] -0.649917080      -0.206994349 -0.6833247401      -0.353010511
## [532,]  0.017311104      -0.508215934  0.4933841652       0.307518995
## [533,] -0.726609975      -0.419370780 -0.1531591894      -0.339168736
## [534,]  0.177407522       0.710362296  0.5160131826      -1.112093456
## [535,] -0.287063822      -0.243353925 -0.9888164752      -0.064547919
## [536,]  0.820189847       1.524269104  0.2848739333      -0.457100660
## [537,]  0.724803059       0.328510691 -0.3277258951      -0.024129936
## [538,] -0.636016493       0.246359349 -0.1580082645       0.872263417
## [539,] -1.304682669      -1.743528697  0.2541631240       0.854545945
## [540,]  0.321686031      -0.982868129 -0.1790209236       1.254296409
## [541,] -0.443325595      -0.691078735 -0.9241621397      -0.144276543
## [542,]  0.631812924       0.089663272  0.4626733558       1.016217878
## [543,] -0.532481085      -0.077682053 -0.2889332938      -0.796500984
## [544,] -0.795633580      -0.532861337 -0.6914065321      -1.080534209
## [545,] -0.644644444      -0.702184379 -1.0534708106       0.053937675
## [546,] -0.801864878      -0.652132914 -0.4182419648      -0.798161997
## [547,] -1.094544138      -1.381302134 -0.3552039877      -0.551224730
## [548,] -0.450036223      -0.475811794 -0.3390404038       0.600410955
## [549,] -0.856508565      -1.158428588 -0.5637142195      -0.262762138
## [550,] -1.007785300      -1.246969478  0.2557794824      -0.425541413
## [551,] -1.304682669      -1.743528697 -0.7156519079      -0.998037229
## [552,] -0.555009623      -0.767905452  0.4335789049      -0.200750986
## [553,] -0.889917908      -0.754974223 -0.7980861856      -1.057833698
## [554,] -0.921553727      -1.353461958 -0.7528281507      -0.554546756
## [555,] -0.135595355      -0.755734883 -0.8546587291      -0.638151078
## [556,] -0.346021485      -0.355018896 -1.0906470535      -0.061779564
## [557,] -1.256509945      -1.403969819 -1.0324581516      -0.914986579
## [558,] -1.304682669      -1.743528697 -0.6881738153      -0.789303261
## [559,]  0.450625960      -0.062468841 -1.0389235851      -0.216253774
## [560,]  0.435287381      -0.274997404 -1.2749119096       0.186818716
## [561,] -0.669090304      -0.149184146 -1.0518544522      -0.040740066
## [562,] -1.304682669      -1.743528697 -2.1574435885      -1.378409208
## [563,]  4.303485261       1.840703900  1.9206286204       3.153387948
## [564,]  1.858419910       2.123669631  0.0456528921       0.818557330
## [565,]  0.663928073       1.627718941 -1.3589625456      -0.708467295
## [566,]  0.236365185       0.733182113 -0.5313870518      -0.973122034
## [567,]  0.326479336       0.413704674 -1.1035779206      -0.318129238
## [568,]  3.194793601       2.287972314  1.9173959036       2.217683953
## [569,] -1.304682669      -1.743528697 -0.0480958943      -0.750546291
## attr("scaled:center")
##          mean texture          mean perimeter          mean area
##          1.928965e+01          9.196903e+01          6.548891e+02
##          mean smoothness        mean compactness        mean concavity
##          9.636028e-02          1.043410e-01          8.879932e-02
##          mean concave points      mean symmetry      mean fractal dimension
##          4.891915e-02          1.811619e-01          6.279761e-02
##          worst radius            worst texture            worst perimeter
##          4.051721e-01          1.216853e+00          2.866059e+00
##          worst area              worst smoothness        worst compactness

```

```
##          4.033708e+01          7.040979e-03          2.547814e-02
##      worst concavity      worst concave points      worst symmetry
##          3.189372e-02          1.179614e-02          2.054230e-02
## worst fractal dimension          sd radius          sd texture
##          3.794904e-03          1.626919e+01          2.567722e+01
##          sd perimeter          sd area          sd smoothness
##          1.072612e+02          8.805831e+02          1.323686e-01
##      sd compactness          sd concavity      sd concave points
##          2.542650e-01          2.721885e-01          1.146062e-01
##          sd symmetry      sd fractal dimension
##          2.900756e-01          8.394582e-02
## attr(,"scaled:scale")
##      mean texture          mean perimeter          mean area
##          4.301036e+00          2.429898e+01          3.519141e+02
##      mean smoothness      mean compactness      mean concavity
##          1.406413e-02          5.281276e-02          7.971981e-02
##      mean concave points      mean symmetry      mean fractal dimension
##          3.880284e-02          2.741428e-02          7.060363e-03
##      worst radius          worst texture          worst perimeter
##          2.773127e-01          5.516484e-01          2.021855e+00
##      worst area          worst smoothness      worst compactness
##          4.549101e+01          3.002518e-03          1.790818e-02
##      worst concavity      worst concave points      worst symmetry
##          3.018606e-02          6.170285e-03          8.266372e-03
## worst fractal dimension          sd radius          sd texture
##          2.646071e-03          4.833242e+00          6.146258e+00
##          sd perimeter          sd area          sd smoothness
##          3.360254e+01          5.693570e+02          2.283243e-02
##      sd compactness          sd concavity      sd concave points
##          1.573365e-01          2.086243e-01          6.573234e-02
##          sd symmetry      sd fractal dimension
##          6.186747e-02          1.806127e-02
```

```
# 3. Tính toán ma trận hiệp phương sai và tìm giá trị riêng, véc-tơ riêng.
cov_matrix = cov(data_scale)
eig = eigen(cov_matrix)
eig$values
```

```
## [1] 12.674845404  5.356948646  2.817709444  1.976853337  1.646104670
## [6]  1.206899648  0.664782198  0.476595747  0.395339025  0.345539427
## [11]  0.293199144  0.260236830  0.241310397  0.156366009  0.093908811
## [16]  0.078160043  0.057512058  0.051866926  0.044536250  0.031057639
## [21]  0.029747999  0.027208196  0.023954934  0.017030144  0.015466013
## [26]  0.007979153  0.006665744  0.001481967  0.000694198
```

```
eig$vectors
```

```
##          [,1]          [,2]          [,3]          [,4]          [,5]
## [1,] -0.10468254  0.07558047 -0.063357594  0.600828668  0.0536831424
## [2,] -0.22284120  0.24012723  0.010694332 -0.046656431 -0.0416870797
## [3,] -0.21573658  0.25657474 -0.027214969 -0.058840559 -0.0152897125
## [4,] -0.15318292 -0.17808174  0.103833985 -0.161295017  0.3647212937
## [5,] -0.25040719 -0.13434904  0.073987483 -0.031929016 -0.0122787788
```

```

## [6,] -0.26632742 -0.03783924 -0.002361816 -0.020418863 -0.0884574847
## [7,] -0.26476411 0.05938718 0.026338004 -0.068971194 0.0405624887
## [8,] -0.14891449 -0.18256927 0.039774427 -0.067784137 0.3075498277
## [9,] -0.08084874 -0.37030315 0.021107834 -0.044343264 0.0475133471
## [10,] -0.20723628 0.13160347 -0.267215993 -0.104933934 0.1480797015
## [11,] -0.02208649 -0.08806388 -0.374656406 0.360939945 0.1977001556
## [12,] -0.21333375 0.11548235 -0.265447689 -0.095479352 0.1148703193
## [13,] -0.20208916 0.17909441 -0.214552842 -0.115818104 0.1203450210
## [14,] -0.02307931 -0.20869736 -0.309773592 -0.042474739 0.2352948063
## [15,] -0.18317354 -0.22432491 -0.155549437 0.032293173 -0.2779015712
## [16,] -0.16472457 -0.18900245 -0.177119876 0.003339536 -0.3530364984
## [17,] -0.19237540 -0.11766986 -0.224963956 -0.071869115 -0.1960104703
## [18,] -0.05063187 -0.18553177 -0.289409632 -0.041956879 0.2569284935
## [19,] -0.11561219 -0.27999749 -0.212742677 -0.008884428 -0.2602266860
## [20,] -0.22374873 0.24690439 0.049090900 -0.021513383 -0.0008143162
## [21,] -0.10619523 0.06150123 0.043511986 0.630103377 0.0972806740
## [22,] -0.23344035 0.22730503 0.050071386 -0.019662431 -0.0125862034
## [23,] -0.22090420 0.24709895 0.013547727 -0.032631560 0.0215221216
## [24,] -0.13814450 -0.16379540 0.259577949 -0.020048122 0.3244769229
## [25,] -0.22057403 -0.12764672 0.236079593 0.091660545 -0.1218011582
## [26,] -0.23781284 -0.07893983 0.173285615 0.073871052 -0.1895938417
## [27,] -0.25588831 0.03175764 0.171058861 -0.008978110 -0.0463084523
## [28,] -0.13140790 -0.13412742 0.271119831 0.035222918 0.2464845674
## [29,] -0.14599282 -0.27026904 0.232002212 0.079922247 -0.0924072364
##      [,6]      [,7]      [,8]      [,9]      [,10]
## [1,] 0.0326699441 -0.009690817 0.13002464 1.559750e-01 0.218680655
## [2,] -0.0187579497 0.115630647 -0.01725135 -2.355440e-01 0.144155036
## [3,] 0.0001316294 0.054338930 0.03597861 -2.036120e-01 0.122526199
## [4,] 0.2880645060 0.134956379 -0.28913345 -2.646235e-03 -0.074541403
## [5,] 0.0139522041 -0.035393011 -0.14999948 -1.814782e-01 0.061948030
## [6,] 0.0082838748 0.118022265 -0.07397464 -5.873957e-05 -0.134564634
## [7,] 0.0511788183 0.155983745 -0.15217711 -1.457401e-01 0.052882647
## [8,] -0.3551706729 0.096038837 -0.23314544 3.523726e-01 0.516153093
## [9,] 0.1207871961 -0.315693950 -0.17456160 -7.116706e-02 0.080003844
## [10,] 0.0226414431 -0.293088293 0.02048332 2.477383e-01 -0.099029570
## [11,] 0.0313374104 0.074766329 -0.47343245 -2.953745e-01 -0.243773851
## [12,] -0.0048118135 -0.295047021 -0.01389221 2.088787e-01 -0.152736437
## [13,] 0.0394836085 -0.326786341 0.08394919 2.235287e-01 -0.140872891
## [14,] 0.3458923918 0.228924203 0.57470642 -1.146598e-01 0.188470906
## [15,] -0.0691517779 -0.029743162 0.11883671 -1.442315e-01 0.081590748
## [16,] -0.0569380659 0.226193686 0.05673057 3.399025e-01 -0.224068711
## [17,] 0.0309695551 0.387432067 -0.11195979 2.460896e-01 0.063227849
## [18,] -0.4879817379 0.066520071 0.22253405 -3.543239e-01 -0.256456994
## [19,] 0.0539596093 -0.213780650 0.01464142 -1.168944e-01 0.378551260
## [20,] -0.0018414860 0.017769282 0.04311378 -1.235204e-01 0.112762740
## [21,] 0.0505905359 -0.006861870 0.03540060 1.121132e-01 0.006270415
## [22,] -0.0106807962 0.009143062 0.03097788 -1.275152e-01 0.088330536
## [23,] 0.0226624546 -0.057679137 0.07971748 -8.975585e-02 0.096616724
## [24,] 0.3705739486 0.109743546 0.20465029 9.995576e-02 -0.158357901
## [25,] -0.0485562650 -0.139654802 0.08477465 -1.360721e-01 -0.139520079
## [26,] -0.0297358865 0.076710338 0.07014608 1.011927e-01 -0.338671071
## [27,] 0.0295829868 0.182946298 -0.03784962 1.014391e-02 -0.060998853
## [28,] -0.4981570118 0.021510094 0.22761046 6.649806e-02 -0.048793859
## [29,] 0.0803066267 -0.389635420 0.05092356 -9.185255e-02 0.013328102

```

##		[,11]	[,12]	[,13]	[,14]	[,15]
##	[1,]	-0.306076940	-0.246417103	0.205245481	0.021998767	0.102948430
##	[2,]	0.033987918	-0.056104714	0.048210530	-0.058082294	0.037052276
##	[3,]	0.125038652	-0.083617991	0.071944625	-0.021510530	-0.016011564
##	[4,]	-0.149154862	-0.308016088	0.047263001	-0.445365189	0.115341533
##	[5,]	-0.296856204	0.105250256	0.228919851	-0.013236922	-0.223854133
##	[6,]	0.120832150	-0.066953341	0.389070418	0.182615295	0.141229934
##	[7,]	-0.060130663	-0.054721381	0.135916700	0.231302108	0.238676526
##	[8,]	0.177723594	0.282993284	0.187934865	-0.029134290	0.069270796
##	[9,]	-0.042589318	-0.234661023	0.108715188	0.379056978	-0.526699578
##	[10,]	-0.037610975	0.027983392	-0.070791374	-0.004441509	0.110625400
##	[11,]	0.352505434	0.294862584	-0.170015697	0.011612881	-0.035072786
##	[12,]	-0.179222643	0.118007750	-0.042183242	0.055274918	0.001932317
##	[13,]	0.038042783	0.029178994	0.053544558	-0.077772758	0.043158052
##	[14,]	0.101859906	0.285140973	0.149416569	0.200436878	-0.021888699
##	[15,]	-0.194249071	0.264330690	0.007126498	-0.495725518	-0.155797605
##	[16,]	0.323647181	-0.241302260	0.159275430	-0.129874209	-0.259362395
##	[17,]	-0.343535971	0.012080338	-0.495745919	0.205998485	-0.074165860
##	[18,]	-0.194132348	-0.318995178	0.013886597	0.043509737	0.112247409
##	[19,]	0.262084093	-0.298234708	-0.234003901	-0.156260450	0.370472865
##	[20,]	0.119221261	-0.058911191	-0.133682680	-0.037437153	-0.157220456
##	[21,]	0.007887556	-0.077581336	-0.079940688	-0.053557430	-0.095594633
##	[22,]	0.065200592	-0.007589356	-0.093655462	-0.025525718	-0.173527204
##	[23,]	0.197538533	-0.068865631	-0.096454855	-0.011328444	-0.300437614
##	[24,]	0.132152514	-0.052234682	-0.206143509	-0.158136078	-0.044079553
##	[25,]	-0.191705126	0.379674171	0.006836590	-0.160835515	0.043950652
##	[26,]	0.171123992	0.096413532	0.215473284	0.072385161	0.197602527
##	[27,]	-0.112771769	0.063369750	-0.254154546	0.272669788	0.179199048
##	[28,]	0.153031626	-0.046934612	-0.256242518	-0.004389774	-0.136177841
##	[29,]	0.117756991	0.033462270	-0.173677266	0.222372350	0.233923268
##		[,16]	[,17]	[,18]	[,19]	[,20]
##	[1,]	-0.168755246	-0.014613828	-0.044995792	0.04162455	-0.298169082
##	[2,]	-0.104688329	0.134010605	0.090891992	0.42137037	-0.035562005
##	[3,]	-0.126785650	0.226424715	0.309451644	0.25654085	-0.113058524
##	[4,]	-0.225879949	0.301651286	-0.215279979	-0.18987652	0.037285306
##	[5,]	0.183463196	-0.083816603	-0.111484925	0.28256588	0.446491561
##	[6,]	0.255769224	-0.004834364	-0.027865549	-0.00426465	-0.046036850
##	[7,]	0.348372991	0.073830111	-0.009241654	-0.16304133	-0.237686110
##	[8,]	-0.179669672	-0.199091322	0.159361943	-0.03202597	0.038011298
##	[9,]	0.005332867	0.010045063	0.092512830	-0.09366839	-0.142920777
##	[10,]	0.053614993	-0.131396657	-0.315754957	-0.02976947	-0.085632388
##	[11,]	-0.028097025	0.040133233	-0.015614531	0.05372299	-0.099047597
##	[12,]	0.052915670	-0.104556954	-0.183459408	0.28488518	-0.152065999
##	[13,]	0.062699867	0.330468735	0.492628890	-0.14784018	0.251595092
##	[14,]	-0.052883873	0.192373546	-0.183325268	-0.05674208	0.102950863
##	[15,]	0.182141753	0.001072999	0.179650779	-0.23332660	-0.431074818
##	[16,]	-0.096837162	0.056749443	-0.039505313	0.14593891	0.050943486
##	[17,]	-0.180730300	0.108167511	0.068573050	0.07367830	0.050599237
##	[18,]	-0.188155708	-0.268643597	0.176143868	-0.06285061	0.118830924
##	[19,]	0.230838558	-0.122170141	-0.105481116	0.03134626	0.218010598
##	[20,]	-0.124787276	-0.158044629	-0.221873081	-0.09547983	-0.003874282
##	[21,]	0.198830122	0.044934838	0.081318503	-0.07968713	0.373541473
##	[22,]	-0.092317400	-0.174861445	-0.227120275	-0.00497096	0.016806250
##	[23,]	-0.154227669	-0.113940727	0.018848959	-0.43215150	0.049077506

```

## [24,] 0.160396966 -0.448334592 0.309887079 0.29039330 -0.161500118
## [25,] -0.157715236 0.023527235 -0.067946481 0.02302533 0.206907864
## [26,] -0.238666501 -0.138979838 -0.099391674 -0.15053519 -0.008716402
## [27,] 0.155667678 -0.051284010 0.156350366 -0.27541116 0.065319148
## [28,] 0.299455921 0.427555123 -0.251119670 0.09799052 -0.139241077
## [29,] -0.393363547 0.180066141 0.062987089 0.10619169 -0.118988350
##      [,21]      [,22]      [,23]      [,24]      [,25]
## [1,] 0.41557215 -0.075111210 0.007817234 -0.094164834 -9.085297e-02
## [2,] -0.10997376 -0.123685944 0.107105312 0.244695565 -7.162564e-05
## [3,] -0.06805468 -0.153014835 0.075381646 0.056495574 2.283603e-01
## [4,] -0.10554078 -0.069228932 0.034588021 -0.058936198 -3.202665e-02
## [5,] 0.22199851 0.117451468 -0.074513256 0.131984990 -3.878163e-01
## [6,] -0.02835969 0.154113838 -0.341434698 -0.109044350 8.534955e-02
## [7,] -0.05763907 0.307011012 0.030177082 -0.106951368 1.794293e-01
## [8,] -0.08071374 0.017976239 0.052422484 -0.017517594 2.366053e-02
## [9,] -0.04898821 -0.273211731 0.090259389 0.121625596 2.144355e-01
## [10,] 0.09846877 0.209129190 0.186058736 0.557987137 2.111900e-01
## [11,] 0.20570533 -0.037549718 0.002904510 -0.019881603 -5.826263e-02
## [12,] -0.34104364 -0.234133032 0.008151475 -0.477986939 -1.001740e-01
## [13,] 0.22949113 -0.038946003 -0.119610648 -0.011748576 -1.369584e-01
## [14,] -0.05065057 -0.044621185 0.063411534 -0.009261239 -3.165714e-02
## [15,] -0.18109679 0.028651090 -0.015113439 0.129176600 -1.658455e-01
## [16,] 0.05436726 0.232494263 0.387948163 -0.152398677 -2.300534e-02
## [17,] 0.01480972 0.042088752 -0.362360478 0.072373435 1.325395e-01
## [18,] -0.08609992 0.087613003 0.032588553 -0.008877054 1.854704e-02
## [19,] 0.10448007 -0.239268900 -0.066048212 -0.081495709 7.929215e-02
## [20,] 0.00782932 0.109401106 0.031549176 0.162003939 -6.084046e-02
## [21,] -0.55140433 0.085252276 0.001628803 0.113404692 1.256735e-01
## [22,] -0.09978737 -0.042633162 -0.031318436 -0.228451681 -1.327416e-01
## [23,] 0.15658211 0.111071365 -0.222442808 -0.210684674 2.054333e-02
## [24,] 0.12493762 0.057187730 -0.094152626 0.003060196 4.793901e-02
## [25,] 0.21162982 0.004479335 0.144970481 -0.230435221 6.102526e-01
## [26,] -0.08795680 -0.476317402 -0.270078847 0.291804728 -9.757051e-02
## [27,] 0.07285771 -0.231096862 0.583272788 -0.044017975 -2.636001e-01
## [28,] 0.14395067 -0.104846636 -0.110608422 0.016961107 -4.409596e-02
## [29,] -0.16282090 0.437130877 -0.021125720 -0.016226882 -2.809493e-01
##      [,26]      [,27]      [,28]      [,29]
## [1,] -0.017774380 0.017568016 -0.001121688 -0.010216814
## [2,] -0.180624360 0.237773125 -0.206388721 0.583525177
## [3,] 0.228413119 -0.449265284 0.201102884 -0.410768536
## [4,] -0.047405493 -0.062851252 -0.002263117 -0.002080587
## [5,] 0.242113236 -0.139306999 0.014568030 -0.054692132
## [6,] -0.588707214 -0.284865022 -0.038103316 -0.014833347
## [7,] 0.462905592 0.378331407 0.007060218 -0.004024608
## [8,] -0.013371401 0.017758904 -0.003128905 -0.006547124
## [9,] -0.084346533 0.120898611 0.002154424 0.009741752
## [10,] -0.011801475 -0.188556815 0.198215911 0.104246914
## [11,] -0.009148117 0.011838432 0.003764045 -0.008284531
## [12,] 0.117610940 -0.086481522 -0.246100077 0.003767813
## [13,] -0.097204808 0.331203476 0.023917736 -0.082400508
## [14,] -0.005609414 0.009510197 -0.009755270 0.002850872
## [15,] -0.059581097 -0.034596409 0.015305063 0.001697750
## [16,] 0.105617245 0.066641479 -0.005561871 0.018337598
## [17,] -0.018695369 0.017825739 0.023521686 -0.023606187

```

```
## [18,] -0.013044664 0.021915000 0.006446467 -0.005419707
## [19,] 0.039996154 0.002539063 -0.018882028 -0.007687113
## [20,] -0.186305812 0.238151239 -0.522619509 -0.536504816
## [21,] 0.028131862 -0.022916660 0.002802489 0.016175660
## [22,] -0.209277015 0.296363305 0.717391509 -0.043450220
## [23,] 0.230498600 -0.328646955 -0.148143031 0.420558120
## [24,] 0.019405992 0.037672487 0.002981515 0.005838221
## [25,] -0.081595723 0.079167833 -0.051062553 0.007282516
## [26,] 0.277131248 0.107494017 0.014736062 -0.012955426
## [27,] -0.169284984 -0.206760504 -0.015676015 0.034844218
## [28,] 0.023685532 -0.026915955 -0.002345601 0.012734645
## [29,] -0.005216742 -0.059547104 0.025558728 0.008554496
```

4. Thực hiện PCA và phân tích kết quả. Sau đó, vẽ các biểu đồ liên quan đến PCA, bao gồm biplot, biểu đồ tỷ lệ phương sai giải thích (Scree plot)

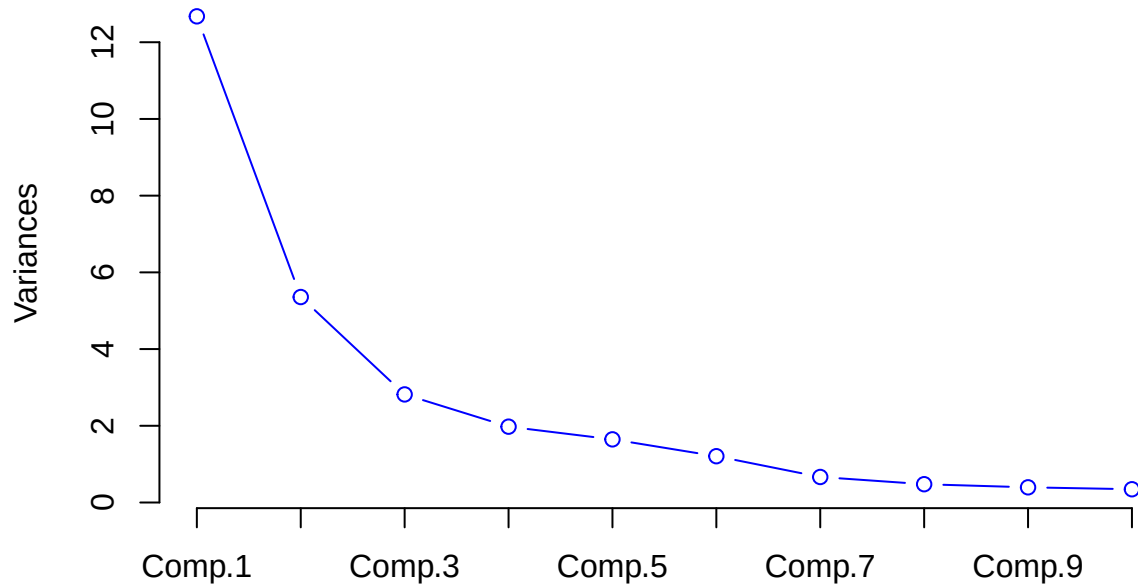
```
pca_wdbc <- princomp(data_scale, cor = T)
summary(pca_wdbc)
```

```
## Importance of components:
##               Comp.1   Comp.2   Comp.3   Comp.4   Comp.5
## Standard deviation 3.5601749 2.3145083 1.67860342 1.40600617 1.28300611
## Proportion of Variance 0.4370636 0.1847224 0.09716239 0.06816736 0.05676223
## Cumulative Proportion 0.4370636 0.6217860 0.71894840 0.78711575 0.84387798
##               Comp.6   Comp.7   Comp.8   Comp.9   Comp.10
## Standard deviation 1.09858985 0.81534177 0.69035914 0.62875991 0.58782602
## Proportion of Variance 0.04161723 0.02292352 0.01643434 0.01363238 0.01191515
## Cumulative Proportion 0.88549521 0.90841874 0.92485307 0.93848545 0.95040061
##               Comp.11   Comp.12   Comp.13   Comp.14
## Standard deviation 0.54147866 0.510134130 0.491233546 0.395431422
## Proportion of Variance 0.01011032 0.008973684 0.008321048 0.005391931
## Cumulative Proportion 0.96051092 0.969484604 0.977805652 0.983197584
##               Comp.15   Comp.16   Comp.17   Comp.18
## Standard deviation 0.306445445 0.279571176 0.239816718 0.227743114
## Proportion of Variance 0.003238235 0.002695174 0.001983174 0.001788515
## Cumulative Proportion 0.986435819 0.989130992 0.991114167 0.992902682
##               Comp.19   Comp.20   Comp.21   Comp.22
## Standard deviation 0.211036135 0.176231776 0.172476083 0.1649490715
## Proportion of Variance 0.001535733 0.001070953 0.001025793 0.0009382137
## Cumulative Proportion 0.994438414 0.995509367 0.996535160 0.9974733741
##               Comp.23   Comp.24   Comp.25   Comp.26
## Standard deviation 0.1547738150 0.1304995926 0.1243624260 0.0893261027
## Proportion of Variance 0.0008260322 0.0005872463 0.0005333108 0.0002751432
## Cumulative Proportion 0.9982994063 0.9988866526 0.9994199634 0.9996951066
##               Comp.27   Comp.28   Comp.29
## Standard deviation 0.0816440083 0.0384963224 2.634764e-02
## Proportion of Variance 0.0002298532 0.0000511023 2.393786e-05
## Cumulative Proportion 0.9999249598 0.9999760621 1.000000e+00
```

Biểu đồ tỷ lệ phương sai giải thích (Scree plot)

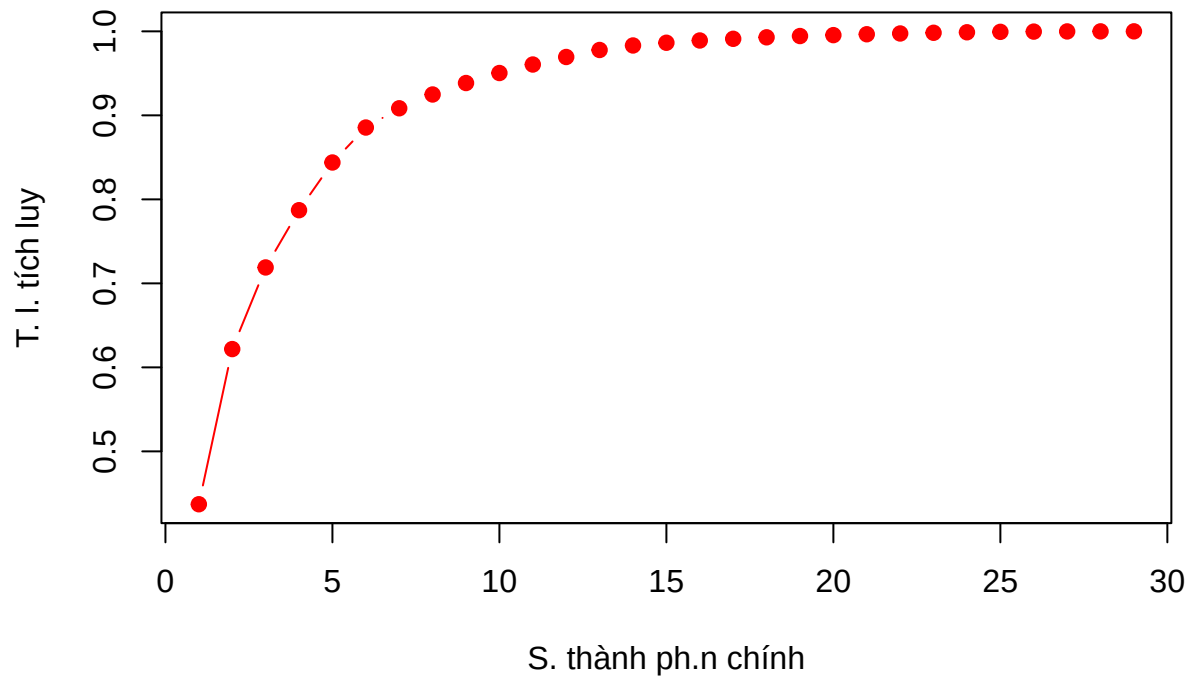
```
screepLOT(pca_wdbc,
  type = "l",
  col = "blue",
  main = "Biểu đồ tỷ lệ phương sai của các thành phần chính (Scree Plot)")
```

Biểu đồ t. l. phương sai c.a các thành phần chính (Scree Plot)



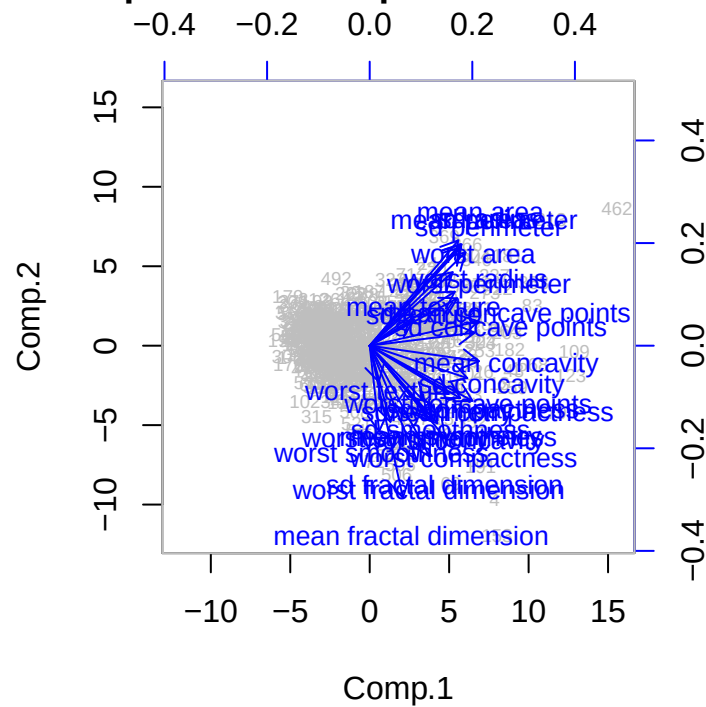
```
# Biểu đồ tỷ lệ phương sai tích lũy
var_explained <- pca_wdbc$sdev^2 / sum(pca_wdbc$sdev^2)
cum_var <- cumsum(var_explained)
plot(cum_var, type = "b", pch = 19, col = "red",
     main = "Biểu đồ tỷ lệ phương sai tích lũy",
     xlab = "Số thành phần chính",
     ylab = "Tỷ lệ tích lũy")
```

Bi.u d. t. l. phương sai tích lũy



```
# Biplot cho 2 thành phần đầu tiên  
biplot(pca_wdbc,  
       scale = 0,  
       cex = c(0.6, 0.8),  
       col = c("gray", "blue"),  
       main = "Biplot hai thành phần chính đầu tiên")
```

Biplot hai thành phần chính đầu tiên



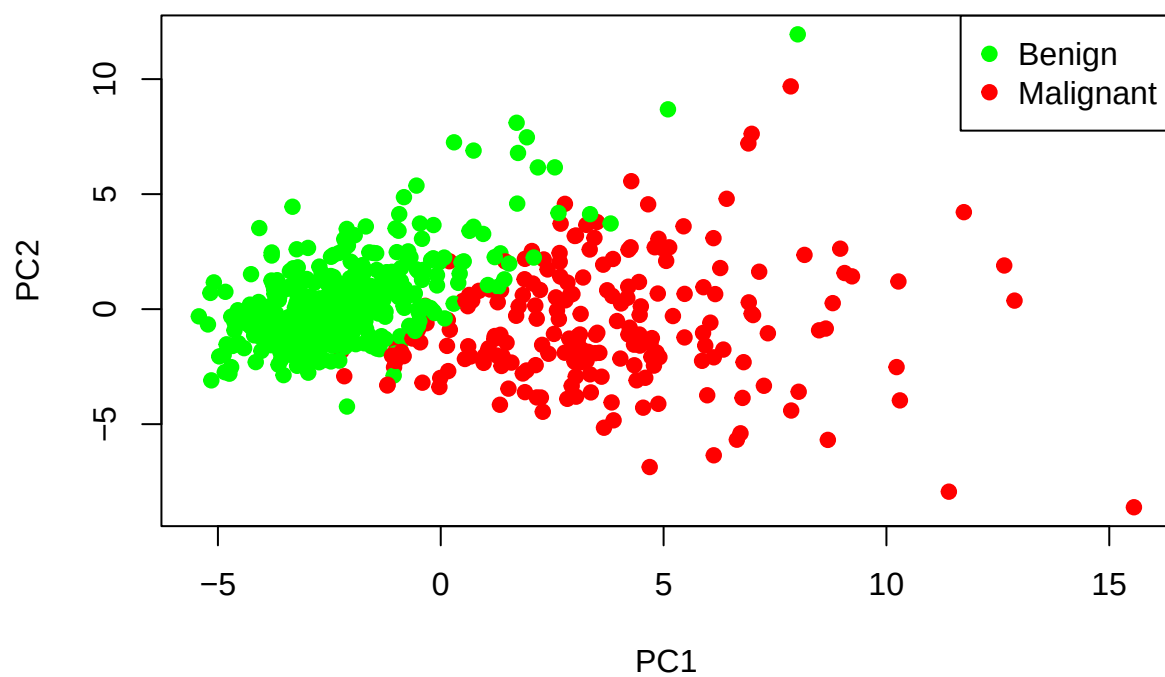
```
# Biểu đồ phân tán (scatter plot) theo PC1 và PC2
pca_wdbc = prcomp(data_scale, center = TRUE, scale. = TRUE)

scores = data.frame(pca_wdbc$x[, 1:2],
                    Diagnosis = wdbc$diagnosis) # giữ nguyên nhãn 'B' và 'M'

# 3 Vẽ biểu đồ phân tán
plot(scores$PC1, scores$PC2,
     col = ifelse(scores$Diagnosis == "M", "red", "green"),
     pch = 19,
     xlab = "PC1", ylab = "PC2",
     main = "Biểu đồ phân tán PC1 và PC2 (Màu theo Diagnosis)")

legend("topright",
     legend = c("Benign", "Malignant"),
     col = c("green", "red"),
     pch = 19)
```

Bi.u d. phân tán PC1 và PC2 (Màu theo Diagnosis)



#5. *Viết nhận xét về kết quả phân tích và ý nghĩa của các thành phần chính:*